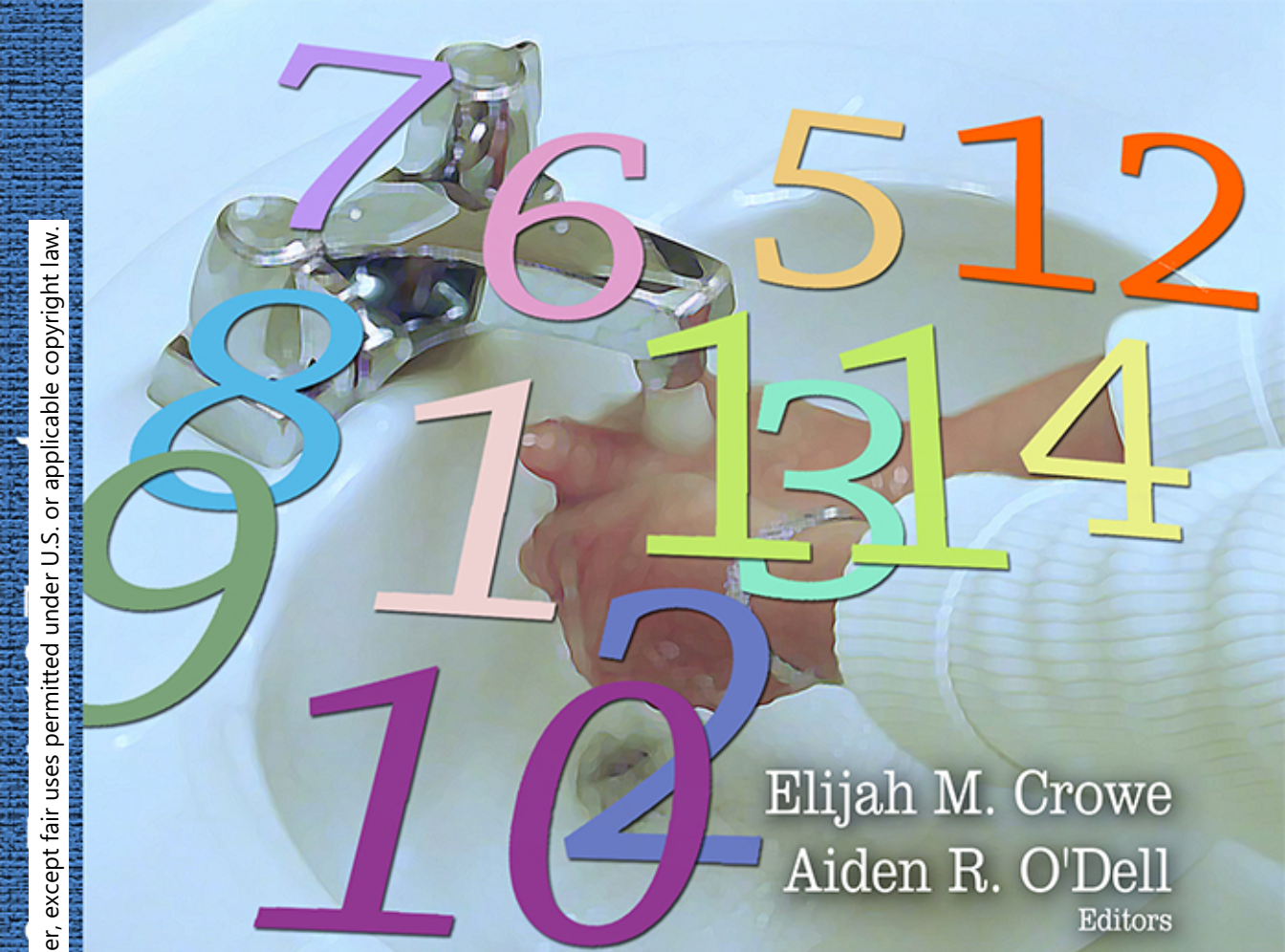


Elijah M. Crowe
Aiden R. O'Dell
Editors

Obsessive-Compulsive Disorder

Symptoms, Prevalence and Psychological Treatments

Health Psychology Research Focus

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OBSESSIVE-COMPULSIVE DISORDER

SYMPTOMS, PREVALENCE AND PSYCHOLOGICAL TREATMENTS

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HEALTH PSYCHOLOGY RESEARCH FOCUS

OBSESSIVE-COMPULSIVE DISORDER

SYMPTOMS, PREVALENCE AND PSYCHOLOGICAL TREATMENTS

ELIJAH M. CROWE
AND
AIDEN R. O'DELL
EDITORS

New York

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Preface

Clinical Obsessive-Compulsive Disorder (OCD) is distinguished from "normal" obsessions and compulsions primarily by the frequency and intensity of the symptoms, and the extent to which they interfere with daily functioning. In this book, the authors present current research in the study of the symptoms, prevalence and psychological treatments for OCD. Topics discussed include the role of Comorbid Personality Disorders (CPDs) in Obsessive-Compulsive Disorder (OCD); the adaptive significance of and natural expression of obsessive and compulsive-like symptoms; OCD in children; cognitive techniques for OCD treatment; the dimensionality of OCD symptoms; internet-based cognitive behavior therapy for OCD; the phenomenology, correlates and psychosocial impact of obsessive-compulsive love; the importance of quality of life in patients with OCD; and an investigation into the dimensions of OCD symptoms in patients with Tourette Syndrome.

Chapter I – The present chapter will critically review the available research on the role of Comorbid Personality Disorders (CPDs) in Obsessive-Compulsive Disorder (OCD). The first section will cover evidence on the prevalence of concurrent axis II conditions in OCD. There is a variety among studies in the reported findings on this topic, with some studies indicating such prevalence as high as 88% and other as low as 9%. Such inconsistency may be attributed to both theoretical and methodological variations across studies. Many studies differ in terms of theoretical definition of personality psychopathology (focus on personality dimensions vs. CPD diagnoses), instruments for CPD assessment (semi-structured interviews vs. self-report measures), research designs (cross-sectional vs. longitudinal), sample selection (inpatient vs. outpatient samples), focus on specific CPDs rather than number of CPDs. This section will also systematically review research on demographic and clinical features related to OCD with CPDs. For example, consistent findings suggested that OCD patients with CPDs tend to be to a higher extent impaired in overall functioning. Conversely, contradictory results have been found on the association between OCD severity and CPDs, with some studies showing a significant association and other not. This section will also focus on moderators of the association between OCD and specific CPDs. For example, a greater rigidity was found to mediate the relation between OCD severity and Obsessive-Compulsive Personality Disorder (OCPD). The second section of the chapter will cover research on the prognostic role of CPDs in the outcome of cognitive-behavioural therapy (CBT) for OCD. The role of CPDs in OCD outcome raises questions on the necessity to adapt treatment planning to the comorbid status. A noticeable body of research indicated that the presence of CPDs may be associated

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with a poorer prognosis for CBT. However, not all studies consistently found such association. Variations across studies could explain such inconsistency. Some studies assessed personality psychopathology as traits and other as categorical CPD diagnoses. Moreover, the role of CPDs will be critically reviewed in relation to differences across trials in terms of treatment formats (for example low vs. high intensity treatments) or therapeutic components. Consistently, some evidence showed that OCD patients with CPDs may benefit from augmentation treatments, like a time-intensive CBT format. Personality factors that negatively impact treatment could also be specific to some CPDs. Recent trials focused on specific CPDs or personality traits, with some results indicating OCPD and schizotypal traits as predictors of non-response. A greater perfectionism and a poorer insight could account for such negative outcome of OCPD and schizotypal personality respectively. These findings may underline symptom patterns that could be targeted by tailored interventions. In conclusion, starting from all the available evidence on CPDs in OCD, implications for clinical practice and directions for future research will be highlighted.

Chapter II – Obsessive compulsive disorder (OCD) is a serious and debilitating condition that affects approximately 1 – 4% of the population. Yet, many individuals that do not meet the criteria for clinical OCD nevertheless report having experienced obsessive-like thoughts and compulsive-like behavior at some time in their lives. Indeed, clinical OCD is distinguished from “normal” obsessions and compulsions primarily by the frequency and intensity of the symptoms, and the extent to which they interfere with daily functioning. Therefore it appears that obsessions and compulsions serve some adaptive purpose, and that OCD might represent a severe dysregulation of these "normal" (adaptive) thought and behavioral patterns. Why do obsessions and compulsions exist? This chapter will compare the phenomenology of “normal” and pathological obsessions and compulsions, in both human and animal populations. In humans, obsessive-compulsive-like symptoms often appear at certain developmental and reproductive stages, such as prepubertally in both genders, premenstrually in women, and, in new parents, associated with the birth of a child. Interestingly, these stages are also associated with an increased risk for the onset or exacerbation of clinical OCD. The authors will discuss theories on the physiological underpinnings, as well as possible adaptive significance, of obsessive-compulsive-like symptoms during these life stages. Although in animals “obsessions” (if they exist) are impossible to observe, many animals certainly display compulsive-like behavior, in the form of stereotyped, stimulus-bound behavioral patterns, such as nest building and courtship displays, as well as rigid habits. In the case of captive animals, exaggerated and stereotyped forms of normal behaviors such as grooming and locomotion represent a pathological condition surprisingly similar to human OCD. By understanding the physiology and adaptive significance of “normal” obsessions and compulsions, the authors might achieve a better understanding of this devastating neuropsychiatric disorder.

Chapter III – Pediatric Obsessive-Compulsive Disorder (OCD) is a prevalent and debilitating condition for many young patients. OCD dictates compulsive behavior and rituals that complicate young patients’ emotional, familial, academic, and social lives. This chapter reviews the literature on symptoms as well as offering both neurobiological and psychosocial explanatory models. Additionally, the extant treatment outcome research is briefly summarized and a modular approach to cognitive behavioral therapy (CBT) is explained. Finally, two cases are presented to illustrate the modular approach to CBT.

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Chapter IV – The exposure and response (ritual) prevention (ERP) was the first treatment revealed to be a truly effective therapy for the treatment of Obsessive Compulsive Disorder (OCD) symptoms, considered as one of the first-line treatments for such disorder back in the seventies. However, a significant percentage, circa 30% of the patients, didn't show any improvement. Many of them were presented with difficulties in joining the activities, and dropouts were very common. Misjudgments and dysfunctional beliefs that were observed almost as a rule in OCD patients not only could explain such difficulties, but have an important role in the appearance and maintenance of symptoms. Correcting them, therefore, could improve the treatment effectivity. Whether the addition of cognitive techniques indeed improves the results of the ERP therapy or not is still an open question, though the clinical impression is that they can be very useful and even relevant for the treatment of certain groups of OCD patients. It is believed that the cognitive approach, for those patients, clears the ground and makes the application of behavioral techniques possible. Though there could be some controversies on the theoretical field between the behavioral model, that proposes the exposition and response prevention, and the cognitive model, that proposes the correction of dysfunctional beliefs for the treatment of obsessive-compulsive symptoms (OC), in the clinical practice, the cognitive therapy is no longer seen as a competing model to the ERP therapy, and gradually cognitive techniques have been incorporated to behavioral techniques that are still thought to be the fundamental strategies for the treatment of OC symptoms. And indeed the designation of Cognitive-Behavioral Therapy (CBT) has been used more frequently. In the present chapter, the authors will describe the most used cognitive techniques for OCD disorder. They will also emphasize the patient's psychoeducation about the cognitive model of the disorder, the role of the thoughts and dysfunctional beliefs in the origin and maintenance of symptoms, their capacity to identify and record them and the use of cognitive techniques to correct them, with the aim to make the treatment more effective. The authors will mention situations in which the techniques could be useful and therefore referred, as well as situations in which those techniques could be either counterproductive or not useful.

Chapter V – Obsessive-compulsive disorder (OCD) has a heterogeneous symptom presentation and a dimensional nature, according to current cognitive formulations. However, although numerous factorial analyses have examined the disorder's content dimensions, there is no agreement about the dimensions that characterize OCD symptoms, due in part to the measures used. Recently, different instruments have been developed that offer a promising dimensional alternative (i.e., DY-BOCS, DOCS, INPIOS, ITTIS) to previous self-reports, and alternative structural proposals. The present paper reviews existing data about OCD heterogeneity, emphasizing results provided by dimensional instruments, and it updates data about the OCD dimensional structure and severity using the Obsessional Intrusive Thoughts Inventory (INPIOS). The INPIOS was specifically designed to assess the content and frequency of 48 obsessional intrusive thoughts, as well as the unpleasantness, difficulty to control and interference of the most upsetting intrusions/ obsessions experienced. Responses of 79 OCD participants are analyzed and compared to responses of a matched non-clinical group. Although both OCD and community subjects experienced intrusions representative of all types of obsessional contents, differences appeared regarding the frequency, unpleasantness, difficulty to control, interference and prevalence of some dimensions.

Chapter VI – From a phenomenological standpoint, eating disorders (ED) and hoarding disorder (HD) have several features in common, and both conditions are closely associated

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with impulse control disorder (ICD) and obsessive-compulsive disorder (OCD). So far, no studies seem to have used specific measures for assessing HD to investigate the presence of hoarding symptoms in patients with ED. There is also a paucity of research on HD in Italy, and even less attention has been paid to its association with ED. The aim of the present study was to identify features of hoarding in a sample of patients with ED, comparing them with a group of self-identified HD individuals. Judging from the literature, the two groups could be expected to have similar levels of compulsive acquisition and obsession with order and symmetry, and similar anxiety and depression symptoms. The authors' study sample of patients with ED included 15.15% who scored higher than the cut-off for hoarding psychopathology in the Saving Inventory-Revised (SI-R). On the subscales of the SI-R, 12% of the ED patients exceeded the clinical cut-off in the Clutter and Acquisition domains, and 9% in the Discarding domain. A comparison between the authors' ED and HD groups showed, however, that the self-identified hoarders scored higher than the ED patients both on the subscales of the SI-R, and on the scales designed to measure their obsessive-compulsive behavior, while the ED patients scored higher for anxiety and depression. The authors' initial hypotheses were therefore not confirmed, even though the patients with ED showed some features of hoarding. Further research is needed, focusing on specific types of ED (e.g. bulimia and binge eating), and analyzing in more depth those domains related to impulse control that seem to characterize both ED and HD.

Chapter VII – Cognitive behaviour therapy is the psychological treatment of choice for obsessive compulsive disorder. Traditionally, cognitive behaviour therapy is delivered via face-to-face sessions with a specialist clinician. It often involves significant portions of therapist-assisted exposure. However, there are many barriers to accessing appropriate treatment and there is a clear need for more effective treatment dissemination. Can CBT be successfully delivered without seeing the patient? Can OCD be properly treated over the Internet? This body of research has been steadily increasing over the last decade, and this review provides the latest evidence for the efficacy, cost-effectiveness, and acceptability of online treatments for OCD. The outcomes of online programs will be compared to those found for traditional face-to-face cognitive behaviour therapy. The clinical advantages and disadvantages of Internet-based treatments will also be discussed and exemplified with examples drawn from the online clinic that the authors operate. This article concludes with recommendations for future research.

Chapter VIII – The construct of obsessive-compulsive love has not been clearly established despite its widespread presence in media outlets such as film, music, and literature. To help clarify this construct and how it affects individuals, this study investigates the phenomenology of obsessive-compulsive love as well as its association with various emotional states, impairments, and relationship conflict. Participants (N = 445) completed measures assessing obsessive-compulsive symptoms, emotional dysregulation, anger, stress, anxiety, depression, loneliness, psychosocial functioning, and relationship conflict with other-sex peers. Additionally, they completed a newly developed measure of obsessive-compulsive love. Results suggest obsessive-compulsive love is relatively common and functionally impairing. In addition, it is associated with a host of difficulties, such as obsessive-compulsive symptoms, emotional dysregulation, anger, stress, anxiety, depression, and loneliness. Lastly, obsessive-compulsive love was found to impact conflict in other-sex relationships indirectly through the influence of emotional dysregulation and anger.

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Collectively, these results suggest that obsessive-compulsive love is relatively pervasive, personally distressing, and problematic for relationships.

Chapter IX – Quality of life (QOL) assessment and improvement are an increasingly important aspect of health care, particularly in the field of mental health. QOL in those with OCD is significantly impaired when compared to that of the general population and of patients with other psychiatric and medical disorders. Therefore, it is important to not only address the relentless symptoms but also this populations' overall well-being. Likewise, QOL in OCD appears to be largely affected by comorbid conditions, suggesting that practitioners should take comorbid disorders into account during treatment planning. While QOL in OCD can improve with medications and with both individual and group psychotherapy, it does not seem to improve to the levels enjoyed by the community. Thus, QOL assessment in both clinical and research settings should be assessed at baseline and during follow up so that the authors can examine the disease burden, monitor treatment efficacy, and strive for true relief. Thus, treatment providers should endeavor to reach symptom remission while also striving for improvements in functioning and quality of life.

Chapter X – Tourette syndrome (TS) is a chronic neuropsychiatric disorder defined by multiple motor and vocal tics, and characterized by frequent comorobidity of obsessive-compulsive disorder (OCD) and obsessive-compulsive (OC) symptoms. OCD comorbid with chronic tic disorders including TS is specified as tic-related OCD. Dimensional approach of OC symptoms is recognized to be important not only in patients with typical OCD but also in patients with tic-related OCD. It is suggested that OC symptoms of aggression and symmetry dimensions are associated with tic-related OCD. In this study, the authors described OC symptom dimensions in TS patients and investigated relationship between OC symptom dimensions, other clinical features including tics, anxiety and depression, and neuropsychological performance with focusing on aggression and symmetry dimensions. OC symptoms of miscellaneous dimension were found most frequently, followed by those of symmetry and aggression dimensions. When change percent of Dimensional Yale-Brown Obsessive Compulsive Scale (DY-BOCS) severity score in each dimension from the worst ever period to the current survey was calculated, aggression dimension showed almost no change, and contamination dimension showed largest decrease, followed by symmetry and miscellaneous dimensions. When global functioning was compared between patients with OC symptoms of aggression dimension as prominent OC symptoms, patients with OC symptoms other than aggression dimension, and patients without OC symptoms, it was lowest in the patients with aggression dimension. DY-BOCS severity scores for aggression dimension correlated significantly with State Trait Anxiety Inventory (STAI) trait scores in adolescents and adults with TS. When findings of neuropsychological tests were compared between patients with OC symptoms of aggression dimension, patients without OC symptoms of aggression dimension, and controls, the patients with aggression dimension performed most poorly on perseverative errors of the Wisconsin Card Sorting Test (WCST). In contrast, OC symptoms of symmetry dimension had almost no significant impact on clinical features including tics, anxiety and depression, and neuropsychological performance. In TS patients, OC symptoms of aggression dimension were associated with low global functioning, trait of anxiety especially in adolescents and adults, and poor performance on the WCST. It is suggested that OC symptoms of aggression dimension have an important role for TS and/or tic-related OCD.

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Chapter I

Obsessive-Compulsive Disorder and Comorbid Personality Disorders

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Abstract

The present chapter will critically review the available research on the role of Comorbid Personality Disorders (CPDs) in Obsessive-Compulsive Disorder (OCD).

The first section will cover evidence on the prevalence of concurrent axis II conditions in OCD. There is a variety among studies in the reported findings on this topic, with some studies indicating such prevalence as high as 88% and other as low as 9% (Black et al., 1993; Pigott et al., 1994). Such inconsistency may be attributed to both theoretical and methodological variations across studies. Many studies differ in terms of theoretical definition of personality psychopathology (focus on personality dimensions vs. CPD diagnoses), instruments for CPD assessment (semi-structured interviews vs. self-report measures), research designs (cross-sectional vs. longitudinal), sample selection (inpatient vs. outpatient samples), focus on specific CPDs rather than number of CPDs.

This section will also systematically review research on demographic and clinical features related to OCD with CPDs. For example, consistent findings suggested that OCD patients with CPDs tend to be to a higher extent impaired in overall functioning (Denys et al., 2004). Conversely, contradictory results have been found on the association between OCD severity and CPDs, with some studies showing a significant association (Aubuchon and Malatesta, 1994) and other not (Matsunaga et al., 1998).

This section will also focus on moderators of the association between OCD and specific CPDs. For example, a greater rigidity was found to mediate the relation between

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OCD severity and Obsessive-Compulsive Personality Disorder (OCPD) (Wetterneck et al., 2011).

The second section of the chapter will cover research on the prognostic role of CPDs in the outcome of cognitive-behavioural therapy (CBT) for OCD. The role of CPDs in OCD outcome raises questions on the necessity to adapt treatment planning to the comorbid status.

A noticeable body of research indicated that the presence of CPDs may be associated with a poorer prognosis for CBT (Keeley et al., 2008). However, not all studies consistently found such association. Variations across studies could explain such inconsistency (Dreessen and Arntz, 1998). Some studies assessed personality psychopathology as traits and other as categorical CPD diagnoses. Moreover, the role of CPDs will be critically reviewed in relation to differences across trials in terms of treatment formats (for example low vs. high intensity treatments) or therapeutic components. Consistently, some evidence showed that OCD patients with CPDs may benefit from augmentation treatments, like a time-intensive CBT format (Dèttore et al., 2013).

Personality factors that negatively impact treatment could also be specific to some CPDs. Recent trials focused on specific CPDs or personality traits, with some results indicating OCPD and schizotypal traits as predictors of non-response (Moritz et al., 2004; Pinto et al., 2011). A greater perfectionism and a poorer insight could account for such negative outcome of OCPD and schizotypal personality respectively. These findings may underline symptom patterns that could be targeted by tailored interventions.

In conclusion, starting from all the available evidence on CPDs in OCD, implications for clinical practice and directions for future research will be highlighted.

1. Introduction

Personality comorbidity in obsessive-compulsive disorder (OCD) is the rule rather than the exception (Brown et al., 2001), and OCD has been found to be the anxiety disorder[1] with the highest prevalence of Comorbid Personality Disorders (CPDs), regardless of the type (Friborg et al., 2013). The relevance of CPDs in OCD is reflected in analyses of healthcare costs showing that patients with concurrent Axis II disorders represent a markedly higher economic burden to the healthcare system than patients with OCD alone (Soeteman et al., 2008).

Clinical impression has associated for a long time CPDs with more severe Axis I disorders, and CPDs have been commonly thought to complicate the therapeutic process, thus resulting in a poorer treatment outcome (Dreessen et al., 1994).

However, research to date has undermined the impact of untreated co-occurring axis II conditions in outcome trials (Sookman and Steketee, 2010), as such conditions have been part of the exclusion criteria for a long time in randomized controlled trials (RCTs) (Brown and Barlow, 1992). Treatment-seeking OCD patients in naturalistic healthcare settings typically present with multiple co-occurring diagnoses, particularly concurrent Axis II conditions (Friborg et al., 2013).

Only recently, a growing body of evidence has allowed to clarify some clinical variables which can account for the co-occurrence of CPDs in OCD (e.g. Coles et al., 2008). Several

[1] This definition is according to DSM-IV-TR not the new version of DSM-5, in which OCD is situated in another chapter: Obsessive-Compulsive and Related Disorders.

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research efforts have been conducted to demonstrate the evidence base of psychological and pharmacological treatments for OCD (Eddy et al., 2004; Gava et al., 2009).

Despite such treatments have been consistently found to be effective for OCD, a relevant group of people suffering from OCD do not fully respond to such treatments, do not comply with therapist's instructions or prematurely drop out of treatment (Olatunji et al., 2013). CPDs could be associated to some therapeutic processes involved in treatment failure (Brandes and Bienvenu, 2006). Starting from these considerations, research trials have started to focus on the effects of CPDs on outcome, also examining potential mechanisms that moderate a negative treatment response.

Disentangling personality patterns from OCD features may facilitate the matching of different patients to different therapeutic modalities and the identification of patients who may be more difficult to treat with standard treatments for OCD (Fricke et al., 2006).

2. OCD with CPDs: Prevalence and Clinical Features

2.1. Prevalence of CPDs in OCD

One way to investigate the association between OCD and CPDs has consisted of examining the presence of one or more full diagnoses of CPDs in OCD. Personality disorders (PDs) have been found to be three times more prevalent in samples of OCD patients than in the general population (Denys et al., 2004). Clinical studies examining the association between CPDs and OCD showed that a mean prevalence of approximately 50% of the OCD patients met the criteria for at least a full CPD diagnosis (Denys et al., 2004; Joffe, Swinson, and Regan, 1988; Mataix-Cols et al., 2000; Mavissakalian, Hamann, and Jones, 1990; Piggott et al., 1994; Skodol et al., 1995). Almost similar prevalence rates have been found for the presence of more than two CPDs (e.g. Bejerot et al., 1998). However, there is a considerable variety among the studies with regard to the reported prevalence of CPDs in OCD. Such prevalence has been estimated to be as high as 88% in some studies and as low as 9% in other (Baer et al., 1992; Black et al., 1993; Crino and Andrews, 1996). According to a British epidemiologic study (Torres et al., 2006), the prevalence of any screen-positive Personality Disorder (PD) in patients with OCD was 74%.

Such inconsistency among research findings could be attributed to a range of factors, including differences in the types of samples used (community individuals, patients with mild or severe OCD symptoms, patients seeking treatment or patients on treatment). For example, in a study conducted on a sample of 173 inpatients with severe, treatment-resistant OCD (Dèttore, Pozza, and Coradeschi, submitted), about 28% of the participants had one or more CPDs.

A recent meta-analysis (Friborg et al., 2013) on 125 papers published since 1980 examined the association between CPDs and anxiety disorders. Results demonstrated that the risk of having at least one CPD, regardless of the type, was the highest for OCD among the anxiety disorders with a mean of 52% (Friborg et al., 2013).

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However, only a relatively small amount of research did provide reliable findings, since several of these studies used samples with small sizes (N<100) (Mataix-Cols et al., 2000; Skodol et al., 1995).

In the largest cross-sectional study on prevalence rates of CPDs conducted on 420 outpatients with a primary DSM-IV diagnosis of OCD (Denys et al., 2004), 36% of the outpatients met the criteria for at least a current CPD.

Cluster C CPDs have been found to occur most frequently in OCD, in about 25% of patients of clinical samples (Denys et al., 2004), and these findings were reliably supported either for Western (Bejerot et al., 1998; Mavissakalian et al., 1993; Thomsen and Mikkelsen, 1993) and non-Western countries, such as Japanese samples (Matsunaga et al., 2000). Such prevalence rates were supported either for outpatients with mild OCD (Denys et al., 2004) and for samples of inpatients with severe OCD (Dèttore et al., submitted).

Some explanations for this stronger association between OCD and cluster C CPDs have been proposed. Some authors have suggested that cluster C CPDs could be the most prevalent cluster in OCD as this type of CPDs is characterized by behaviours that can be seen as maladaptive attempts to cope with anxiety (Siever and Davis, 1991; Skodol et al., 1995). Other researchers have proposed that the pervasive nature of OCD symptoms may influence so many areas of daily functioning that they confound the assessment of CPDs, particularly cluster C CPDs, thus leading to an overestimation of co-occurrence rates for this cluster of CPDs (Black and Noyes, 1997; Ricciardi et al., 1992; Stein, Hollander, and de Caria, 1994). Likewise, reassurance seeking, which is considered a functional equivalent of checking in OCD, may be an essential feature of dependent PD (Summerfeldt, Huta, and Swinson 1998).

In the recent meta-analytic review performed by Friborg and colleagues (2013) comorbid cluster C PDs were observed more than twice as often as cluster A or cluster B PDs. Prevalence rates found at a clusters level in OCD seem to reflect those prevalence patterns typically occurring in the anxiety disorders, since in some reviews (Bienvenu and Stein, 2003; Brandes and Bienvenu, 2009) the proportions are highest in the avoidant, dependent, and OCPD (cluster C), and smallest in the schizoid, schizotypal, and paranoid (cluster A).

The degree of any CPD has been found to vary from 66% (Stein et al., 1993) to as low as 4% (Joffe et al., 1988). OCPD has been consistently found to be the most prevalent Axis II condition in several studies conducted on large OCD clinical samples (Denys et al., 2004; Maina et al., 2008; Mataix-Cols et al., 2000). Several clinical studies conducted on current prevalence rates showed that OCPD was the most prevalent CPD in OCD, with rates of approximately 10% (e.g. Denys et al., 2004), and these findings appear to be confirmed across different types of samples, including outpatients with mild OCD severity (Mataix-Cols et al., 2000), and medication-non-responders inpatients with severe OCD (Dèttore et al., submitted).

A strong association between OCD and OCPD is supported by findings that OCPD occurs more frequently in individuals with OCD than in those with other Axis I conditions or specific anxiety disorders, such as panic disorder and social phobia (Crino and Andrews, 1996; Diaferia et al., 1997; Gordon et al., 2013; Skodol et al., 1995). These rates are substantially higher than prevalence results of OCPD co-occurrence found in community samples, which range between 0.9 to 2.0% (Samuels et al., 2000).

Such consistent findings seem to support a specific association between OCPD and OCD. However, it should be argued that in the meta-analysis conducted by Friborg and colleagues (2013) OCD patients did not have a noticeably higher occurrence of a comorbid OCPD

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compared with the other cluster C PDs, specifically the avoidant PD, despite this absence of difference depended to some extent on the diagnostic method used based on clinical interviews rather than questionnaires.

Research provided evidence somewhat consistent on prevalence rates of cluster A and cluster B CPDs, which seem to be the less common CPDs in OCD. In the study conducted by Denys and colleagues (2004), OCPD was the most prevalent CPD in OCD, followed by dependent CPD, with a rate of about 7%.

Conversely, avoidant, histrionic, antisocial, paranoid, and schizoid CPDs were less common than have been reported in research on general population (Bjil, Ravelli and van Zessen, 1998). However, in this study all the patients were recruited from a secondary mental health setting specialized for anxiety disorders, thus the prevalence rates of some CPDs could have been underestimated as some patients with some impairing CPDs could attend other mental health settings (Denys et al., 2004).

Association between OCD and schizotypal personality disorder (SPD) has been consistently found in several researches (Poyurovsky and Koran, 2005). Intercorrelations between schizotypal and OCD dimensions has been reported in non-clinical samples of students through self-report measures of schizotypy (Dinn et al., 2002; Lee and Telch, 2005; Roth and Baribeau, 2000; Tallis and Shafran, 1997), as well in treatment-seeking samples (Norman et al., 1996; Rossi and Daneluzzo, 2002; Yamamoto et al., 2012).

However, the rate of occurrence of SPD in OCD doesn't appear to be clarified, since the observed rates range from 0 to 50% (Baer et al., 1990; Baer and Jenike, 1992; Jenike et al., 1986; Matsunaga et al., 2000; Rasmussen and Tsuang, 1986; Samuels et al., 2000). For example, large epidemiological studies have shown concurrent SPD to be substantially higher in outpatient samples with mild OCD severity (Denys et al., 2004) than in OCD severe inpatients (27% vs. 0.70%; Dèttore et al., submitted), suggesting that the presence of SPD could not necessarily increase OCD severity.

Substantial differences in gender distribution across samples could explain such a variety of findings.

In a study on 94 Japanese OCD patients Matsunaga and colleagues (2000) found that schizotypal CPD was significantly more frequent among males than females. This result was also replicated for OCPD, despite only at a trend level. Thus, the pervasive pattern of interpersonal deficits marked by a strong discomfort in relation to close relationships, a feature typical of SPD, could explain research findings which showed that male OCD patients tend to have greater functional impairment, odd rituals, and earlier age of onset of OCD (Lensi et al., 1996).

In the study of Matsunaga and colleagues (2000) females were, however, more likely than males to have cluster B (particularly borderline CPD) and dependent CPDs (Matsunaga et al., 2000).

This result is consistent with findings, which suggest that female OCD patients are generally more prone to be dependent on others, even using OCD symptoms to manipulate others (Narita et al., 1974). However, this result has not been confirmed in others studies conducted in Western samples (Castle, Deale, and Marks, 1995).

It could be argued that gender-related differences in personality patterns of presentation could explain these differences found in OCD patients, since cluster A PDs appear to be more common in males, whereas dependent and borderline PDs are more frequently diagnosed in females. Consistent with this reasoning, borderline CPD has been found more frequent among

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women with OCD than SPD in a sample of 140 OCD patients, despite SPD was found to be more frequent than borderline CPD in the overall sample (Dadashzadeh et al., 2013).

2.2. CPDs and Clinical Correlates in OCD

Given this large variability in the CPDs prevalence estimates some recent studies have been conducted on the investigation of clinical correlates of concurrent CPDs that could account for such variability. Research on other anxiety disorders, such as panic disorder, has demonstrated that the association of one or more CPDs with the anxiety disorder tends to be related to higher Axis I symptom severity, and greater level of functional impairment, specifically more severe anxiety, depression, and earlier age of onset (Ozkan and Altintag, 2005).

Consistent findings suggest that patients with OCD and CPDs do not seem to have more severe OCD symptoms compared to those without CPDs (e.g. Cavedini et al., 1997). These findings have been replicated also in samples of patients with severe OCD. Tenney and colleagues (2003) assessed patients with severe OCD and reported that those individuals with one or more CPDs had higher overall levels of functional impairment compared with those without a CPD. Dèttore and colleagues (submitted) reported that in treatment-resistant OCD inpatients either the presence of one or more CPDs and the number of CPDs was not associated with greater OCD severity, despite it was positively correlated with more impairing anxious and depressive symptoms. These latter findings appear in line with some literature reporting that, although it does not increase OCD severity, the presence of CPDs seems to increase risk for other Axis I complaints, for example being associated with a greater number of other Axis I disorders (Fricke et al., 2006).

Most of the available empirical literature focused on the co-occurrence of OCPD or SPD in OCD. Studies have been conducted with the aim to identify sources of heterogeneity in the prevalence rates across samples. A line of research has focused on the clinical correlates of the association of OCD with specific CPDs.

Some evidence suggests that patients with OCD and associated SPD display a more deteriorative course and greater cognitive impairment than counterparts without SPD (McDougle et al., 1990, 2000; Ravizza et al., 1995; Sobin et al., 2000; Spitznagel and Suhr, 2002). OCD patients with SPD tend to present with greater functional impairment (Poyurovsky and Koran, 2005). Poyurovsky and colleagues (2008) compared OCD patients with concurrent SPD with OCD patients without SPD matched on socio-demographic characteristics. Although no significant difference was found between the two groups on OCD symptom severity and other clinical correlates of severity like symptoms subtypes, age at onset of OCD, and lifetime Axis I disorders, the patients with concomitant SPD showed higher patterns related to impairment, as they were more likely to be unmarried or unemployed, had poorer levels of insight, more severe negative schizophrenia symptoms, and lower levels of general functioning (Poyurovksy et al., 2008). In addition, other studies using different OCD symptom severity measures provided further indications that OCD with SPD displays a more severe OCD symptomatology (Dadashzadeh et al., 2013; Fonseca-Pedrero et al., 2010; Lee, Cougle, and Telch, 2005), despite not all studies did show a significant relation between concurrent SPD and OCD severity (Aubuchon and Malatesta, 1994; Stanley, Turner, and Borden, 1990). Evidence that OCD patients with SPD have a greater functional

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impairment could reflect a more general pattern typical of personality disorders, in which individuals with cluster A PDs have higher degree of disability than those with cluster C (Skodol et al., 2002).

Other researchers have suggested that the severity of an Axis I disorder, like OCD, may confound personality assessment, leading to inflated estimates of personality disorder comorbidity (Loranger et al., 1991; Widiger and Francis, 1987).

Recently, some researchers sought to determine which SPD symptoms could be most related to OCD. In a clinical sample of 61 outpatients with OCD Yamamoto and colleagues (2012) found that obsessions severity (assessed through the Yale-Brown Obsessive Compulsive Scale), but not compulsions severity, correlated with the cognitive-perceptual and disorganized factors of the Schizotypal Personality Questionnaire (SPQ; Raine, 1991), whereas the interpersonal factor was not related to OCD. These findings could be interpreted according to the fact that OCD patients with elevated scores on SPD measures experience a reduction of cognitive inhibition which would lead to the frequent entry into obsessions (Yamamoto et al., 2012). This explanation appears also in line with results of a study on a large student sample (Lee et al., 2005), from which it emerged that schizotypal traits, particularly magical thinking, were related to the Likelihood factor of Though Action Fusion (TAF). Such findings were interpreted as evidence that magical thinking could be a risk factor for likelihood TAF, which in turn could lead to developing OCD symptoms (Lee et al., 2005). However, future longitudinal studies are required to establish the causal links between SPD factors and OCD processes.

Given that stronger OCD severity associated to OCD with concurrent SPD, some researchers have proposed a subtype of OCD with schizotypal traits, which could be considered distinct from OCD without such traits (Sobin et al., 2000). These authors refer to a type of OCD with similar traits to the psychotic spectrum disorders. Such traits have been recognized as magical thinking and autogenous obsessions (Sobin et al., 2000).

In conclusion, investigation on prevalence rates and clinical correlates of CPDs in OCD require additional research work, and firm conclusions from the available data cannot be drawn, due to the limited number of studies on prevalence rates and the small sample sizes of studies focusing on clinical correlates. In addition, some limitations related to the recruitment strategies should be considered in that most of the studies did not use a multi-site design or were conducted in specialized clinics for anxiety disorders, thus findings could have poor generalizability to community, to individuals suffering from OCD who do not seek treatment. Moreover, most of the studies were conducted in secondary mental health settings, thus OCD patients could not be representative of customers who utilize primary healthcare services. Furthermore, in those studies examining the relation between certain CPDs and OCD cluster A and cluster B CPDs were underrepresented; this shortcoming could have limited statistical power to detect significant relationships (Mataix-Cols et al., 2000). More investigation through longitudinal studies is required to improve understanding of the role of specific CPDs in the clinical course of OCD. This area of research could have growing directions, as some recent long-term longitudinal studies (Ansell et al., 2011) demonstrated that patients with cluster B, particularly borderline PD, were more likely to relapse after having reached a remission status at 7-year follow-up.

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2.3. OCD Dimensions and CPDs

The wide range in the prevalence rates of CPDs in OCD patients reported in previous studies could be also explained by personality differences across OCD dimensions.

For example, to test this hypothesis one study was conducted on a non-clinical sample using the Personality Diagnostic Questionnaire-Revised (PDQ-R; Hyler and Rieder, 1987) and the MOCI as measures of personality traits and OCD dimensions respectively (Rosen and Tallis, 1995). Findings showed that personality impairment was related only to checking; OCPD traits significantly correlated with checking, slowness, and doubting dimensions, whereas the dependent CPD correlated with washing, slowness, and doubting dimensions (Rosen and Tallis, 1995).

Consistent evidence suggests that hoarding dimension could be related to the presence of CPDs in OCD. Indeed, hoarding has been found to be correlated with the association of one or more CPD to OCD, positively related with the number of CPDs (Mataix-Cols et al., 2000). Hoarding dimension has also shown to significantly predict any cluster C CPD, and specifically it has been found to more strongly correlate with OCD and avoidant PD (Mataix-Cols et al., 2000). Other studies have suggested that the hoarding dimension may be related also to other Axis II disorders, like SPD, dependent, and paranoid PD (Frost et al., 2000). One pilot study conducted on a clinical sample found an association between symmetry and hoarding OCD symptoms and obsessive-compulsive personality traits (Baer, 1994). These findings were partially supported by another study (Mataix-Cols et al., 2000), which reported a significant relationship between cluster C CPDs, specifically OCPD and avoidant CPD, and hoarding symptom measured using a factor-analyzed scores of the Y-BOCS Symptom Checklist. In this study diagnoses were based on diagnostic criteria of the DSM-III-R, in which "difficulty to throwing away useless objects" was one of the nine criteria for OCPD diagnosis. However, the correlation between hoarding and OCPD was not uniquely attributable to the overlap with this criterion, since the authors found that other specific diagnostic criteria were related to hoarding, such as "unreasonable insistence that others submit to his/her way of doing things or reluctance to allow others to do things", "Indecisiveness", and "Overconsciousness, scrupulousness, inflexibility" (Mataix-Cols et al., 2000). In another study (Horesh et al., 1997) conducted in Israel on 51 patients with a primary OCD, washers patients were 32 times more likely to have CPDs than checkers. Such higher levels of impairment, perhaps due to personality comorbidity, could explain why in some trials OCD patients with higher levels of compulsive washing have been found to have a poor response to CBT or medications than those with stronger levels on compulsive checking (Clark, 2003). Despite in this study the authors used a semi-structured interview to assess CPDs, OCD dimensions were not assessed by a standardized measure; in addition, the small number of participants limited the study. Moreover, none of these studies examining the relationship between CPDs and OCD on clinical samples partialled out the effects of general distress. It could be argued that anxiety and depression levels influence this relation; OCD is such a debilitating disorder that it confounds the assessment of personality, therefore this could lead to the finding that several different PDs coexist with OCD (Ricciardi et al., 1992).

In a recent study on a large sample of OCD patients Ecker, Kupfer, and Gönner (in press) reported a significant relation between obsessive-compulsive personality traits and incompleteness-linked OCD dimensions (ordering and checking), suggesting that this relation could be explained by incompleteness feelings, as this relation did not result significant after

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controlling for the effects of incompleteness feelings. This link could be due to the effects of not just right experience, which could play as a common affective-motivational underpinning for ordering/checking dimensions and obsessive-compulsive personality traits (Ecker et al., in press).

2.4. The OCD Onset Time and CPDs

Some studies suggested that an Axis I disorder which had its onset during childhood might predate a CPD in the adulthood (e.g. Ramklint et al., 2003). Overall, some research examined the prevalence of CPDs during adulthood in relation to the onset of OCD, suggesting that the onset of OCD during childhood could increase the risk for CPDs during adulthood. One study (Grant et al. 2007) reported that adult patients with OCD onset before age 18 (early-onset group) were more likely to have Axis II comorbid conditions than those with onset at ages 19-29 (young-onset group) and those with onset after age 30 (late-onset group), despite this difference did not reach the significant level. In another study Hemming and colleagues (2004) found a significantly higher prevalence of schizotypal and borderline CPDs in adults with OCD onset before age 15 compared with those with a later onset. However, the strongest limitations of this study were that the authors evaluated the prevalence rates of only these two PDs diagnoses. In addition, the choice of age 15 could be considered too high as a cut-off point to determine early onset of OCD (Rosario-Campos et al., 2001). In a more recent investigation on a large sample of adult patients with a primary OCD (Maina et al., 2008), individuals with an early onset of OCD ($\leq$ 10 age) were more likely to have an OCPD in the adulthood than the counterparts with late onset, and this result did not emerge for the other PD diagnoses. These findings seem to suggest that the relationship between early OCD and OCPD is uniquely a result of lower global functioning and impaired relationships during a developmental stage crucial for the personality maturation. Conversely, it could be hypothesized that the development of OCPD works as a personality style in order to cope with the need for perfection and control imposed by OCD (Maina et al., 2008). An alternative explanation could be that child-onset OCD is a specific subtype sharing a common aetiology or developmental pathway with OCPD (Maina et al., 2008).

However, it should be noted that both the studies used cross-sectional designs, and were based on the retrospective reports of age at onset, and this could not accurately reflect true onset of OCD symptoms.

Two long-term longitudinal studies on children with OCD have investigated the onset of PDs during adulthood. Thomsen and Mikkelsen (1993) following for 27 years 47 children with a OCD diagnosis reported that all the cluster C PDs were the most prevalent during the adulthood. Wewetzer and colleagues (2001) followed 55 children for a mean duration of approximately 11 years, reporting that a OCPD was significantly more diagnosed in the sample of the patients with OCD than that with patients without OCD. Despite the longitudinal design of these studies, it could be also argued that early-onset OCD is an earlier manifestation of genetic or environmental influences that also affect the risk for certain PDs, or, alternatively, early OCD symptoms could be, in some cases, prodromal phenomena of PDs (Brandes and Bienvenu, 2006).

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2.5. Personality Traits Associated to OCD

Recently, some researchers have highlighted the importance of conceptualizing PDs through a dimensional approach (Krueger et al., 2007; Trull and Durrette, 2005). A number of authors have evidenced the problematic features of the DSM-IV PDs model (American Psychiatric Association, 2000), including an atheoretical approach to diagnostic comorbidity, poor convergent and discriminant validity, poor reliability, and presumed discontinuity between normal and abnormal personality in the absence of evidence for such a discontinuity (Clark, 2007; Widiger and Trull, 2007). Specifically, with regard to the issue of presumed discontinuity, arbitrary cut-off are highlighted as a further point of criticism creating artificial categories that disregard the continuum between normal and abnormal personality (Widiger, 1993). While a noticeable body of research has examined the relationship between OCD and CPDs adopting a categorical diagnostic approach, less evidence has been provided on personality features in OCD through a dimensional approach (Wetterneck et al., 2011).

Some research has shown that OCD patients tend to score higher than healthy controls on self-report measures of normal personality dimensions such as neuroticism (Samuels et al., 2000), and trait anxiety (Hoehn-Saric et al., 1995).

The majority of the studies in this field used non-clinical samples, such as those formed by undergraduates. The rationale for the use of non-clinical sample in OCD research relies on findings, which suggest that obsessive-compulsive phenomena are quite common in the general population. For example, approximately 80% of non-treatment-seeking individuals experience obsessions similar to those of OCD patients (e.g. Berry and Laskey, 2012), and about 30% of community individuals report high levels of compulsive checking behaviours (Gershuny et al., 2000).

Within the research context on personality traits involved in OCD the dimensional unified psychobiosocial model of personality has been widely used (Cloninger, Svrakic, and Przybeck, 1993). This dimensional mode includes four temperament factors which are considered to remain stable throughout life, and three character factors which are believed to develop in response to life experiences (Cloninger et al., 1993). The four temperament factors are novelty seeking, harm avoidance, reward dependence, and persistence, while the three character factors are composed of self-directedness, cooperativeness, and self-trascendence (Cloninger et al., 1993). Studies using the Temperament and Character Inventory (TCI; Cloninger et al., 1993) in patients with OCD have shown consistent results of higher harm avoidance scores in OCD patients than normal controls (Pfohl and Black, 1991; Kusunoki et al., 2000; Bejerot et al., 1998; Kim, Kang, and Kim, 2009). In contrast, results on other subscales of the TCI were not as consistent. Some researchers have reported lower self-directedness (Kim, Kang, and Kim, 2009 Lyoo et al., 2001), cooperativeness (Bejerot, Knorring, and Ekselius, 2000), or higher reward dependence in OCD patients (Pfohl, Black, and Coryell, 1991). It seems difficult to draw definitive conclusions about the specificity of some personality factors in OCD since most of these personality patterns appear common characteristics of all types of personality disorder (Cloninger et al., 1993). Moreover, the patterns of higher harm avoidance and lower self-directedness have been found also in other Axis I conditions, like major depressive disorder (Kusunoki et al., 2000).

Results of the relationship between temperament and character profiles and OCD symptoms severity have also been controversial. In some studies lower self-directedness and higher harm avoidance scores have been shown to correlate with OCD symptom severity (e.g.

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Cruz-Fuentes et al., 2004). More recently, however, a study failed to detect a significant correlation between any of the TCI subscales and the Y-BOCS scores (Alonso et al., 2008).

In the context of research on the relation between OCD and concurrent SPD, a dimensional approach to study the role of schizotypy has been followed, based on the evaluation of positive, negative and disorganized symptoms on a continuum (Siever and Davis, 2004).

Within the dimensional approach to the investigation of PDs, personality psychopathology has been examined also in terms of extremes on continuums of normal personality traits (Costa and McCrae, 2010; Widiger and Mullins-Sweatt, 2010).

Gershuny and colleagues (2000) compared OCD patients with checking, anxious patients without checking, and non-clinical controls on a measure of the big five personality factors, the Goldberg 5-Factor Model of Personality Questionnaire (GPQ; Goldberg, 1992). Results showed that OCD checkers had higher levels of emotional instability than anxious controls and non-clinical participants. In addition, OCD patients showed higher levels of conscientiousness than anxious individuals, and this results was attributed by the authors to the fact that OCD patients tend to have also higher levels of perfectionism (Gershuny et al., 2000).

More recently investigations on personality traits in OCD have used the Five Factor Model of Personality framework (FFM; Costa and McCrae, 1992). The FFM framework consists of five higher-order domains, each representing six specific lower-order facets, and is significantly related to various forms of psychopathology in adults (Malouff, Thorsteinsson, and Schutte, 2005). Studies with the FFM NEO-PI-R measure (Costa and McCrae, 1992) in community and treatment-seeking adults with OCD have demonstrated higher Neuroticism and lower Extraversion scores for OCD patients relative to controls (Bienvenu et al., 2004; Clark, 2005; Rector et al., 2002). Other preliminary evidence suggested that OCD may be associated with higher Agreeableness and lower Conscientiousness levels (Rector et al., 2002; Samuels et al., 2000).

Recent work developed by Wu and colleagues (Wu, Clark, and Watson, 2006) further suggests that, beyond general trait measures, specific maladaptive trait measures may be necessary to provide a more exhaustive description of OCD. These authors identified a more specific pattern of personality psychopathology in OCD adults compared with general outpatients, demonstrating that besides elevated Neuroticism and lower Extraversion scores, which are common to various forms of psychopathology (Clark, 2005; Clark, 2007), OCD patients are also characterized by extreme scores on several maladaptive lower-order dimensions, including lower scores on manipulativeness, mistrust, inhibition, self-esteem, and entitlement.

Some research has investigated the association between personality traits and OCD symptoms in children and adolescents samples. Ivarsson and Winge-Westholm (2004) with cluster analysis in a sample of 83 children and adolescents with OCD described two temperamental subgroups: the uninhibited and the inhibited groups. In the latter subgroup the authors also hypothesized the existence of an autistic-like subgroup characterized by low levels of sociability and presence of repeating compulsions (Ivarsson and Winge-Westholm, 2004).

In a recent study (Marchesi et al., 2008) using the Tridimensional Personality Questionnaire (TPQ; Cloninger, 1987), high-scoring adolescents on a OCD measure were

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characterized by a compulsive temperament with high harm avoidance and persistence levels and low levels of novelty seeking.

More recently, Aelterman and colleagues (2011) examined the association between OCD symptoms and personality traits in a large sample of non-clinical children and adolescents. Findings showed that participants with higher levels of OCD symptoms had significantly higher levels of emotional instability and introversion, assessed through the Dimensional Personality Symptom Item Pool (DIPSI; De Clercq, De Fruyt, and Mervielde, 2003).

In addition, children and adolescents with higher OCD symptoms scores showed significantly elevated levels of neuroticism, and lower levels of extraversion , assessed through the Hierarchical Personality Inventory for Children (HiPIC; Mervielde and De Fruyt, 1999).

2.6. The Relation between OCD and Obsessive Compulsive Personality Disorder

A significant room for research has been dedicated to the relation between OCD and OCPD given the high comorbidity rates and the consistent body of theoretical interpretations of such co-occurrence (Coles et al., 2008). Some researchers have suggested that not only there is a relationship between Axis I and Axis II disorders, but that the various diagnoses are woven together (Wetterneck et al., 2011). As a result, for example, patients with psychotic disorders tend to be more frequently diagnosed with a cluster A CPD (Oldham et al., 1995; Pfohl et al., 1991). Based on this assumption, OCD patients are expected to be diagnosed more frequently with cluster C CPDs, particularly OCPD (Starcevic et al., 2012).

The idea of an association between OCD and OCPD has been the subject of debate during the last 100 years (Coles et al., 2008). According to the psychodynamic theories (Janet, 1904; Freud, 1908/1963), the two disorders could be considered as not only associated, but as causally linked.

Janet (1904) described the development of frank obsessions and compulsions as being preceded by a prodromal state which he defined as “psychastenic state”, characterized by a sense that actions are performed imperfectly, a strong focus on order, indecisiveness, and restricted emotional expression. In accordance with Janet’s clinical observations, Freud (1908/1963) proposed the construct of anal character, a typology presenting with obstinacy, orderliness, and parsimony. Such character was regarded by the author as a necessary precursor of the development of OCD (Freud, 1908/1963). These suggestions have been subsequently integrated into the modern definitions of OCPD (Mancebo et al., 2005). Given this hypothesized causal link, therefore, OCD would be more likely to occur with OCPD than any other PD (Krockmalik and Menzies, 2003).

Although several studies found a strong association between OCD and OCPD, indicating OCPD as the most frequent CPD in OCD, indeed a great variety of OCPD comorbidity rates has been found. Research on the association of OCPD to OCD has shown heterogeneous findings, with studies reporting prevalence rates ranging from 2% (Pollack, 1987; Sciuto et al., 1991; Steketee, 1988) to over 60% (Mavissakalian et al., 1990; Mavissakalian et al., 1993).

This heterogeneity in research findings on prevalence rates of OCPD in OCD compared to other Axis I disorders could be attributed to the adoption of diagnostic criteria based on

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different classification systems (Eisen et al., 2006). Reviewing the literature over the evolution of the OCPD definitions, Pfohl and Black (1991) highlighted that studies using DSM-III-R diagnosed OCPD at approximately double the rate of studies using the DSM-III criteria. Indeed, several studies using the stringent DSM-III criteria, which required four of five OCPD criteria to be present, found low rates (0-6%) of OCPD in OCD patients (Baer et al., 1990; Black et al., 1989; Joffe et al., 1988), despite also using DSM-III criteria for OCPD, Eisen and Rasmussen (1991) reported that 19% of OCD participants met criteria for OCPD.

Investigations adopting the more lenient DSM-III-R criteria, which require five of nine OCPD criteria to be present, found frequencies of OCPD as high as 30% in samples of OCD patients (e.g. Diaferia et al., 1997). Nevertheless, studies using DSM-IV OCPD criteria and structured clinical interviews on large clinical samples reported rates ranging from 2-9% (Denys et al., 2004) to 30% (Maina et al., 2008), and these prevalence patterns appear to be more elevated compared to community samples, which resulted of about 2-3% (Karno et al., 1988). Considering jointly prevalence data from studies using DSM-III and DSM-III-R OCPD criteria, compared with those using DSM-IV or DSM-IV-TR, some noteworthy differences emerge. Studies using the DSM-III and DSM-III-R criteria for OCPD have found OCPD comorbidity rates varying from 6 to 3% in clinical samples (Baer et al., 1990; Diaferia et al., 1997; Mancebo et al., 2005). More recent clinical studies using DSM-IV criteria have found OCPD comorbidity rates ranging from 23 to 32% (Albert et al., 2004; Coles et al., 2008; Garyfallos et al., 2010; Pinto et al., 2006; Samuels et al., 2000).

The association between OCD and OCPD has been supported also by studies conducted on clinical samples with a primary PD. For example, McGlashan and colleagues (2000) found that 21% of their participants with DSM-IV OCPD met criteria for OCD. Similarly, Eisen and colleagues (2006) reported that in a sample of 733 patients with PDs, of the 262 individuals who had an OCPD diagnosis 20% had also OCD, and OCD was substantially more frequent among patients with OCPD compared to those with other PDs.

A possible explanation for this variety of prevalence rates of OCPD in OCD could be related to the role of specific OCPD symptoms in OCD, as not all the OCPD symptoms could be equally associated to OCD. Examining the relation between OCD and OCPD at a symptom level seems to have some relevance since patients with a OCPD diagnosis can present with a variety of different combinations of four or more of the eight diagnostic criteria (163 possible combinations) (Wetterneck et al., 2011).

Consistent with reasoning, Eisen and Rasmussen (1991) assessed the DSM-III-R individual criteria for OCPD in 114 OCD patients, showing that the majority of patients endorsed more likely perfectionism and indecisiveness; in contrast, rigidity and excessive devotion to work were seen less frequently. More recently, Eisen and colleagues (2006) found that three of the eight DSM-IV OCPD criteria, specifically hoarding, perfectionism, and preoccupation with details were significantly more common in patients with OCD than in those without OCD. Conversely, frequencies of the remaining criteria (rigidity, miserliness, reluctance to delegate, inflexible morality, and excessive devotion to work) were not found to differ significantly between individuals with and without OCD, with small magnitudes of the between-groups differences (Eisen et al., 2006). Interestingly, different OCPD symptoms were associated with different anxiety disorders (Eisen et al., 2006). For example, OCD was found to be associated with higher odds of meeting the hoarding, perfectionism, and preoccupation with details criteria, whereas GAD was associated with significantly decreased odds of meeting the preoccupation with details and reluctance to delegate tasks criteria, and

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social phobia and panic disorder with agoraphobia were not associated with any of the OCPD criteria (Eisen et al., 2006). Such interactions between OCPD symptoms and anxiety disorders could highlight the role played by transdiagnostic factors. For example, the two OCPD symptoms, found to be associated also with GAD (namely preoccupation with details, and reluctance to delegate tasks), could be associated with the construct of intolerance of uncertainty, that has been shown to be common to both GAD and OCD (Comer et al., 2004; Gentes and Ruscio, 2011).

Thus, the fact that perfectionism seems to be an OCD-specific criterion could also explain why OCD patients with OCPD have a better treatment response to CBT designed for OCD as it has been suggested by early trials (Fals-Stewart and Lucente, 1993). However, this hypothesis appears only tentative due to the small number of studies on the relation between comorbid OCPD and CBT outcome, and the fact that evidence on this relation is yet unclear, given that this positive relation has not been reported in all the available trials (e.g. Pinto et al., 2011). In addition, the findings reported by Eisen and colleagues (2006) appear in line with the explanation according to which the strong relationship between OCPD and OCD could be primarily due to the criteria of hoarding, perfectionism, and preoccupation with details. It could be argued that these three OCPD symptoms are associated with psychopathology in general, and not specific to OCD. Indeed, perfectionism has been considered a transdiagnostic entity, since it has been shown to be associated with many forms of psychopathology (Egan, Wade, and Shafran, 2011; Flett and Hewitt, 2002).

With regard to the longitudinal associations between OCD and OCPD, research has provided inconsistent findings. Although OCPD appears to be neither a necessary nor sufficient factor in the development of OCD, some early evidence suggested that OCD could be more frequently associated with premorbid OCPD patterns than others personality patterns (Pollack, 1987). Examining the course of development of OCD, Rasmussen and Eisen (1998) used retrospective assignments of childhood traits in 90 adults with OCD. Results showed that a significant proportion of the OCD patients endorsed having perfectionism, hypermorality, and excessive devotion to work prior the onset of OCD, thus suggesting that this constellation of traits could be a precursor of the adult OCPD (Rasmussen and Eisen, 1998). Different findings were reported in the Collaborative Longitudinal Personality Disorders Study by Shea and colleagues (2004), who did not identify a clear association between the course of OCD and the course of OCPD.

However, to date there is a lack of prospective studies in the literature investigating the unique contributions of specific OCPD symptoms as developmental antecedents in the OCD onset.

A further test of specificity of OCPD symptoms to OCD would be the evidence that changes in those OCPD symptoms specific to OCD predict changes in OCD symptoms over time. In the literature there is to date a study performed by Shea and colleagues (2004), who found that improvement in OCPD did not significantly predict improvement in OCD symptoms. This finding seems not to confirm a putative specificity in the OCPD/OCD relation. However, this study did not assess the predictive contribution of specific OCPD symptom changes in the outcome, thus future large studies are needed to add knowledge on the linkage between OCPD and OCD (Eisen et al., 2006).

Some researchers have proposed that OCPD may be a subtype of OCD. Indeed, some data have suggested that OCPD tends to occur more frequently in individuals with OCD than in those with other Axis I disorders. For example, lower rates of OCPD comorbidity have

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been shown in panic disorder (11.4%), and in major depressive disorder (13.8%) (Diaferia et al., 1997; Gordon et al., 2013).

Some early studies reported a strong association between obsessive compulsive personality traits and OCD symptoms (Ingram, 1961). However, a later review of the available evidence on the comorbidity between OCD and OCPD highlighted that OCPD was not associated to a greater extent with OCD than other PDs, suggesting that OCD and OCPD could be separate diagnostic conditions that can co-occur (Pollack, 1987).

Although the long-standing theoretical interest in the relation between OCD and OCPD, indeed there is a relative dearth of empirical research on this topic, and specifically little evidence has been provided to date on clinical correlates of this comorbidity pattern (Grant, Mooney, and Kushner, 2012).

One strategy to shed some light on the relationship between OCD and OCPD has been to systematically compare OCD individuals with and without OCPD (Starcevic et al., 2012). However, research on this topic yielded mixed results. For example, with regard to OCD symptom severity most studies did not find a significant difference in OCD severity, assessed via the total Y-BOCS scores, between OCD patients with and without OCPD (Diaferia et al., 1997; Garyfallos et al., 2010; Pinto et al., 2011). In contrast, one more recent study (Lochner et al., 2011) showed that OCD patients with comorbid OCPD had a more severe symptomatology than those without OCPD, whereas other studies reported that patients with OCPD had more severe compulsive symptoms although they did not have a more severe overall OCD (Albert et al., 2004; Coles et al., 2008). This could be also related to the fact that for example in one of these studies (Coles et al., 2008) relative to the OCD patients without OCPD, those with OCPD were more likely to meet criteria for a comorbid anxiety disorder and an avoidant PD. In addition, it could be considered that a comorbid OCPD may introduce a "double burden" in terms of intensified needs for order and perfection when performing OCD-related rituals in order to reduce anxiety (Friborg et al., 2013).

Some conflicting results have been found with regard to the differences between OCD with and without OCPD in terms of functional impairment and course of OCD. Four studies found OCPD was associated with greater disability or impaired functioning (Coles et al., 2008; Denys et al., 2004; Garyfallos et al., 2010; Lochner et al., 2011). However, functional impairment in individuals with comorbid OCPD does not appear to be related to OCD, as in two of these studies (Coles et al., 2008; Garyfallos et al., 2010) such patients presented with more impairment without having more severe OCD.

Nevertheless, in those studies different measures were used, and this issue could make it difficult to compare findings (Starcevic et al., 2012). In addition, in a large sample of inpatients with severe OCD (Dèttore et al., submitted) the presence of concurrent OCPD was associated with significantly higher anxious and depressive symptoms, but such differences were not found on OCD severity. Moreover, differences in depression and anxiety did not emerge between inpatients with comorbid OCPD and those with CPDs other than OCPD (Dèttore et al., submitted).

Some inconsistency has been found on the differences between OCD with and without OCPD in terms of OCD age of onset, with some studies reporting no significant difference between these two conditions (Albert et al., 2004; Diaferia et al., 1997; Lochner et al., 2011), and other studies showing an earlier onset for individuals with concurrent OCPD (Coles et al., 2008; Garyfallos et al., 2010).

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Some research has consistently shown that symmetry and hoarding OCD symptoms were more frequently associated with OCD with comorbid OCPD (Coles et al., 2008; Garyfallos et al., 2010; Lochner et al., 2011).

Some studies investigating the patterns of co-occurrence with other Axis I mental disorders in OCD patients with and without OCPD have provided conflicting results. One study reported that any mood disorder was significantly more frequent among individuals with OCPD (Coles et al., 2008), whereas other studies did not confirm this relationship (Garyfallos et al., 2010; Lochner et al., 2011).

Coles and colleagues (2008) reported that any anxiety disorder was significantly more frequent among OCD patients with OCPD, thus confirming a more general evidence that the presence of a concomitant PD could represent a vulnerability factor for developing other Axis I disorders (Stein, Hollander, and Skodol, 1993); however, such relationship did not emerge in another study (Garyfallos et al., 2010).

There is also a growing body of evidence highlighting the relationship between OCD and OCPD based on familiality. Frost and colleagues (1994) reported that parents of participants with subclinical OCD symptoms tend to be more-risk-aversive and, specifically, fathers are more critical and perfectionistic. In a family study (Samuels et al., 2000), the first-degree relatives of OCD probands were twice as likely to have OCPD as compared to the relatives of control probands (11.6% vs. 5.8%, respectively). Moreover, several studies have shown increased frequencies of OCPD traits in the parents of paediatric OCD probands compared with the parents of healthy children (Calvo et al., 2009; Lenane et al., 1990; Swedo et al., 1989). However, other researchers have suggested that similarities in phenomenology can make it difficult to differentiate between OCD and OCPD. For example, perfectionism, while a criterion for OCPD, can be regarded as a symptom of OCD if it involves order, symmetry, and arranging (Mancebo et al., 2005).

In conclusion, accumulating empirical evidence on the relation between OCD and OCPD to date does provide some indications that these two conditions could be closely linked. However, research on OCPD specificity in OCD has not yet offered unequivocal evidence, despite most of the studies point in the direction that OCPD is more frequent among individuals with OCD symptoms, either in clinical and in community samples. In order to better understand the underpinnings of such discrepancies in the studies, further researches conducted on large clinical samples with different forms of psychopathology are required, across different cultural sites. It seems also necessary more research based on longitudinal design to clarify the causal link between the two disorders. Indeed, previous research has shown that OCPD could play a role as risk factor for new onset or chronicity of panic disorder (Alnæs and Torgersen, 1999). A recent multi-site longitudinal study (Ansell et al., 2011) found that OCPD was associated with increased risk for new onset of OCD, but also generalized anxiety disorder (GAD) and agoraphobia, and higher risk for GAD relapse after 7 years.

3. The Role of CPDs in Treatment Outcome of OCD

Cognitive behaviour therapy (CBT) with exposure and response prevention (ERP) is the empirically-supported psychotherapy of choice for OCD (American Psychiatric Association,

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2007; Gava et al., 2009; Olatunji et al., 2013), and Serotonin Reuptake Inhibitors (SRIs) are the most effective pharmacological treatment (Eddy et al., 2004). Several randomized controlled trials (RCTs) have found that CBT combined with medications is no more effective than CBT alone (e.g. Foa et al., 2005). Improvement tends to be more sustained with ERP compared to pharmacotherapy, whereas adding ERP to medications significantly improves treatment efficacy, and reduces risk for relapse compared with medications alone (Simpson et al., 2005; Simpson et al., 2008).

RCTs on CBT for OCD report a rate of 70-80% of clinical improvement (Clark, 2003). Given such evidence base, the Expert Consensus Guidelines strongly recommended that patients with OCD receive CBT alone or in combination with SRIs as the first line treatment (March et al., 1997).

However, such rates of clinical improvement has been considered an overestimation of clinical change observed in routine psychotherapy practice, since most of the RCTs include only patients with moderate levels of OCD symptomatology, in absence of comorbid disorders (Brown and Barlow, 1992; Dèttore, Pozza, and Coradeschi, 2013). Despite RCTs warrant more accurate internal validity with strict inclusion criteria, it is argued that the results obtained are of limited relevance to ordinary clinical practice as in such trials patients are often recruited through the mass media or participate in the research project on their own initiative, thus they could be highly motivated to change (Hollon and Wampold, 2009). This issue has strong relevance since readiness for change has shown to be one of the strongest predictors of clinical improvement in CBT for OCD (de Haan et al., 1997; Vogel et al., 2006). Most of the available outcome trials have been conducted in academic mental heath settings specialized for anxiety disorders treatment, whereas little research has been conducted on naturalistic healthcare services, like primary healthcare or private practice settings.

Available evidence suggested that approximately 50% of OCD patients do not optimally respond to CBT even when combined with pharmacotherapy (Sookman and Steketee, 2010). This proportion includes patients who refuse to participate or drop out of ERP (20%), do not improve (25%), or have relapsed at follow-up (Couttraux, Bouvard, and Milliery, 2005). Moreover, in the few studies where it is reported only 25% of those OCD patients who achieve improvement do experience a full recovery status at the end of treatment (Fisher and Wells, 2005). Clinical impression suggests that CPDs could increase risk for such a negative outcome, either in terms of early treatment discontinuation, lower compliance with homework, and a poorer response after a first-line treatment course or even relapse after immediate clinical improvement (Keeley et al., 2008; Young, Klosko, and Weishaar, 2003). In effect, CPDs have been found to be predictive of a negative response for several Axis I conditions either with regard to psychological and pharmacological treatments (Newton-Howes, Tyrer, and Johnson, 2006; Olatunji, Cisler, and Tolin, 2010).

The investigation on the predictive value of CPDs with regard to treatment response for OCD represents an important field of research, since understanding which CPDs predict a worse treatment outcome could inform the development of personalized interventions to be incorporated in evidence-based treatment protocols (Keeley et al., 2008). However, relatively few studies have been conducted on the relationship between CPDs and outcome in OCD psychotherapy research (Keeley et al., 2008).

A line of investigation has focused on the prognostic role of the presence of one or more CPDs associated with OCD (Keeley et al., 2008). This research approach has been developed starting from the literature data which suggested that the co-occurrence of one or more CPDs

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was associated with greater Axis I disorder severity and a more disabilitating impairment (e.g. Denys et al., 2004). In this vein, Steketee, Chambless, and Tran (2001) demonstrated that the presence of a CPD, regardless of the type (anxious, dramatic or odd), predicted treatment failure after CBT. In another study Dreessen, Hoekstra, and Arntz (1997) found that OCD associated with at least a CPD, assessed on a categorical or on a dimensional basis, was not predictive of dropout, and negative response to CBT at post-treatment or follow-up assessment.

In this study such results were similar when completer or intent-to-treat-analyses were applied (Dreessen et al., 1997). This finding appears relevant, as it could be argued that patients with CPDs could leave treatment early when it does not work; therefore, it would be too optimistic conclude that personality psychopathology does not play a role in treatment response while not evaluating treatment effects also including in the analyses non-completers patients (Dreessen et al., 1997).

More recently, a meta-analysis (Olatunji et al., 2013) summarized evidence from 13 randomized controlled trials on efficacy of CBT for OCD, and concluded that the presence of one or more Axis II conditions was not associated with a negative outcome. However, it should be argued that the use of moderator analyses, as it occurs in meta-analyses, provides information of limited relevance with regard to the investigation of clinical predictors of treatment response, as they only provide correlational data, that cannot be reliably interpreted in causal terms, whereas causal explanations can be reached through individual-level meta-analyses (Viechtbauer, 2007). In addition, it could be considered that in this meta-analysis non-randomized trials on patients with severe OCD symptoms were excluded; this limitation could have impeded to detect a significant effect of CPDs, which typically are most frequent in effectiveness open trials (Oldfield, Salkovskis, and Taylor, 2011). Furthermore, in this meta-analysis the researchers used a methodology in which the effects of CPDs were assessed classifying studies based on the presence of over 50% of patients with CPDs in the sample, but this strategy seems to be somewhat arbitrary.

Starting from some evidence reporting a worse CBT outcome for OCD patients with CPDs, some authors suggested that such patients could require a tailored therapeutic approach, based on more intensive levels of ERP (Oldfield et al., 2011). In addition, there is a lack of research on the role of CPDs in populations of patients with severe OCD, despite Axis II disorders are more common among patients with severe OCD symptomatology (Denys et al., 2004; Sookman and Steketee, 2010), and among inpatients rather than outpatients (Bejerot et al., 1998). For severe patients with CPDs inpatient treatment can be a tailored strategy, since hypercriticism from family members and interpersonal vicious cycles in the patient's living environment seem to work as specific factors for the maintenance of OCD symptoms (Pace, Twaites, and Freeston, 2011). In a recent trial Dèttore and colleagues (2013) evaluated the efficacy of a tailored protocol based on time-intensive ERP in a sample of 49 treatment-resistant OCD inpatients. The treatment consisted of daily ERP sessions delivered for two hours in the morning and two hours in the afternoon for five days a week in combination with SRIs. Results showed that the presence of one or more CPDs was not associated with dropout, and did not negatively affect outcome, suggesting that therapeutic gains for OCD patients with CPDs could be attributed to the inpatient treatment. In this format of treatment, personality disorder-related schemas may remain dormant, because the inpatient is away from the living context that usually activates those schemas (Dèttore et al., 2013). Thus, such a treatment could attenuate the negative effects of those schemas, thus enhancing the treatment

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progress (Dèttore et al., 2013). This aspect appears also relevant as severe OCD patients tend to have people around them who unintentionally accommodate their OCD requests, thereby contributing to the maintenance of symptoms (Oldfield et al., 2011; Osgood-Hynes, Riemann, and Björgvinsson, 2003). Moreover, a time-intensive treatment format, in which psychoeducation is delivered daily during the whole first week of treatment, could facilitate socialization to the treatment rationale, thereby increasing motivation to change also for patients difficult to treat (Dèttore et al., 2013).

The therapeutic benefit attributable to a time-intensive residential treatment could be also in that such an intervention offers more psychological support by the therapist between exposure sessions; this additional ingredient, despite non-specific, could motivate complex patients with Axis II and interpersonal problems by the construction of a stronger help alliance than in weekly outpatient treatment (Dèttore et al., 2013). However, given the small number of patients and the open research design, no firm conclusions can be drawn in that intensive CBT is a personalized strategy for resistant OCD patients with CPDs, since intensive and weekly CBT have shown to be equally effective for OCD in some other trials (Storch et al., 2008). It should be, however, considered that such trials assessed the dosage of CBT on samples with mild to moderate levels of symptoms.

Given this inconsistency of findings on the negative effects of CPDs on outcome of OCD, it has been suggested that such putative effects could be overestimated due to lore of clinicians (Dreessen and Arntz, 1998). Such discrepancy of evidence with the clinical impression could be explained by the fact that clinicians who assess the personality of the patient during or at the end of treatment are not blind to treatment outcome (Dreessen and Artnz, 1998). Consequently, therapeutic failure might increase false positive personality disorders diagnosis. In addition, a self-fulfilling prophecy might explain a poor treatment outcome when the therapist has negative beliefs about the prognosis of a patient who has been diagnosed with a concurrent personality disorder (Dreessen et al., 1997). Thus, it seems reasonable that clinician's expectations can in turn influence those of the patients, thereby adversely affecting treatment gains, as some research demonstrated that patients with CPD tend to have negative expectations about treatment outcome (Dreessen and Arntz, 1998).

Nevertheless, this inconsistency between research and practice could be affected by some methodological limitations of most of the studies investigating the predictive role of CPDs. The most important shortcomings concern the large variety of instruments used to assess personality psychopathology that could have reduced the comparability of results (Fricke et al., 2006).

Moreover, many of those studies did not use a semi-structured interview for the assessment of personality disorders (Dreessen and Arntz, 1998), which is considered the gold-standard for the assessment of personality psychopathology (Widiger and Francis, 1987). Nevertheless, this argument has recently raised some criticism since it appears in contrast with findings of some more recent studies. For example, in a very large sample of patients with Axis I disorders, including individuals with OCD, van den Hout, Brouwers, and Oomen (2006) found that, compared to the more usual open clinical interviews, the use of semi-structured interviews tended to be associated with larger proportions of patients diagnosed with one or more CPD. It could be hypothesized that the inclusion of many cases with relatively moderate severity levels of Axis II complaints has hidden the actual negative effects of CPDs (van den Hout et al., 2006). Nevertheless, in the same study the authors did not find evidence for this reasoning, since even when CPDs were diagnosed through open

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clinical interviews, their presence did not seem to affect improvement of Axis I problems after CBT, regardless of the type of Axis I disorder (van den Hout et al., 2006).

In some studies PDs were assessed after the treatment (Dreessen et al, 1997). We can assume that if therapy with a particular type of patient is presumed to be related to a failure, it might cause the interviewer to assign more Axis II diagnoses than if the rater is blind either to OCD symptoms severity and therapeutic gains (Dreessen et al., 1997).

Few studies used a longitudinal design, which is considered a better methodological strategy for evaluating clinical improvement (Dreessen et al., 1997). Patients with CPDs may have higher OCD pre-test scores than those without CPDs. If so, patients with CPDs may have higher symptoms rates after treatment, but accounting into baseline scores, they might have improved parallel to patients without Axis II complaints (Dreessen and Arntz, 1998).

Starting from these conflicting results on CPDs and OCD outcome, Chambless and colleagues (1992) have argued that examining the effects of all PDs as a group might obscure specific comorbidity effects, since Axis II conditions could not have uniform effects on outcome.

To date, there is only a small amount of research on the role of specific CPDs in the outcome of OCD (Olatunji et al., 2013). In a case-control study (Ferrão et al., 2006) OCD patients who had achieved a significant symptom improvement after a first-line treatment (pharmacotherapy or CBT or both) were compared on CPDs with those who were non-responders to such treatments (Ferrão et al., 2006). Findings showed that no specific CPDs were significantly more associated with non-responder status (Ferrão et al., 2006).

Most of research has been conducted on the predictive value of SPD. Data in this field seemed to consistently indicate that schizotypal CPD is associated with a poorer outcome of OCD treated with CBT. Minichiello, Baer, and Jenike (1987) reported markedly lower rates of response to behavioural treatment based on ERP in OCD patients with SPD. These findings were further supported by results from de Haan and colleagues (1997) of a poorer treatment response for OCD patients with cluster A CPDs, despite this relationship disappeared when controlling for Axis I disorders symptoms.

It has been suggested that the presence of specific CPDs might require a more comprehensive treatment approach, addressing not only OCD symptoms, but also targeting clinical aspects related to functioning and relapse vulnerability, for example combining CBT with additional medications (Reich, 2003). Consistent with this explanation, some research focused on the role of specific CPDs, evaluated on a categorical basis, and assessed the efficacy of modular treatment protocols, a form of treatment which can be adapted as a function of the patients' clinical characteristics (Steketee et al., 2011). Fricke and colleagues (2006) evaluated a multimodal case-formulation-driven CBT protocol for OCD complicated with CPDs. The protocol was delivered in combination with medications for a mean of 2 months, and consisted of ERP sessions, cognitive restructuring interventions combined with training groups for assertiveness, stress-management and problem-solving. A flexible approach was adopted by which treatment components were delivered as a function of the patients' needs (Fricke et al., 2006). Authors reported that patients with CPDs benefited equally from treatment in comparison with those patients without any Axis II comorbidity, maintaining symptom improvement at a six-month follow-up (Fricke et al., 2006). In addition, patients with CPDs did not need for longer treatment duration to achieve OCD symptom improvement than the counterparts without CPDs (Fricke et al., 2006). In contrast with previous research (Moritz et al., 2004), SPD was not significantly associated with a

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poorer prognosis, despite a tendency was found for these patients to improve less than the other patients (Fricke et al., 2006). Such a tendency could, however, suggest the importance of additional CBT modules aimed at targeting positive schizotypal symptoms or the use of augmenting strategies, for example based on low-dose atypical neuroleptics (Fossati and Borroni, 2008; Moritz et al., 2004). It could be also hypothesized that the negative impact of SPD on OCD outcome could be moderated by a poorer insight.

Indeed, some research (Alonso et al., 2007; Catapano et al., 2010; Matsunaga et al., 2002) has demonstrated that OCD patients with schizotypal CPD tend to have lower scores on insight self-report measures, such as the Brown Assessment of Beliefs Scale (BABS; Eisen et al., 1998). This interpretation seems also consistent with previous data (Foa et al., 1999) pointing out that OCD patients with fixity of beliefs (i.e. those OCD individuals subgroup who recognize to a lower degree that their symptoms are unreasonable) tend to not respond to ERP.

Although these promising results for multimodal protocols tailored for OCD with CPDs, one important question remains unsolved regarding the long-term maintenance of treatment gains. The cited-above study (Fricke et al., 2006) used a follow-up assessment at six months that could be yet short to evaluate whether CPDs increase susceptibility to relapse after treatment. Indeed, this hypothesis appears relevant as in that study there was a trend, not significant, for patients with CPDs to be more non-responder at follow-up than those without Axis II disorders (Fricke et al., 2006). Consistently, the presence of residual symptoms even at a subclinical level tends to confer susceptibility to relapse (Sookman and Steketee, 2010). Similarly, in another trial (Hansen et al., 2007) on 39 OCD patients, both cluster A and cluster B CPDs predicted a poorer long-term outcome (at one year) after CBT, but not at immediate post-treatment. It should be noted that in this study both of these patients who underwent ERP and those who received ERP combined with cognitive therapy, did not maintain therapeutic gains (Hansen et al., 2007). This result appears noteworthy as both treatment arms contained also relapse prevention modules, and, importantly, cognitive therapy can be also designed to increase resistance to stress-triggered relapse (Sookman and Steketee, 2010). Thus, these findings could suggest that complex cases of OCD with concurrent cluster A or B PDs require additional strategies, perhaps targeting problems other than OCD. This issue seems to highlight the need for extra interventions implemented during follow-up periods in order to enhance treatment gains maintenance, like the motivational interviewing-based techniques (Miller and Rollnick, 2002). However, this issue appears crucial and the effects of CPDs on long-term outcome require further investigation since some research suggests that OCD patients who are non-responders at post-treatment tend to become responders at follow-up (de Haan et al., 1997). It remains unsolved whether the influence of CPDs on relapse is attributable to the general effects of personality pathology or, alternatively, to an interaction between OCD features and PDs. This raises further examination since research on other Axis I disorders has shown that the presence of a CPD can render the patient more susceptible in case of relapse; for example it can make a substance abuser more likely to relapse in presence of craving or negative emotional states (Ball, 2005).

In a more recent trial Steketee et al. (2011) evaluated clinical predictors of immediate and long-term (at one-year follow-up) improvement for 39 OCD patients who underwent a modular CBT protocol, which consisted of Beckian cognitive therapy and some tailored interventions, such as planning adaptive activities to fill time formerly occupied by the obsessions and the compulsions, and relapse prevention training. Although this protocol

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resulted effective for many other CPDs, schizotypal personality traits predicted decreased follow-up outcomes, highlighting that such personality traits could exert some effects in relapse (Steketee et al., 2011).

Overall, these findings suggest that SPD is associated with a worse treatment response, thus requiring a more comprehensive approach including also components for non-OCD complaints, such as schizotypal symptoms.

Despite these consistent findings, however the association between SPD and OCD has not been demonstrated a common pattern in OCD populations, as studies report SPD having much lower prevalence rates compared with other personality conditions, like OCPD, and Dependent personality disorder (Dèttore et al., submitted; Samuels et al., 2000; Tukel et al., 2002), and, more generally, cluster A CPDs being more infrequent in OCD samples than in the general population (Denys et al., 2004).

Based on the high rates of comorbidity between OCD and OCPD (e.g. Maina et al., 2008), some research evaluated the prognosis of OCD with comorbid OCPD. OCPD has been found to act as a negative prognostic factor for pharmacological treatments (Ravizza et al., 1995). In a small open pilot trial Cavedini and colleagues (1997) reported that the presence of comorbid OCPD predicted a poorer response to a 10-week standardized treatment based on serotonergic medications. Comorbid OCPD in OCD has been found to be associated with a significant greater number of CPD than OCD without OCPD, and to a more significant association with other Axis II disorders, such as schizotypal, dependent, and avoidant PDs (Diaferia et al., 1997). Therefore, OCPD comorbidity could adversely affect outcome of OCD as it is associated to a more complex maladaptive functioning that reasonably would act as a non-specific factor slowing treatment response or requiring augmentation pharmacotherapy (Cavedini et al., 1997). However, research did not provide unequivocal evidence on the relation of comorbid OCPD to greater OCD severity, and in general to a more impaired functioning (Albert et al., 2004; Pinto et al., 2011). Moreover, in other trials comorbid OCPD was not associated with a negative prognosis for pharmacological treatments in OCD patients (Baer et al., 1992). Overall, such inconsistent findings could be attributed to the use of small sample sizes and uncontrolled designs; thus, larger randomized controlled trials are required to investigate the potential adverse effects of comorbid OCPD in the outcome of pharmacological treatments for OCD.

Only a recent study focused on the specific role of OCPD in the outcome of cognitive behaviour therapy for OCD (Pinto et al., 2011). The authors found that both OCPD diagnosis and greater OCPD severity at baseline predicted worse outcome after a protocol consisting of weekly sessions based on exposure with response prevention. It could be hypothesized that the interpersonal dysfunction associated with OCPD (Costa et al., 2005) may interfere with working alliance between therapist and patient that is typically request by the ERP therapy. This could negatively affect adherence to ERP assignments (Pinto et al., 2011). Accordingly, patients with OCPD traits have been described as having difficulty to trust (Gibbs and Oltmanns, 1995).

It could be, therefore, that clinical features of patients with OCPD have a negative impact on outcome by adversely affecting the therapeutic relationship. Such an interpretation could be consistent with some previous research on clinical predictors of non-response for OCD, suggesting that problems linked to the therapeutic relationship do play a role in resistant to change cases of OCD (e.g. Keijsers, Hoogduin, and Schaap, 1994). In addition, research demonstrated that OCD co-occurring with OCPD has greater severity and, specifically, more

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severe associated depressive symptoms (Gordon et al., 2013). Therefore, individuals with comorbid OCPD could be particularly prone to experience regrets, disappointments, and anger towards themselves due to their perfectionistic standards. Thus, when such patients experience suboptimal performances, they may get depressed (Gordon et al., 2013). Accordingly, OCD patients with concurrent OCPD could have a negative response to CBT due to their more severe levels of depression.

Severe levels of comorbid depression have been recognized as factors responsible for therapeutic failure in CBT for OCD, since hyperactivation due to higher levels of depression could impede between-session habituation to obsession-eliciting situations (Abramowitz and Foa, 2000; Abramowitz et al., 2000). As a consequence of these findings Abramowitz (2004) recommend that Beckian cognitive techniques validated for depression should be integrated into the ERP-based protocols for OCD in order to target comorbid depression. This tailored treatment strategy could also improve dropout rates, as some research demonstrated that the combination of ERP with cognitive therapy could improve patients' adherence to treatment, thereby preventing early treatment discontinuation (e.g. Hansen et al., 2007).

However, other studies (e.g. Ansell et al., 2011) reported that OCD patients with OCPD had increased vulnerability for worry and GAD, thus suggesting that the negative effects of the unattainment of high-standard goals typical of OCPD could be non-specific, and patients with OCPD could fail to benefit from treatment also as they have higher vulnerability for worry when their rigid standards are unmet.

An alternative explanation could be that given the stronger association of hoarding symptoms to OCD with OCPD (e.g. Eisen et al., 2006), higher hoarding severity could explain why OCD patients with concurrent OCPD do not respond to CBT, since hoarding has been consistently found to predict negative outcome for OCD (Black et al., 1998; Maher et al., 2011).

Some transdiagnostic features, common also to some axis I conditions and not specifically associated with PDs, could interfere with treatment response of OCD. For example, in a further analysis of the data Pinto and colleagues (2011) found that the unique OCPD symptom for predicting a negative ERP outcome was perfectionism. Perfectionism has been consistently found as a trait associated with poorer treatment outcome for depression (Blatt et al., 1995; Blatt et al., 1998). OCD patients with perfectionist traits could avoid or do not adhere to ERP homework assignments for fear of not doing them perfectly, or could withdraw efforts if they believe that treatment progress is suboptimal (Pinto et al., 2011). As suggested by some authors, an alternative explanation for that OCPD is more treatment resistant could be that OCPD is an OCD subtype characterized by more ego-dystonic OCD features (Starcevic et al., 2012). However, no firm conclusions can be drawn from the available evidence on the role of OCPD in the outcome of OCD, as some research has indicated that cluster C CPDs could be associated with a better outcome (Steketee, 1990). In addition, concurrent OCPD has been found to be related to a higher level of insight, that could be a facilitating factor for treatment success (Catapano et al., 2010).

A further limitation across the published outcome trials regards limited data on the effects of CPDs in terms of rates of remission and recovery status (Fisher and Wells, 2005; Hollander and Zohar, 2004). Accordingly, although patients with a post-treatment Y-BOCS score under the clinical cut-off or would not meet diagnostic criteria, they could still suffer from noticeable symptoms which can adversely impact their quality of life, thus contributing to a greater susceptibility to relapse (Sookman and Steketee, 2010). The presence of comorbid

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Axis II conditions or personality traits could impede the achievement of a full recovery status at the end of a treatment pathway (Sookman and Steketee, 2010). This issue seems also to be particularly relevant for more complex cases of OCD with concurrent Axis II conditions, since for these patients maladaptive schemas and interpersonal deficits can increase vulnerability to the relapse-activating effects of stressful life events (Sookman and Steketee, 2010).

Given the large heterogeneity across treatments implemented in the studies, it seems difficult to conclude which CPDs are associated to a worse outcome. The lack of comparative trials does not allow to establish which treatments patients with CPDs most benefit, and which ones they do not respond to.

The use of only categorical personality variables in research on the influence of personality comorbidity on outcome has been criticized (Widiger, 1992). Moreover, the categorical approach to personality assessment has been considered to be dependent on state and acute clinical expression (McCrae and Costa, 1997). Several authors recommend the dimensional approach, and dimensional personality indicators have been found to be more reliable than categorical diagnoses, the thresholds of which are largely arbitrary (Rees, Anderson, and Egan, 2005).

It could be assumed that personality traits affect treatment outcome before the minimum set of criteria for a personality disorder diagnosis is obtained, or conversely, that many more criteria are needed than just the minimum set of criteria (Rees, Anderson, and Egan, 2005). Some personality features that negatively impact outcome for OCD could be common to several Axis II conditions (Dreessen and Arntz, 1998). This interpretation seems consistent with those studies reporting a relation between outcome and CPDs at a cluster level (e.g. Steketee et al., 2001).

In addition, CPDs could play a negative influence on outcome for OCD by symptom patterns, which are specific to certain CPDs (Arntz et al., 1999). Alternatively, more specific behaviours could affect resistance to change in severe OCD, like those defined by VanDyke and Pollard (2005) as treatment-interfering behaviours, such as the attempts to change the focus of sessions to issues not on the treatment plan.

Thus, an alternative area of investigation on the role of CPDs in OCD outcome focused on those patients who had low and high scores on personality disorder traits measures (Steketee, Chambless, and Tran, 2001).

In this vein, Sartory and Grey (1982) examined personality traits that were related to poorer outcome in treatment with clomipramine and/or behaviour therapy. Using the Eysenck Personality Inventory (EPI; Eysenck and Eysenck, 1969), they found that patients high in extraversion did better in treatment than those with low extraversion (Sartory and Grey, 1982). Specifically, results suggested that individuals with these traits could habituate to feared cues more rapidly, resulting in shorter treatment (Sartory and Grey, 1982). Contrary to the available evidence that OCD patients with odd rituals and schizotypal personality tend to have a poorer prognosis (e.g. Minichiello et al., 1987), the authors did not find a significant relationship between outcome and psychoticism.

In a more recent retrospective study, Corchs and colleagues (2008) found that those OCD patients who had not responded to a previous trial with evidence-based therapies (SRIs or CBT) had lower scores on self-directedness, a pattern of temperament and character measured by the Temperament and Character Inventory (TCI; Cloninger et al., 1994). According to the model of temperament proposed by Cloninger (Cloninger et al., 1993), self-directedness

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refers to the ability of an individual to control, regulate and adapt his/her behaviour as a function of the situation features and his/her personally relevant goals. Accordingly, individuals with higher self-directedness believe that their success depends on their own efforts and, therefore, they tend to act as effective problem-solvers (Cloninger et al., 1993). In addition, individuals with higher self-directedness tend to be more capable to engage in behaviors with delayed but greater reinforces (Cloninger et al., 1993).

The result that OCD patients with lower self-directedness had a poorer treatment response could be therefore explained by their higher susceptibility to engage in behaviour more immediately reinforced (like the compulsions) than those with immediate costs and risks in terms of discomfort, and with delayed but greater advantages (Corchs et al., 2008). The importance of examining the role of specific subset of symptoms has been addressed by Moritz and colleagues (Moritz et al., 2004) in a trial on the predictive value of comorbid schizotypal symptoms in the CBT outcome of OCD. In this study the authors found that schizotypal personality symptoms predicted negative outcome for OCD patients who underwent a modular protocol of CBT based on exposure and response prevention combined with additional components aimed at meeting individuals' therapeutic needs such as social skills or stress-management trainings (Moritz et al., 2004). Specifically, it emerged that positive schizotypal features, especially suspiciousness and magical ideation, were associated with greater risk for non-response (Moritz et al., 2004). However, non-responders did not have significantly higher scores on negative disorganized features measures.

In this vein in a more recent trial Haaland and colleagues (2011) investigated the predictive value of early maladaptive schemas in the outcome of CBT for OCD. In the study eighty-eight OCD patients were assigned to an individual or group CBT arm, and maladaptive schemas were assessed before and after treatment through the Young Schema Questionnaire-Short Form (YSQ-SF; Young et al., 2003). Results demonstrated that higher pre-treatment scores on the abandonment schema predicted a poorer treatment response (Haaland et al., 2011). It has been suggested that individuals with this schema constantly expect to lose close relationships (Young et al., 2003). Consequently, such individuals tend to avoid intimate relationship as a coping style to prevent others from leaving (Young et al., 2003). Based on these assumptions, it could be hypothesized that patients with this schema activated often have difficulties in forming positive therapeutic relationship, especially during the first stages of treatment.

Accordingly, the activated abandonment schema could interfere with treatment progress especially for OCD patients with comorbid borderline personality disorder, since such schema has been consistently found to be more strongly associated with this CPD (Arntz, Dietzel, and Dreessen, 1999; Ball and Cecero, 2001; Joveev and Jackson, 2004). Since maladaptive schemas seem to play a role in OCD treatment progress by negatively impacting working alliance, additional interventions, based on schema therapy, for improving therapeutic relationship could be indicated for such patients (Sookman and Steketee, 2010). Accordingly, patients with maladaptive schemas could resist to exposure and unpleasant emotions to the extent that they endorse negative emotional schemas (e.g. "Painful emotions are to be avoided") (Leahy, 2007). In two small samples of patients with resistant OCD and CPDs Sookman and Pinard (1999) and Sookman and colleagues (2003) evaluated a long-term schema-based CBT delivered through individual weekly sessions. Sample sizes were 7 and 32 respectively and the treatment mean duration was 10 months. Results showed promising

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findings in that over one third of the patients achieved a recovery status (Sookman and Pinard, 1999; Sookman et al., 2003).

Another important finding provided by the study of Haaland and colleagues (2011) was that self-sacrifice schema predicted a better outcome, suggesting that patients with this schema activated could have a stronger compliance with treatment assignments due to their predisposition to meet the expectations of significant others, including those of the therapist (Haaland et al., 2011).

This explanation appears consistent with findings of previous research, which reported a positive relationship between dependent personality traits and OCD treatment outcome (Fals-Stewart and Lucente, 1993; Hansen et al., 2007; Steketee, 1990). More generally, patients with cluster C CPDs seem to benefit best from a stricter adherence to completion of the agreed-upon hierarchy, and from having more available time dedicated to within-session exposure exercises (Hansen et al., 2007).

Conversely, in contrast with previous research which suggested that general personality psychopathology did affect negatively treatment response, in the study conducted by Haaland and colleagues (2011) higher pre-treatment total scores on the YSQ-SF were not related to outcome, and these results rather supported that specific symptom patterns could moderate the negative influence of specific CPDs on OCD treatment outcome. However, due to the small sample size, the author could not examine whether specific schemas interacted with treatment modalities, in that for example group CBT could be more or less effective for patients with particular schemas rather than others.

A further area of research concerns the effects on personality produced by CBT designed for OCD (McKay et al., 1996). Early evidence suggested that comorbid personality features changes can be associated with OCD improvement (Ricciardi et al., 1992). Seventeen patients diagnosed with OCD and assessed with the SCID-II at pre and post-treatment, participated in a 4-month trial of behavioural, pharmacological, or combined treatment (Ricciardi et al., 1992). Ricciardi and colleagues (1992) reported that changes in Axis II diagnoses might be related to OCD symptom improvement. However, the small sample size, and the fact that the type of behavioural or pharmacological treatment were not sufficiently described, neither were mentioned how many patients received each type of treatment, strongly limit the interpretations of the results.

In another study on twenty-one OCD patients McKay and colleagues (1996) found that the number of CPDs from pre to post-treatment significantly decreased after behaviour therapy specifically delivered for OCD.

Other researchers focused on changes in clinical correlates of CPDs that could be related to OCD improvement in CBT designed for OCD. In an open pilot trial Albucher, Abelson, and Nesse (1998) investigated in a sample of 17 OCD patients whether behaviour group therapy for OCD could enhance increase in the use of adaptive defense mechanisms, assessed by the Defense Style Questionnaire (DSQ; Bond et al., 1989), an 88-itm self-report measures evaluating defenses by tapping conscious derivates of unconscious processes. Results showed that improvements in OCD symptoms after treatment were linked to a greater use of healthier defenses, such as humor, altruism, and sublimation, and a significant decrease in a less adaptive defense mechanism that could be specific to OCD, the undoing (Albucher et al., 1998). However, the small number of participants does not allow drawing reliable conclusions on these CBT effects at a defense mechanisms level. In addition, the short-term duration of therapy (7 weeks) raises the possibility that the increased use of mature defenses

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could be due to a non-specific factor, for example the effects of group interactions. Moreover, the lack of repeated assessments during the intervention prevents to establish whether OCD improvement predated defenses, or, conversely, clinical change was driven by defense mechanisms change.

As pointed out (Vogel et al., 2006), motivation to change plays a relevant role in the therapeutic process. It could be that the co-occurrence of a CPD, either at a diagnostic or trait level, interferes with treatment progress through a motivational pathway.

Tyrer and colleagues (2003) have suggested a different classification of personality disorder based on the readiness to change. These authors have hypothesized that certain personality disorders are characterized by a higher wish to change and greater insight, and those PDs were classified as Type S PDs (treatment seeking personality disorders), whereas other PDs would have lower degree readiness to change or seek help and were classified as Type R PDs (treatment rejecting personality disorders). In their study these authors found Type S PDs were more frequently patients with cluster C PDs, whereas those defined as Type R PDs were more likely to have a cluster A PD diagnosis. This could explain why some research demonstrated that cluster C CPDs are more likely to be associated with better outcome for OCD, and cluster A (such as schizotypal CPD) typically predict a worse treatment response.

Conclusion

Studies on comorbidity and symptoms patterns suggest that OCD is far from being a homogeneous disorder (Mataix-Cols et al., 2000). Specifically, a noticeable heterogeneity in findings on prevalence rates of CPDs in OCD has been found across studies (Friborg et al., 2013). Such heterogeneity could be primarily attributed to some extent to substantial differences in the recruitment strategies of samples, and in the clinical characteristics of samples themselves.

There is accumulating evidence that the relation between OCD and CPDs could be moderated by some clinical factors, including OCD severity and specific OCD dimensions (Horesh et al., 1997). Investigation of clinical moderators of the relation between OCD and the presence of CPDs has relevance for the development of tailored treatment approaches. For example, the evidenced association between CPDs and some OCD dimensions could add knowledge on the use with OCD associated to CPDs of specialized treatments specifically developed for OCD dimensions.

Most of the researches have been conducted following cross-sectional designs, and yet there is a paucity of examination to disentangle the causal link between CPDs and OCD. Future studies should examine whether CPDs have a role in the onset of OCD or, alternatively, OCD predates the development of CPDs, or the two categories of disorders share common etiological factors. For example CPDs could be the consequence of having OCD.

Recent research efforts using dimensional personality models has clearly demonstrated that certain personality traits are very common among individuals with OCD and also among non-clinical participants. These traits appear to influence OCD severity and to be associated to specific OCD dimensions (Bienvenu et al., 2004).

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Despite the promising advantages of a dimensional approach, mostly based on the use of self-report measures, this type of assessment tools does not establish persistence, pervasiveness and maladaptivity or ask respondents to differentiate personality traits from the effects of the Axis I disorder or from an emotional state (Ball, 2005). Thus, a more comprehensive approach to personality assessment would be recommended in research on OCD and personality which integrates semi-structured assessments with self-report measures, thus resolving state vs. trait ambiguities (Zimmerman 1994).

The prognostic role of specific CPDs in OCD outcome need for more systematic research. To date, evidence on the negative effects of having one or more CPDs is not conclusive. Moreover, such an assumption seems to be more dependent on a clinical lore than on an empirical base. Evidence of recent studies conducted according to a more valid methodology including the use of pre-post-test longitudinal designs and pre-treatment assessment of personality, actually does not seem to support this assumption, with some recent studies showing no relation between OCD outcome and CPDs (Fricke et al., 2006). In addition, recent data suggest the usefulness of high intensity treatments for such patients as a tailored dosage of CBT can improve response also for those patients with CPDs who have not responded to previous treatments (Dèttore et al., 2013).

The importance of Axis II diagnoses in the outcome of OCD could have been overlooked in the past years, and the present systematic evaluation seems to suggest that clinicians should focus also on other factors, such as symptoms patterns (e.g. lower levels of insight) or theory-driven vulnerability factors (e.g. obsessive beliefs) or even other diagnostic factors, like other Axis I conditions (e.g. comorbid severe depression). Consistently, future treatment approaches should be dedicated to target these negative predictors by including in the evidence-based protocols (i.e. ERP) other strategies (i.e. schema therapy).

Inconsistent results on the role of CPDs in treatment response suggest the importance of a more complex research approach, which can identify the role of specific CPDs or specific symptom patterns within each CPD. To date outcome trials focused only on the effects of OCPD and SPD, either from a categorical diagnostic and dimensional point of view (Pinto et al., 2011; Moritz et al., 2004). Despite these PDs have been found to be the most frequent among OCD patients (Denys et al., 2004), or very common, many OCD patients often present with other CPDs, such other cluster C PDs (with dependent PD appearing more prevalent and more specific to OCD than avoidant). There is also a noticeable room for future research on the predictive role of cluster B PDs, particularly borderline PD, as in some studies this CPD has been found to be associated to OCD to a extent comparable to the previous cited PDs, and has shown to be associated with marked interpersonal deficits (Arntz et al., 1999). In addition, it could be suggested that emotion regulation deficiency, that typically occurs in this type of PD (Arntz et al., 1999), might interfere with OCD treatment, which is typically demanding and could not be enough effective for patients with weak tolerance for strong emotions.

Consistent findings seem to demonstrate that concurrent SPD tend to increase risk for a negative treatment response in OCD (e..g. Moritz et al., 2004; Steketee et al., 2011). However, clinical mechanisms responsible for such a failure need for additional investigation. It is not sufficiently clear whether SPD exert a negative impact on outcome being associated with increased pre-test OCD severity. Based on this assumption, SPD could increase vulnerability to non-response exclusively due to a non-specific effect of general psychopathology.

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However, this does not appear to be supported since trials on modular treatments do not seem to improve response for OCD with SPD, despite including augmentation strategies centered on OCD severity (Steketee et al., 2011). Only two small studies (Sookman, et al., 2003; Sookman and Pinard, 1999) tested an integrative protocol for OCD associated to CPDs targeting either OCD symptoms and personality problems, based on ERP in combination with schema therapy. Results of these preliminary trials appear promising. Consistent with this line of research, future studies should introduce also outcome measure for personality-related complaints, such as emotion regulation skills indicators.

To date, a limited number of studies with small sample sizes has been conducted to examine moderators of negative outcome for OCD with SPD. A dimensional approach could allow clarifying specific schizotypal symptoms responsible for such a treatment failure. Research to date lacks also process studies, which could help to understand what mediators could drive change for OCD with SPD.

Findings on the role of OCPD appear inconsistent, with some studies evidencing that OCD with OCPD benefit more than OCD alone from CBT (Fals-Stewart and Lucente, 1993), and other reporting a negative relationship between concurrent OCPD and outcome (Pinto et al., 2011). Some of these inconsistencies could be related to the assessment of OCPD on a diagnostic basis or through a dimensional approach. Such dimensional approach appears promising, as specific OCPD symptoms seem to be linked with outcome (Pinto et al., 2011). Future research should establish whether OCPD symptoms predicting outcome are those most specific to OCPD or transdiagnostic features. This issue could help to develop treatments specifically targeted for OCPD or to include in the existing treatment transdiagnostic components derived from disorders that share symptom criteria with OCPD (i.e. perfectionism).

This discrepancy on results of outcome of OCD with CPDs raises also the possibility to adopt empirically-driven strategies, not based on previous research. Using a different treatment approach, not informed by evidence of predictors, research is required to examine the feasibility of stepped care for OCD with CPDs. This alternative form of treatment refers to a self-correcting algorithm that is intended to provide the least restrictive and most effective treatment for each patient, potentially reducing time, cost, and associated patient and clinician burden of treatment (Gilliam et al., 2010). According to these novel therapeutic approaches, OCD complicated by CPDs could be treated with low intensity available treatments as for mild OCD (Gilliam et al., 2010). For those cases with more complex symptoms, not responding to low intensity options, more intensive treatments could be offer in order to optimize treatment response. This strategy could help to save costs and time for patients and clinicians since not all the OCD clients with CPDs are expected to not change with lower intensity treatments, as data presented in this review have shown (e.g. Dreessen et al., 1997).

A more reliable picture of the role of CPDs in OCD outcome should also be provided by future randomized clinical trials, given the actual lack of this design in research on CPDs. It could be assumed that OCD patients with CPDs who complete treatments in the available open trials were self-selected and, more importantly, selected their treatment. More information is also required with regard to the influence of CPDs in help-seeking behaviours. CPDs could lead the individual to delay seeking treatment or seeking a specific category of treatment. Such kind of investigations could highlight the importance of programs to increase mental health literacy in clients with CPDs. Comparative trials are needed as actually there is

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insufficient knowledge about which CPDs benefit from what treatments since some substantial differences in assessment procedures of PDs, levels of OCD severity, and other clinical characteristics of patients at baseline can be found across different studies. Thus, specific CPDs could be prescriptive factors of certain treatments. Other variables connected to treatment delivery could be involved in CPD response. For example, treatment duration may be crucial, in that patients with CPDs might require a longer duration to obtain sustained improvement.

Available studies do not provide sufficient information on the maintenance of gains for OCD with CPDs; therefore, future trials should also apply long-term follow-up assessments given the interpersonal aspects linked with personality psychopathology.

Given the relevance attributed to the quality of life in emotional disorders and the shift towards a broader concept of health (World Health Organization, 1986), future studies should examine which specific areas of quality of life are most affected by the presence of a CPD. However, it could be useful to examine protective factors, which facilitate a better adjustment to OCD problems despite the presence of CPDs. Such a research direction would inform treatment planning and prevention programs by suggesting the addition of interventions aimed at improving quality of life in OCD patients with CPDs.

More longitudinal studies are required to examine the direction of the relationship between OCD and CPDs, and to identify potential developmental personality predictors of OCD onset or, conversely, OCD patterns that predate the development of personality psychopathology.

In conclusion, investigation on CPDs in OCD is a challenging research field, and needs for further efforts to understand why some individuals suffering from OCD also present with CPDs, in order to be able to provide them with the most appropriate treatments, with optimal efficacy at minimal costs, both from the patients' and the healthcare service's point of view.

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Chapter II

The Adaptive Significance of and Natural Expression of Obsessive- and Compulsive-Like Symptoms

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Absract

Obsessive compulsive disorder (OCD) is a serious and debilitating condition that affects approximately 1 – 4% of the population. Yet, many individuals that do not meet the criteria for clinical OCD nevertheless report having experienced obsessive-like thoughts and compulsive-like behavior at some time in their lives. Indeed, clinical OCD is distinguished from "normal" obsessions and compulsions primarily by the frequency and intensity of the symptoms, and the extent to which they interfere with daily functioning. Therefore it appears that obsessions and compulsions serve some adaptive purpose, and that OCD might represent a severe dysregulation of these "normal" (adaptive) thought and behavioral patterns. Why do obsessions and compulsions exist? This chapter will compare the phenomenology of "normal" and pathological obsessions and compulsions, in both human and animal populations. In humans, obsessive-compulsive-like symptoms often appear at certain developmental and reproductive stages, such as prepubertally in both genders, premenstrually in women, and, in new parents, associated with the birth of a child. Interestingly, these stages are also associated with an increased risk for the onset or exacerbation of clinical OCD. We will discuss theories on the physiological underpinnings, as well as possible adaptive significance, of obsessive-compulsive-like symptoms during these life stages. Although in animals "obsessions" (if they exist) are impossible to observe, many animals certainly display compulsive-like behavior, in the form of stereotyped, stimulus-bound behavioral patterns, such as nest building and courtship displays, as well as rigid habits. In the case of captive animals, exaggerated and stereotyped forms of normal behaviors such as grooming and locomotion represent a pathological condition surprisingly similar to human OCD. By understanding the physiology and adaptive significance of "normal" obsessions and

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compulsions, we might achieve a better understanding of this devastating neuropsychiatric disorder.

1. Introduction

1.1. Obsessive-Compulsive Disorder (OCD)

Persons with OCD experience intrusive and persistent thoughts that cause distress (obsessions), and compulsions, which are performed as an attempt to alleviate this distress. Typically, these symptoms are egodystonic, that is, the person is consciously aware that they are abnormal, yet cannot control them. Several sub-classifications of OCD have been based on age of onset (juvenile vs adult onset OCD), comorbidity with other neuropsychiatric disorders (tic vs non-tic associated OCD), the specific content of obsessive-compulsive symptoms (washer vs checker OCD), and the core motivations that underlie the expression of the symptoms (harm avoidant vs feelings of incompleteness; for review, see Hoffman, 2011). Functional neuroimaging of individuals with OCD has consistently revealed abnormal activity within corticostriatal-thalamocortical circuits, involving the orbitofrontal cortex, anterior cingulate cortex, insula, striatum and thalamus. These studies are consistent with the existence of distinct symptom dimensions, or subtypes, as contamination/washing, checking, and hoarding subgroups show slightly different patterns of brain activity (Mataix-Cols et al., 2005). It has been proposed that hyperactivity of error detection and action monitoring functions of the anterior cingulate cortex might be related to OCD associated with feelings of incompleteness and pathological doubt (Maltby et al., 2005), while hyperactivity of the orbitofrontal cortex and insula are especially associated with emotion-laden and disgust-related obsessions and compulsions (Mataix-Cols et al., 2005). At the circuit level, emphasis has been placed on a possible imbalance in the activities of the so-called “direct” and “indirect” pathways of the basal ganglia, which are considered to modulate positively (direct pathway) and negatively (indirect pathway) thalamocortical and motor output (Smith et al., 1998). However, the purpose of this paper is not to discuss specifically the pathophysiology of OCD; rather, it will explore the expression and phenomenology of obsessions and compulsions in nonclinical, “normal” populations. The intended purpose of this approach is twofold: first, to emphasize that neuropsychiatric symptoms, including obsessions and compulsions, often manifest themselves along a continuum of severity in the general nonclinical population and are often more prominent during certain life events and physiological states; and, second, to argue that the study of normal behaviors, as well as their neural substrates and regulatory mechanisms might lead to important insights on the pathophysiology of OCD. This paper is divided into two main parts: the first part examines the phenomenology of obsessive-compulsive symptoms in nonclinical populations, and the second part emphasizes how the study of normal aspects of animal behavior might lead to important insights on the pathophysiology of obsessive-compulsive symptoms.

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1.2. "Everybody Obsesses about Something, Right?"

Numerous studies have shown that the lifetime prevalence of OCD is within the range of 0.7 to 2.5% (reviewed in Gibbs, 1996). However, it has become clear that the experience of obsessions and compulsions is quite prevalent among individuals of the general population, that is, in persons that do not fulfill the diagnostic criteria for OCD.

The notion that obsessive symptoms are components of normal psychological processes was tested in a seminal paper by Rachman and De Silva (1978), in which these authors investigated whether intrusive thoughts and impulses reported by unaffected individuals were distinguishable from those of persons suffering from OCD. They presented a list of 58 obsessions derived from 40 normal individuals, along with 23 from 8 individuals suffering from OCD to a panel of 6 judges (5 psychologists and one psychiatric nurse) who were asked to try to identify those that came from OCD subjects. Some examples of these intrusive thoughts and impulses were: (1) Thought of having been poisoned by chemicals; (2) Thought of harm to or death of close friend or family member; (3) Thought: "Did I commit this crime?", when hearing the report of a crime; (4) Impulse to jump on the tracks when a train is approaching; (5) Impulse to strangle children and sometimes adults; (6) Impulse to physically and verbally attack someone. The panel was not much better than chance at identifying OCD obsessions, but could identify normal obsessions fairly well. These results were interpreted to suggest that the content of normal and abnormal obsessions did not differ markedly. By contrast, Rachman and De Silva provided evidence showing that OCD obsessions differed from normal ones in their frequency, duration, and intensity. In other words, what seemed to distinguish OCD from "normal" was not the content or presence of obsessive-compulsive symptoms themselves, but rather their severity and negative effect they had on daily functioning.

However, more recent studies (Rassin and Muris, 2007) have taken a second look at the Rachman and De Silva obsessions, and questioned whether normal and abnormal obsessions really are indistinguishable with respect to content. They presented the same obsessions that were studied in the Rachman and De Silva article to a contemporary panel of 11 psychotherapists, as well as one comprising 90 undergraduate psychology students. In contrast to the earlier study, both panels were significantly better than chance at distinguishing nonclinical and OCD obsessions. As a follow up study, Rassin et al. (2007) asked a sample of 133 undergraduate students to rate the extent to which they had experienced each of the obsessions listed in the original Rachman and De Silva study, as well as complete a self-report measure of OCD symptoms. They found that (1) nonclinical obsessions were endorsed approximately twice as frequently (that is, these obsessions were more prevalent in the general population), and (2) that the extent to which abnormal obsessions were endorsed correlated significantly with self-reported OCD symptoms. These studies seem to indicate that there are subtle differences in the content of normal and abnormal obsessions, although at present it is not clear exactly what these differences might be. To illustrate the difficulty in determining exactly how OCD obsessions differ in content from normal ones, of the short list of obsessions given above, numbers 1, 3, and 5 were from individuals with OCD, while 2, 4, and 6 were from subjects that did not suffer from this disorder.

Gibbs (1996) reviewed published studies on the prevalence of obsessive-compulsive symptoms in the general population. The studies cited in this review indicated that the

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prevalence of "subclinical" OCD (generally defined as obsessive-compulsive symptoms that do not result in significant distress) in samples of the general population was in the range of 1.3 – 19%. In the 19 studies reviewed by Gibbs, the vast majority (80 – 99%) of adults reported having had experienced intrusive thoughts, images, or impulses. Moreover, 14% reported having experienced 3 or more obsessions in the past month that were accompanied by some distress.

Compulsive-like behavior appears to be somewhat less prevalent, being present in approximately 10 – 27% of the individuals in the general population. Interestingly, in adult samples, individuals that were experiencing symptoms but that had not sought professional help tended to have either obsessions or compulsions (not both), whereas those that had sought professional treatment for their symptoms most frequently reported experiencing both obsessions and compulsions.

2. Obsessive-Compulsive Symptoms Are Associated with Certain Life Stages and Events

The fact that obsessive-compulsive symptoms are quite common in the general population suggests that they might have served (and perhaps still serve) some important adaptive purpose. Some clues as to the adaptive significance of these symptoms might be gained by considering the context in which obsessive compulsive symptoms occur in otherwise "normal" individuals. Common experience and scientific studies concur that obsessive-compulsive-like symptoms are more prevalent during certain life stages and events, namely, during early childhood, while falling in love, or becoming a parent. Can considering cognitive and physiological characteristics peculiar to these life stages, as well as the specific content of obsessions and compulsions typically experienced during these times, tell us something about the nature of OCD?

2.1. Obsessive-Compulsive Symptoms during Early Childhood

As many parents will attest, young children can be quite obsessive-compulsive. Beginning around the age of 2 -3, and tapering off around age 6, it is normal for children to express rigidity in their routine as well as ritualized and repetitive behaviors, such as insisting on the same bedtime story told in the same fashion, lining up objects in a symmetrical manner, and an insistence for their belongings and environment to be "just right" (Evans et al., 1997; 1999; 2004; Pietrefesa et al., 2007; Zohar and Felz, 2001). The developmental period during which obsessive-compulsive-like behaviors are prominent, mirrors that during which the capacity for executive function (e.g, the ability to shift and maintain a cognitive set and inhibit a prepotent behavioral response) is developing. Indeed, obsessive compulsive-like behaviors during early childhood (4 – 6 years old) have been associated with decreased response inhibition and set shifting abilities (Pietrefesa et al., 2007). Thus, normative expression of obsessive compulsive-like behaviors during early childhood is proposed to arise as a result of ongoing developmental changes in brain circuitry that subserve executive function, namely, the dorsolateral prefrontal, anterior cingulate, and orbitofrontal cortices and

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their associated basal ganglia circuits (Evans et al., 2004). Yet, these ritualistic behaviors, at least in children younger than about 6 years old, might also serve as an adaptive, involuntary, and anxiety-reducing response to normal fears, and in older children they might represent a more voluntary expression of a previously adaptive strategy that no longer serves to reduce anxiety (Evans et al., 1999; Pietrefesa et al., 2007). Notably, deficits in response inhibition and set shifting are possible endophenotypes of OCD (reviewed in Evans et al., 2004; Lennertz et al., 2012; Rajender et al., 2011); moreover, altered neuronal activity of cortical-basal ganglia thalamocortical circuitry has been consistently reported in persons with OCD (Mataix-Cols et al., 2003; 2005; 2008; Maia et al., 2008).

Obsessive-compulsive-like symptoms in children are associated with certain childhood-typical beliefs. "Magical thinking", which is the belief in the power of wishing or ritualized acts for achieving a desired outcome, or in the existence of supernatural beings or special powers, is a common characteristic of young children that often persists in some form into adulthood. These beliefs (both in children and in adults) can give a sense of personal control over circumstances that are in reality uncontrollable, thereby serving as a psychological mechanism in the face of anxiety and uncertainty. One could imagine that such mechanism might be especially relevant to young children, who have very little control over events that transpire in their daily lives. Indeed, there are important correlations among measures of magical beliefs, anxiety, and obsessive compulsive symptoms in children, although the causal relationships among them remain to be clarified. Thus, magical beliefs, and in particular the belief in the power of wishing, were found to be positively associated with compulsive-like behavior (Evans et al., 2002). In a separate study, although magical thinking in boys (but not girls) was positively correlated with compulsive-like behaviors as well as with anxiety, anxiety measures alone statistically accounted for 59% of the variance in compulsive-like behavior, whereas magical thinking did not contribute unique variance to this measure (Simonds et al., 2009).

One form of magical thinking that has been extensively studied in the context of OCD is "thought action fusion" (TAF). In essence, TAF is the belief that thinking about a carrying out a certain action is morally equivalent to actually doing it, or that thinking about the occurrence of a particular event makes that event more likely to happen. Evans et al. (2011) examined the beliefs about TAF, and their relationship to anxiety and obsessive compulsive symptoms, in children aged 7 – 14. They found that within this age group, the older children were less likely than younger ones to endorse belief in TAF. In younger children, compulsive-like behavior was best predicted by measures of anxiety, while in adolescents TAF predicted both anxiety and compulsive-like behavior. Thus, although TAF and obsessive-compulsive like behavior is prominent in young children and declines as adolescence approaches, older children who tend to be more ritualistic are also more anxious and report stronger beliefs in TAF (Evans et al. 2011). Thus, the relationship between anxiety, magical thinking and ritualistic behavior seems to suggest that ritual and repetitive behavior might be mechanisms employed to adapt to fearful and uncertain circumstances, and which are associated with certain beliefs, such as TAF.

Taken together, these studies indicate that on the one hand, the normative expression of ritualistic and obsessive-compulsive-like behavior might be ascribed to maturational changes in brain circuitry, which is reflected in deficits in set shifting and response inhibition. On the other hand, these behaviors might be considered an adaptive response to normal childhood fears, or as a result of childhood-typical beliefs, such as the belief in magic and TAF.

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Interestingly, magical thinking and its corresponding ritualistic behavior can persist into adulthood and co-exist comfortably with normal capacity for logical thought. Examples include common superstitions such as "knocking on wood" and carrying a good luck charm in order to prevent undesirable events, as well as the elaborate rituals associated with many religious and cultural traditions. A common theme that unites many of these rituals is that certain thoughts or actions can influence the outcome of events, despite the absence of a logical cause-effect relationship. Studies of nonclinical adult populations in cultures that are more, or less, likely to have superstitious beliefs and engage in magical thinking (Icelandic and Australian, respectively), have shown that measures of magical thinking were significantly correlated with obsessive-compulsive symptoms, and especially with compulsive checking. In the Australian sample, magical thinking was correlated with checking, but not contamination and washing symptoms, even when measures of TAF and superstitious beliefs were statistically held constant (Einstein and Menzies, 2004). In the Icelandic sample, magical thinking and measures of stress and anxiety each significantly predicted obsessive-compulsive symptoms (Dögg Helgadóttir et al., 2012). Similarly, in an adult Turkish nonclinical sample, magical beliefs were significantly correlated with checking and obsessive thoughts (Yorulmaz et al., 2011).

2.2. Love Sickness

Leckman and Mayes (1999) and Fisher et al. (2006) have described the striking similarities between clinical obsessions and compulsions, and the experience of romantic love. Similarly to an individual with OCD, the lover experiences intrusive, time-consuming thoughts and preoccupations about the beloved. The lover is often tormented by persistent doubts about whether one's feelings are reciprocal; these doubts can grow into irrational fears of rejection and intense feelings of jealousy and despair. Similar to the experience of pathological compulsions, the lover's daily routine and usual priorities are turned upside down in order to be in contact with the beloved and to increase his or her beauty and desirability to the beloved.

Perhaps not surprising to anyone who has been there, the experience of being in love represents a period of increased vulnerability to develop pathological obsessive-compulsive symptoms centered around relationship themes, leading to distress, relationship dysfunction and disability, a condition that has been recently termed "relationship obsessive-compulsive disorder" (ROCD; Doron et al., 2013).

ROCD has been conceptualized within the context of cognitive models of OCD, which propose that obsessive-compulsive symptoms arise when normal obsessions (intrusions) are consciously appraised in a maladaptive manner, in the context of certain "obsessive" (cognitive or metacognitive) beliefs. Maladaptive, or negative appraisals include interpreting the thought as threatening or personally significant, or obligating oneself to control the thought in order to prevent some feared outcome. Obsessive cognitive beliefs include an exaggerated sense of personal responsibility for external events, perfectionism and intolerance of uncertainty, and the overestimation of threat. Obsessive metacognitive beliefs (essentially, beliefs about one's own thought processes) include those on the importance of thoughts and the need to control them, as well as TAF beliefs (see Abramowitz et al., 2007; Myers et al. 2008). Doron and Kyrios (2005) proposed that a person's tendency to react

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negatively to a given intrusion is increased if the intrusion threatens the individual's perceived competence in certain "sensitive domains". A "sensitive domain" is an area highly valued by the individual (examples include morality, job competence, social acceptability, interpersonal relationships), yet at the same time the individual feels incompetent within that domain. According to this model, compulsive behavior arises as an attempt to compensate for self-perceived deficits within the sensitive domain. Within the context of these cognitive models, ROCD has been proposed to arise due to a specific "double relationship-vulnerability": the co-occurrence of a strong fear of abandonment (attachment anxiety) and sensitivity in the domain of interpersonal relationships, such that the individual's self-worth is contingent on his or her competence in this domain (Doron et al., 2013).

2.3. Parental Preoccupations

For parents, pregnancy and childbirth also represent a life stage where obsessive-like thoughts and compulsive-like behaviors are quite commonly experienced. Leckman et al. (1999) highlighted the similarities between "normal" parental preoccupations and behaviors, and the obsessions and compulsions of OCD. Across the perinatal period, both mothers and fathers reported experiencing recurrent intrusive thoughts related to something bad happening to the baby (80 – 90% of parents), often due to their negligence, or, of things not being "just right" for the baby. A significant proportion of parents reported having intrusive thoughts of intentionally harming the infant (30 – 40% of parents). These thoughts were accompanied by compulsive-like behaviors, which included the need to frequently check the baby, and excessive cleaning and arranging. The majority of parents said that they would experience significant distress if they were prevented from performing this checking behavior.

Similar to the experience of falling in love, becoming a parent can be a time of increased vulnerability for the development or exacerbation of clinical obsessive compulsive symptoms. A number of retrospective surveys have indicated that a considerable proportion of adult women that suffer from OCD attribute the onset of their symptoms to pregnancy or the post-partum period (reviewed in Abramowitz et al., 2004). In a small prospective study of 44 women, Chaudron (2010) found that 2 women (12.5%) suffered the onset of clinical OCD symptoms during the first month post-partum. Similar to ROCD, cognitive models have been proposed to explain the onset of clinical obsessive-compulsive symptoms associated with becoming a parent. In a prospective study of 76 expecting parents (female and male), Abramowitz et al. (2007) showed that the presence of obsessive cognitive and metacognitive beliefs during the third trimester of pregnancy predicted the severity of obsessive-compulsive symptoms at 3 months post-partum, and this relationship was partially mediated statistically by the parents' negative appraisals of their intrusive thoughts at 3 weeks post-partum. Even so, the variance in severity of obsessive compulsive symptoms that was statistically explained by obsessive beliefs and that was partially mediated by negative appraisals was relatively small: a substantial proportion of the variance in obsessive compulsive symptoms could not be accounted for by either dysfunctional beliefs or appraisals. Notably, in the Abramowitz et al. (2007) study, the content and prevalence of the intrusive thoughts was very similar to the previous study by Leckman et al. (1999). Thus, 84 – 92 % of the parents reported intrusive thoughts of accidents that would harm the baby, such as suffocation, while 35 – 46% of the parents experienced intrusive thoughts of intentionally harming the baby. Other thoughts had

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to do with losing the baby (46 – 49%), illness (10 – 30%), contamination (54 – 59%), and sexual thoughts about the baby (8 – 13%).

While cognitive models to explain the emergence of obsessive-compulsive symptoms associated with romantic love and childbirth place special emphasis on dysfunctional belief systems and negative appraisals of normally-occurring intrusive thoughts, neurobiological models focus on the function of the neural circuitry that underlie these adaptive psychological and behavioral patterns. Leckman et al. (1999) state: "Our hypothesis remains that EPPB *(early parental preoccupations and behavior)* is associated with the enhanced activity of highly conserved neural pathways that link heightened risk assessment and error detection to harm avoidant behaviors and that these same systems are altered in a similar fashion in OCD." Both romantic love and parental attachment involve the action of the neuropeptide oxytocin within brain circuits associated with reward and motivated behavior (Love, 2013), and some studies have suggested an association between oxytocin and obsessive-compulsive symptoms (Leckman et al., 1994). Fisher et al. (2006) and Feygin et al. (2006) highlight the similarities between brain regions associated with romantic love, parental attachment, and obsessive compulsive symptoms. Thus, an fMRI study of individuals who were in love for from 1 – 17 (mean of 7.4) months showed increased activation in the ventral tegmental area and caudate nucleus when shown a picture of their beloved. In the later stages of being in love (mean of 28 months, a time when the individuals were less passionately "in love", but during which a longer-term attachment is typically forming), the VTA and caudate were activated, along with anterior cingulate cortex and mid-insular cortex, and in the ventral putamen/pallidum (Bartels and Zeki, 2000, 2004; Aron et al., 2005; reviewed in Fisher et al., 2006; Feygin et al., 2006). In similar fMRI studies in which mothers were shown photographs of their own infant, heightened activity was observed in the lateral orbitofrontal cortex and periaqueductal gray, medial insula, anterior cingulate gyrus and caudate nucleus (Bartels and Zeki, 2004; reviewed in Fisher et al., 2006; Feygin et al., 2006). As already mentioned, many of these same regions have been implicated in OCD symptoms.

3. Symptom-Based Classification of OCD: Washing, Checking, and Ordering: Preprogrammed Adaptive Behavioral Programs

Certain common day-to-day events or experiences are particularly effective at provoking obsessive compulsive-like symptoms in otherwise "normal" people. Being exposed to a disgusting stimulus can trigger compulsive-like hand-washing; the sudden doubt about whether or not the stove was turned off can cause one to go back to the house to check; a hopelessly disordered desktop can provoke an anxious urge to put things in order. What is it about these circumstances that elicit obsessive-compulsive-like symptoms?

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3.1. Contamination Obsessions and Washing Compulsions

One of the most common manifestations of OCD is having contamination obsessions and washing compulsions. These symptoms are considered to be closely related to the normal emotion of disgust, which is an adaptive psychological and physiological response that most likely evolved in order to protect the individual from disease, by promoting behaviors that avoid contact with contamination (Curtis, 2011). In humans of all cultures, there are specific cues that consistently elicit a disgust response: dirty environments, bodily wastes and body contents, sick or unhygienic people, dead bodies, certain sexual behaviors, spoiled or unfamiliar foods, and certain animals. The behavioral reactions to these cues are also quite consistent, including the display of a characteristic facial expression, avoidant behaviors such as withdrawal, stopping, or dropping the disgusting object, and shuddering, along with physiological changes including lowered blood pressure and nausea. Disgust is a motivating factor for the expression of certain ritualistic or habitual behaviors, examples of these include bodily washing and domestic cleaning, avoidance of close contact or exchange of bodily fluids with those other than one's mate or kin, and the avoidance of foods that are spoilt, particularly meat. Curtis (2011) presents a fascinating argument that the disgust response has been co-opted into various religious and cultural traditions, manifesting itself as ritualistic cleansing and bathing practices and, in certain cultures, societal acceptance of the separation of individuals into groups or classes that are "clean" and "unclean".

Disgust sensitivity has been consistently associated with non-clinical obsessive-compulsive symptoms. Disgust sensitivity is typically measured by means of a questionnaire, where the individual rates how disgusting they would consider particular experiences to be. In a large sample of undergraduate university students, Tolin et al. (2006) found that disgust sensitivity was positively correlated with symptoms of obsessive-compulsive washing, even after controlling for anxiety and depression scores. Likewise, Thorpe (2011) found that disgust sensitivity was positively correlated with self-reported compulsive washing, as well as with the duration of time that the individual spent washing, after being experimentally exposed to a disgusting stimulus. This latter correlation was mediated by the individual's self-reported expectations of how severe the hypothetical disease would be, should she or he become contaminated by the stimulus; this result suggests that cognitive or obsessive beliefs about threat severity participate importantly in determining the intensity of the behavioral response. Cisler et al. (2011) examined the initial affective response to a contaminated object and the response to objects that had only distal, indirect contact with this contaminated object, in a "chain of contagion" task. In this task, a contaminated object touched was touched by a clean object, which then in turn touched a second clean object, and so forth, along a chain of 12 objects. This study revealed that, while disgust "propensity" (self-reported prevalence of disgust experiences) predicted the initial affective response to the primary disgusting stimulus, delayed attentional disengagement from disgust-provoking cues in an attention task was associated with an increased appraisal of how contaminated objects distal in the contamination chain were. These results suggest that the inability to shift attention away from disgust-provoking stimuli might be an important factor in contamination-related obsessions and compulsions. Nicholson et al. (2012) made an experimental distinction between disgust propensity (defined as an individual's tendency to experience disgust) and disgust sensitivity, or how negatively the individual appraises a given experience of disgust.

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It was found that a measure of disgust sensitivity was correlated with behavioral avoidance of disgusting stimuli and predicted compulsive washing concerns, while disgust propensity predicted obsessing. The authors suggest that these observations concur with cognitive models of obsessive-compulsive symptoms, in that the negative appraisals of intrusive thoughts, rather than the thoughts *per se*, provoke discomfort that the individual feels compelled to neutralize behaviorally.

Disgust sensitivity is related to enhanced activity in specific brain regions, many of which are also associated with contamination and washing symptoms in OCD. In a sample of 37 healthy volunteers, Mataix-Cols (2008) found significant correlations between disgust sensitivity scores (where disgust-provoking images were rated on a disgust scale of 0 – 8) and percent change in activation (BOLD; Blood Oxygen-Level Dependent response) in the right anterior insula, right ventrolateral prefrontal cortex and temporal pole, right putamen-globus pallidus, bilateral dorsal anterior cingulate gyrus, along with visual cortical regions. Disgust sensitivity can vary according to physiological state; for example, in women, sensitivity to facial expressions of others that indicate disgust or danger was heightened during the luteal phase of the menstrual cycle (Conway et al., 2007) and the self-reported frequency of disease avoidant and grooming behaviors was positively correlated with salivary progesterone levels (Fleishman and Fessler 2011). Disgust sensitivity is also enhanced during the first trimester of pregnancy (Fessler et al., 2005). Changes in disgust sensitivity may be related to reproductive-cycle associated changes in obsessive-compulsive symptoms: in a study of 63 healthy women, obsessive-compulsive symptoms were increased during the late luteal phase of the menstrual cycle (Gonda et al., 2008), and increased "unusual cleaning behavior" during the luteal phase was reported in a second group of healthy women (Dillon and Brooks 1992). These perceptual, emotional, and behavioral changes during the female reproductive cycle are proposed to be part of a prophylactic response that would compensate for the immunosuppressive effects of progesterone, thereby behaviorally preventing exposure to pathogens (Fleishman and Fessler, 2011).

3.2. "Did I Turn Off the Stove?"

Many of us probably have had the experience, while on the way to work, of suddenly thinking that the kitchen stove was not turned off or the door was left unlocked. The level of anxiety and uncertainty provoked by this intruding thought might be intense enough that we turn the car around and go back to the house to check, even if we are reasonably certain that we did indeed adequately secure the house. Such checking rituals or compulsions are a commonly experienced obsessive-compulsive symptom.

Cognitive models associate specific domains of obsessive beliefs with checking compulsions. Thus, in a sample of 238 university students, beliefs associated with the metacognitive belief domain of importance and control of thoughts significantly predicted checking symptoms (while the belief domain of perfectionism and intolerance of uncertainty significantly predicted washing; Myers et al., 2008). Interestingly, at least two separate studies (Einstein et al., 2004; Yorulmaz et al., 2011) found significant positive correlations between magical thinking and checking compulsions, but not washing compulsions.

Consistent with an important role for obsessional beliefs in modulating the intensity of obsessive compulsive checking, experimental manipulations that increased the subject's

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perceived responsibility (the individual's perception of the extent of his/her personal influence on negative outcomes), or that increased his/her expectation of failure, promoted checking behavior during an experimental task (Mancini et al., 2004).

Some cognitive models also stress the importance of "meta-memory", or beliefs about one's own memory abilities, in promoting checking compulsions. When nonclinical experimental subjects were misled to believe that they had performed poorly on a recently taken prospective memory task, feelings of doubt, urges to check, and checking behaviors were increased during a subsequent task, compared to those participants that had been given false positive feedback on their performance in the same task (Alcolado et al., 2011; Cutler et al., 2013). Several studies have demonstrated that, paradoxically, the very act of checking repeatedly has the result of diminishing one's confidence in one's own memory, as well as decreasing the vividness and detail of what was being checked, without affecting actual memory accuracy (e.g., Van den Hout and Kindt, 2003; Radomsky et al., 2006; Dek et al., 2010). Thus, cognitive models of checking compulsions emphasize the importance of beliefs about one's own memory ("meta-memory").

By contrast, some biological models of checking compulsions propose that *actual* deficits in memory capacity might underlie compulsive checking. The evidence supporting a relationship between memory deficits and checking is mixed (for comprehensive reviews see Cutler and Graf, 2008; Harkin and Kessler 2011). Cutler and Graf (2008) argue that the absence of evidence in the literature for a clear memory deficit associated with compulsive checking could be due to the predominant focus on the domain of retrospective memory. These authors have demonstrated clear associations between compulsive checking in nonclinical samples and prospective memory deficits (Cutler and Graf, 2007; 2008). They suggest that an individual's repeated experience with failure in the prospective memory domain could result in undermined confidence and the reliance on checking as a compensatory strategy to ensure that the task in question was carried out adequately. Harkin and Kessler (2011) propose that inconsistencies in the literature regarding the association of memory deficits with obsessive-compulsive symptoms might be due to inter-study differences in task difficulty with respect to their demand on executive function (e.g., attentional control and working memory), binding complexity (binding multimodal stimulus features and maintaining these representations over time), and memory load. They suggest that altered performance on memory tasks by persons with checking compulsions is due to general deficits in executive function that are experimentally apparent only when memory task demands are sufficiently challenging on these three domains.

Thus, in considering cognitive and biological models of checking compulsions, we are left with the question of causality: do beliefs such as magical thinking promote checking and loss of confidence in memory, or do actual deficits in executive function lead to loss of confidence in memory and checking, and perhaps also underlie the tendency to engage in magical thinking? In this context, it is important to consider the findings of studies of childhood obsessive-compulsive symptoms: normal developmental deficits in executive function were associated with obsessive-compulsive symptoms (Pietrefesa et al., 2007), and magical thinking is especially prominent during this developmental period (Evans et al., 2002).

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3.3. "And Now, about That Messy Desk…"

Boyer and Liénard (2006), in a paper in which they present a theoretical discussion and explanation for the existence of normal rituals and compulsive behavior, pose an intriguing and insightful question: Why do pathological compulsions tend to focus primarily on such a narrow set of possible actions (i.e., cleaning, checking, and ordering)? They go on to propose that common intrusive thoughts and compulsions represent the evolutionarily conserved psychological and behavioral responses to recurrent threats to fitness in human ancestral environments. Indeed, pathological phobias, obsessions, and compulsions seem out of place in our modern environment: we have phobias of spiders and snakes, not guns; we compulsively check whether we turned off the stove and locked the door, not the pressure in our tires; we react with disgust and quickly wash after having contact with a cockroach or rat, while barely registering the presence of man-made chemical toxins present in the environment. One can easily imagine the possible adaptive significance of checking behavior (to prevent harm to oneself, one's home or one's kin) and washing (to prevent contamination and sickness) for ancestral humans, but what about ordering and symmetry?

The innate preference for symmetry has been documented in diverse species. Human infants as early as 4 months old show preference for symmetrical patterns (Humphrey and Humphrey, 1989). This hard-wired preference for symmetrical stimuli seems likely to be an important motivating factor for ordering and arranging things in our environment. Boyer and Liénard (2006) suggest that, in the ancestral humans' environment, aligning and ordering one's possessions within the home environment would have made the intrusions of another agent (animal or human) clearly visible, thus being a strategy to detect the presence of dangerous intruders. Ordering objects might also have served as a sort of territorial display, an advertisement of human presence or delineation of territory. Moreover, in ancestral environments where resources were scarce and weapons and tools were probably critical for survival and time-consuming to produce, misplacing them, losing them, or being unable to quickly have them at hand at a moment's notice would have severely compromised reproductive fitness. Finally, as we all know, an ordered environment substantially decreases the functional demand on memory processes during day-to-day functioning, freeing up these processes for more pressing contingencies.

Ordering, doubting and washing (but not checking) symptoms significantly predicted by obsessive beliefs about perfectionism (Myers et al. 2008). Perfectionism has been associated with certain "sensory phenomena", which are sensory perceptions that can precede the expression of tics or compulsions. These sensory phenomena include "Just Right" sensations or "Not Just Right Experiences (NJREs; Coles et al., 2003; 2005), and feelings of incompleteness (Miguel et al., 2000; Coles et al., 2003; 2005; Summerfeldt 2004; Lee et al., 2009). "Just Right" sensations and NJREs are sensations that something is not quite right, and can be experienced in the context of distinct sensory modalities (i.e., the perception that something is not visually "just right", the perception that something does not sound "just right"). NJREs are also quite common in nonclinical samples, where their frequency, as well as the distress and intensity of urges that they provoke, are most strongly correlated with checking and ordering symptoms (Coles et al., 2003, 2005), compared to washing, obsessing, hoarding and neutralizing symptoms.

It is proposed that sensations of incompleteness and NJREs arise due to hyperactivity of neuronal circuitry underlying error monitoring, a function that has been ascribed to the

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anterior cingulate cortex. Error monitoring processes can be measured by electroencephalogram recordings of the "error related negativity" (ERN), which is a negative event-related potential (ERP) associated with incorrect responses during cognitive tasks. ERN amplitudes were significantly greater in a nonclinical sample of individuals that scored high for obsessive-compulsive symptoms, compared to a low-scoring sample (Hajcak and Simons, 2002).

Recently, Evans et al. (2012) showed that normal individuals have a cognitive bias to associate photographs of happy facial expressions with adjectives related to symmetry, and expressions of disgust with adjectives related to asymmetry. Moreover, the individuals' checking symptoms significantly predicted the extent of the cognitive bias toward symmetry. This bias was also predicted by ERP responses to the presentation of visual stimuli of non-parallel lines (comprising 20% of the presentations) interspersed randomly among presentations of parallel lines (80% of presentations). Thus, obsessive-compulsive symptoms were significantly associated with a cognitive bias toward symmetry as well as with measurable evoked neural responses to asymmetrical stimuli.

Taken together, these studies begin to clarify possible relationships between the specific content of obsessive-compulsive symptoms, cognitive processes, and possible underlying neural substrates. Thus, checking symptoms seem to be particularly associated with magical thinking and metacognitive beliefs about the importance and control of thoughts, as well as an inflated sense of responsibility for negative outcomes. Subtle memory deficits or deficits in executive function might be related to checking symptoms, a proposal consistent with the observations that dorsal prefrontal activation is associated with these symptoms (Mataix-Cols et al., 2003). NJREs are sensory phenomena associated with perfectionism, feelings of incompleteness, as well as checking and ordering symptoms, and might be related to faulty error monitoring, which is generally ascribed to activity in the anterior cingulate cortex. Washing symptoms are also associated with perfectionism, as well as disgust sensitivity and delayed attentional disengagement from disgusting stimuli, suggesting the importance of the insular cortex and neural circuits underlying attention.

4. What Can We Learn about OCD from Studying Animal Models?

4.1. A Broader View on Useful Animal Models of OCD

Typically, animal models of neuropsychiatric disorders are assessed and validated according to specifically defined criteria: face validity, construct validity, and predictive validity (for recent reviews on animal models of obsessive compulsive disorder (Korff and Harvey, 2006; Hoffman, 2011; Abelda and Joel, 2012). Face validity refers to the outward similarity of the animal's symptoms or behavior to those associated with the neuropsychiatric disorder in question; construct validity assesses how closely the underlying pathology causing the symptoms/behavior in the animal model resembles that of the neuropsychiatric disorder; predictive validity is based on whether pharmacological treatments known to reduce neuropsychiatric symptoms also reduce the symptoms/behavior of the animal model. Although a plethora of animal models (the majority of which are rodent) for neuropsychiatric

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disorders have been proposed that fulfill some or all of these validating criteria, progress in developing new and effective treatment strategies based on these models has been slow. Perhaps the time has come to take a somewhat broader view of what constitutes a useful and valid animal model for studying neuropsychiatric disorders and developing new therapies.

It is by now apparent that neuropsychiatric symptoms occur along a continuum of severity and frequency in the general population: obsessions and compulsions, depression, anxiety, and even psychotic symptoms (van Os et al., 2009) can occur in individuals that do not otherwise fulfill the diagnostic criteria for mental illness. There are two important implications of this fact. First, the persistence of these psychological and behavioral processes in the human population seems to suggest that they have some fundamental adaptive value, giving weight to the utility of evolutionary models for understanding neuropsychiatric symptoms and disorders. A second implication is that useful information on the underlying modulation and causation of neuropsychiatric symptoms can likely be gained from studying normal, adaptive behavioral and psychological processes.

In order to generate hypotheses on how pathological neuropsychiatric symptoms arise, it is helpful to place these symptoms into the context of normal function: how could this symptom arise given what we know about the modulation of the corresponding normal processes? With respect to obsessive-compulsive symptoms, there are certain features and themes that seem particularly salient: behavioral rigidity, repetition, and persistence; repetitive and stereotyped behavior that is performed (at least ostensibly) to reduce anxiety; the particular thematic content of obsessions and compulsions and their prevalence at certain life stages; the inability to stop a motivated behavior. In this light, questions such as the following become more relevant: how neural circuitry underlying stereotyped, species typical behaviors modulates the duration of those behaviors, how hormones and neuropeptides affect neural substrates in order to promote life-stage typical behaviors, and what are the mechanisms that connect the performance of motivated behavior to the quenching of that behavior's underlying motivation. Moreover, certain "endophenotypes" have been associated with OCD. An endophenotype is a stable, quantifiable biological trait that is associated with the neurospsychiatric disorder and that is present non-affected relatives of the affected individual at higher rates than in the general population, indicating genetic heritability (Gould and Gottesman 2005). Endophenotypes might be directly involved in the production of symptoms, confer vulnerability to the development of symptoms, or reflect an underlying neurological dysfunction relevant to the symptoms. Endophenotypes associated with OCD include deficits in response inhibition and in set shifting, both of which implicate the importance of basal ganglia function in this disorder (Taylor 2012). Thus, general questions of how factors such as developmental changes in brain circuitry and gonadal hormones affect endophenotypes such as response inhibition and set shifting might lead to important insight into OCD symptom pathophysiology.

In the following sections, four aspects of animal behavior that seem particularly relevant to understanding normal as well as pathological obsessions and compulsions are highlighted. Three of these, ritualized behavior, displacement behavior, and motivation are normal (non-pathological) components of animal behavior. A fourth component, stereotypy, is generally thought of as purposeless and pathological but could also be considered to be a normal adaptive behavioral response to abnormal environmental conditions.

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4.2. Ritual

Eilam et al. (2006) speak of "repetitive motor rituals" as being normal components of animal behavior. They characterize rituals as specific acts or movements performed at specific times or places, that is, they have a particular spatiotemporal organization. The normal expression of such ritualized behavior is suggested to be adaptive in the following sense: behavioral rigidity in the day to day life of an animal would minimize the use of attention systems for the control of movement, thereby allowing attention to be directed to other more important aspects of the environment, such as the "lookout" for predators. Serruya and Eilam (1996) describe the daily routine of a group of rock hyraxes, which included leaving the home site primarily via a few (apparently arbitrary) routes and stopping to rest at a few specific places, and Konrad Lorenz (1952) wrote that a group of water shrews continued to jump over the place in their habitual path where a blocking stone had been, even after it had been removed (cited in Eilam et al., 2006). This conceptualization of ritual is closely related to the concepts of "routine" or "habit". We all can describe certain routines we carry out daily, purposeful behaviors that we realize with little conscious thought, always carried out in an identical or nearly identical manner. The formation of habits involves neurobiological substrates that underlie implicit learning, which include those that have been also been implicated in OCD (Graybiel, 2008). Implicit learning and the formation of habits have a relatively slow time course, and involve the repetition of a certain behavior or sequence of behaviors, a phenomenon acknowledged in the adage "practice makes perfect".

Another conceptualization of ritualistic behavior might stress the rigidity in form and sequential organization of individual behavioral components. There are many examples in ethology of "instinctive", unlearned ("hardwired, or inherited) adaptive behavioral patterns that are expressed in an automatic way, often in response to specific "releasing" stimuli, and in an almost identical manner among individuals of a given species. Konrad Lorenz concluded, based on his observations and those of others, that the *performance itself* of instinctive behavior might represent a sort of rewarding experience for the animal, and that animals (as well as humans) often appetitively seek out circumstances in which stimuli can release these behavioral patterns (Lorenz 1938). Among his classic studies in this area are meticulous descriptions and analyses of the "egg rolling" (retrieval) motor pattern elicited in the nesting graylag goose in response to detecting an egg outside of the nest. Baxter (2003) describes the territorial and dominance displays of *Anolis* (a lizard), comprising stereotyped strutting, sagittal enlargement of the body, "push-ups", and dewlap extension. Stereotyped food caching in coyotes and wolves comprises picking up the food article and bringing it to the cache site, inspection of the site, pawing or digging at the site and depositing the food there, followed by bouts of pressing the food into the substrate using the snout alternated with bouts of pushing the loose substrate over the food using snout movements directed away from the body (Phillips et al., 1990). Berridge et al. (1987) characterized the stereotyped motor pattern of rodent grooming behavior, comprising sequentially-expressed bouts of individual, repeated behavioral components (paw-licking, circular face-rubbing movements, and body-licking). It is notable that each of these selected examples, from diverse species, involves fundamental themes that are also relevant to OCD symptomatology: parental care, territory concerns, concerns of the security of one's belongings, and bodily cleanliness. There is another common theme, emphasized in the study by Baxter (2003): like obsessive-compulsive symptoms, these stereotyped, "hardwired" behaviors, in cases that have been investigated in

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detail (Berridge et al., 1987; Baxter, 2003) involve basal ganglia circuitry. Also potentially relevant to understanding OCD are the mechanisms and substrates involved in the sense of reward or satisfaction experienced as a result of carrying out a species-typical behavioral pattern.

Boyer and Liénard (2006) put forth a contrasting view of the meaning and function of ritual. They discuss several domains of human ritual, including children's rituals, cultural rituals, as well as obsessive-compulsive behavior and intrusive thoughts associated with certain life stages, and suggest that these diverse examples of ritual share certain fundamental features. These features are: (1) compulsion, or the feeling that negative consequences would occur if rituals were not performed; (2) rigidity in the performance of individual behavioral components and their sequential order; (3) "goal-demotion", in which the performance of normal behavior is disconnected from the achievement of its typical goal; (4) repetition and redundancy of behavioral components; (5) ritualized behaviors are often centered around a few basic themes, such as contamination and purification. They explicitly distinguish ritual from routine, arguing that while routine is performed "automatically", without conscious control, ritual requires deliberate cognitive control over one's actions. The normal expression of ritual, they propose, ultimately is one of anxiety reduction. They propose that, in OCD, it is precisely this requirement for deliberate cognitive control over behavior that "swamps" working memory capacity, thereby effectively (albeit temporarily) suppressing intrusive thoughts. With the obvious exception of the first feature, each of these common features of human ritual can be associated with ritualistic behavior in non-human animals. For example, many animal behavior patterns (some of which were described above) are expressed in a rigid and repetitive manner. Displacement behavior (discussed in the next section) involves the expression of goal-demoted behavioral patterns, often in the context of conflict or anxiety. Finally, as discussed above, animal ritualistic behavior is often associated with certain themes, such as grooming, territory, and reproduction; stereotyped courtship displays, for example, are often described as "courtship rituals".

4.3. Displacement Behavior

Displacement behaviors are normal behaviors, such as grooming, scratching, or yawning that are often exhibited in an abnormal pattern, such as being more intense, incomplete, or of shorter duration. Although the cause of displacement behavior is a topic of debate (Anselme 2008), there is a general consensus that it serves a de-arousing function in certain stressful situations, such as when the animal is frustrated in its attempts to achieve a certain goal, when there is a discrepancy between the actual outcome of events and what the animal was expecting, and in the face of conflicting motivations. It has also been suggested that they facilitate in behavioral switching, that is, during the switch from one motivated behavior to another (Buckley and Semple, 2012). The neural processes underlying these two proposed functions of displacement behavior – de-arousing and behavior switching – are potentially relevant to understanding obsessive-compulsive symptoms, as compulsions are an attempt to neutralize anxiety or distress, and may arise from dysfunctions in behavioral switching processes.

In the study of graylag geese mentioned earlier, Lorenz observed that, after repeatedly being "fooled" into performing the egg-rolling motor pattern in response to an irrelevant

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stimulus (a cardboard "egg"), the goose ceased to initiate this behavior; instead, at irrelevant stimulus presentation she first directed her attention to the cardboard egg as if to retrieve it, but then paused and suddenly initiated an entirely different stereotyped motor pattern, that of gathering nest material. This phenomenon was explained by Lorenz as an "overflow" of excess motivation or arousal into other motivational channels, leading to the eruption of a different instinctive behavioral pattern. Support for the notion that that displacement activity arises due to thwarted motivation comes from observations of male rats with lesions in the medial preoptic area, which cause a severe deficit in the ability to carry out the copulatory motor pattern while leaving sexual motivation intact. These unfortunate rats display increased displacement behavior (drinking, scratching) when exposed to a sexually receptive female (Hansen and af Hagelsrum, 1984).

In certain situations where displacement activities are promoted, such as when an expected reward is not delivered or in the face of anxiogenic stimuli, there is a concomitant elevation of stress hormones (endorphins and plasma corticosterone) (e.g, Dantzer 1981; Le Moal and Simon, 1991, cited in Anselme 2008), as well as increases in dopamine release in the prefrontal cortex, nucleus accumbens, and striatum (Berridge et al., 1999). It is suggested that in such circumstances, displacement behavior might serve (via an uncharacterized neurobiological mechanism) to reduce activity of the hypothalamic-pituitary-adrenal axis (e.g., Dantzer 1981). When exposed to a stress-provoking environment (bright light) rats or mice that were allowed to perform displacement behavior (gnawing) showed decreases in prefrontal cortex dopamine release, but no changes in dopamine release in the nucleus accumbens or striatum (Berridge et al., 1999), indicating that decreased dopamine release in the prefrontal cortex might be associated with the stress-reducing effects of these behaviors. In humans as well as in non-human primates, the expression of displacement behavior (self-directed behaviors such as scratching, self-touching and grooming) is highly associated with anxiogenic circumstances (Troisi, 2002), and appears to serve as an anxiety-reducing mechanism, particularly in men (Mohiyeddini et al., 2013). Displacement behavior during psychiatric interviews of depressed or anxious individuals was significantly increased in those that were also alexithymic (alexythymia is the inability to recognize and describe one's own emotions) (Troisi et al., 2000). Interestingly, alexithymia has been associated with obsessive-compulsive symptoms in several studies (Kang et al., 2012; Carpenter and Chung 2011; Grabe et al., 2006).

4.4. Stereotypy

Like ritual, the terms "stereotypy" or "stereotypic" have been defined in different ways. It is a term that encompasses behavioral characteristics of repetitiveness, rigidity and invariance in expression, as well as intensity and persistence of performance. Although stereotypy is most often used to describe pathological or abnormal behavior, "stereotypic" can also refer to normal behavioral patterns that involve unvarying, often repeated, components, such as web building by an orb weaver spider or the territorial displays of *Anolis*. Pathological or drug (e.g., amphetamine) – induced stereotypy has been described as "…higher rates of activity, but in decreasing number of response categories…" (Lyon and Robbins, 1975, cited in Ridley, 1994). By comparing rats chronically treated with quinpirole (a selective dopamine D2 receptor agonist) with those treated acutely with amphetamine (a dopamine releaser),

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Eilam et al. (2006) propose a distinction between compulsive behavior and stereotypy: compulsive behavior is associated with a specific spatio-temporal organization, that is, the "compulsion" or "ritual" is performed at specific places and times, leaving the remaining behavior flexible and fairly normal, whereas stereotypy involves changes in the content of behavior, such that certain movements disappear and remaining movements are repeated and intensified. Thus, stereotypy is distinguished from ritual and with respect to purposefulness and flexibility: whereas ritualistic or compulsive behavior is relatively flexible and purposeful, stereotypy is rigid and often seems purposeless. Nevertheless, some forms of spontaneously-arising stereotypy, such as in the captive deer mouse, resemble obsessive-compulsive symptoms on a number of levels (Korff et al., 2008; reviewed in Hoffman 2011).

Many animals living in captivity develop stereotypies. For example, large carnivores (felid species, bears) in captivity can be seen to engage in pacing behavior, where the animal repeatedly traces its steps with striking fidelity. Such pathological stereotypy in captive animals is generally considered to be an adaptive coping response of a normal animal faced with the abnormal or deficient stimuli present in the captive environment (Ridley, 1994, Garner and Mason, 2002; Garner et al., 2003, Eilam et al., 2006). Pathological stereotypy in captive animals often occurs circumstances in which stimuli that normally provoke the expression of a normal instinctive behavioral pattern are absent or deficient, or when normal motivation is chronically thwarted. For example, captive giraffes in zoos often develop oral stereotypies, such as licking of nonfood objects. The development of such stereotypies appears to be due to the inability of the animal to fully express the highly motivated and instinctive tongue-manipulation behavioral pattern necessary for foraging in natural conditions: one study reports that when the standard feeders were replaced with ones that required the performance of more naturalistic foraging behavior, oral stereotypies decreased markedly (Fernandez et al., 2008).

In captive orange winged amazon parrots, individuals reared in barren environments developed significantly more stereotyped behavior, including pacing, route tracing, chewing on the cage wires, vacuous chewing, compared to those reared in environments that provided enrichment in the form of alternate perching sites, swinging or climbing opportunities, objects that could be manipulated, and foraging opportunities that required that the animal perform some naturalistic operation before receiving food (Meehan et al., 2004). Sows develop oral stereotypies such as pulling on their tethering chains, chewing on the bars of the stall, or vacuous chewing, when the duration of feeding is restricted. These stereotypies are suggested to be due to meal-induced sensitization of feeding motivation that is not fully satiated due to restricted feeding duration, resulting in "excess" feeding motivation that is spent on non-food objects after the meal is finished (Lawrence and Terlouw, 1993). Stereotypic pacing in captive lions decreased when the animals were switched from a conventional feeding schedule (6 days fed, 1 day fast) to one that more closely resembled their natural "gorge and fast" feeding pattern (3 days gorge fed, 4 days, randomly selected, of fasting) (Altman et al., 2005).

Ridley (1994) distinguished between cage stereotypies (the response of a normal nervous system to abnormal environment) and "deprivation" stereotypies, which are the product of an abnormal nervous system (for example, in circumstances in which the nervous system was deprived of important sensory input during a critical developmental period). Whereas cage stereotypies are often reversible by providing the animal with the proper environmental

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enrichment, deprivation stereotypies are often resistant to treatment and probably reflect permanent underlying neurological deficits, most likely involving basal ganglia circuitry.

Laboratory reared bank voles (but not in those wild-caught as adults and introduced to laboratory conditions) develop "bar mouthing" stereotypies, in which the animal persistently performs sham biting movements along the bars of the cage. The proportion of time spent bar mouthing correlated with rates of behavior initiation and deficits in the ability to extinguish a previously rewarded response, consistent with basal ganglia dysfunction (Garner and Mason 2002). Similarly, deer mice develop stereotypies including patterned running, jumping, and backward somersaulting that are correlated with the number of errors made in a reversal learning task (Tanimura et al., 2008). Stereotypy in deer mice has been associated with alterations in "indirect" basal ganglia circuits that are considered to have an inhibitory function over behavior (Tanimura et al., 2010, 2011). In captive orange winged amazon parrots, stereotypy was correlated with perseverative choice in a gambling task, where on any given trial one of two feeder boxes was rewarded with a probability inversely related to the proportion of times that same box had been chosen in previous trials (Garner et al., 2003). In captive rhesus monkeys, self-directed stereotypy (nail biting, digit sucking, hair plucking) and fine motor stereotypy (licking or biting cage bars, head twirling) were positively correlated with number of sessions required to extinguish a previously rewarded response (Pomerantz et al., 2012). Thus, like obsessive-compulsive symptoms, stereotypy appears to be related to basal ganglia dysfunction and behavioral set shifting.

4.5. Motivation

Motivation is a concept that has been extensively studied experimentally, and many excellent reviews on its neural substrates and neurochemical modulation exist in the literature (Cardinal et al., 2002; Salamone and Correa, 2002; Berridge, 2004). Motivation can be a difficult concept to define and measure, especially when studying non-human experimental subjects that cannot report their inner psychological experiences. A useful conceptualization of the term might be the central psychological processes that promote movement or behavior. In this light, neural processes and substrates that modulate and control motivation are clearly relevant to obsessive-compulsive symptoms, where obsessions are experienced as mental urges to perform a specific behavior and compulsions are ostensibly performed as an attempt to satiate or neutralize these urges and the anxiety that they provoke.

Motivated behavior is considered to have appetitive and consummatory components: appetitive being the actions directed toward achieving the goal in question (for example, a bar of chocolate), whereas consummatory refers to interactions with the goal object (eating the chocolate). Motivation is conceptualized as having "wanting" and "liking" components. Wanting is the activational state that impels or promotes appetitive behavior, and liking refers to the hedonic, or pleasurable, sensory responses that are associated with interaction with goal object. Wanting is suggested to further comprise the sub-components "appetite to consume" and "activation to obtain" (Cardinal et al., 2002; Salamone and Correa, 2002; Berridge, 2004). In the present example, the former would be measured experimentally by the amount of chocolate eaten, whereas the latter would be measured by the amount of work the subject is willing to do to obtain the chocolate. There is a considerable body of evidence that wanting and liking involve distinct neural and neurochemical substrates. Specifically, the activational

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component of wanting appears to depend on dopamine release within the nucleus accumbens (NuAcc), whereas the hedonic characteristics of the goal object have been associated with opioid neurotransmission within this same region (reviewed in Berridge, 2004). In the specific case of eating behavior, inhibitory GABAergic projections from the NuAcc shell to the lateral hypothalamus control the appetite to consume food, such that experimentally inhibiting these projections results in voracious eating without affecting the hedonic properties of food or the activational component of "wanting" (Kelley et al., 2004), but it is unclear whether similar control mechanisms exist for other motivated behaviors. This inhibitory input to the lateral hypothalamus is suggested to be a mechanism that would allow the rapid interruption of ongoing eating in order to quickly switch to a different behavior, such as vigilance or fleeing, should a more pressing circumstance arise (Kelley et al., 2004).

Processes that control motivation seem relevant for understanding obsessive compulsive behavior. As discussed in the previous sections, many common intrusive thoughts and compulsions appear to represent psychological and behavioral responses to recurrent threats that were commonplace in human ancestral environments (Boyer and Lienard, 2006; Feygin et al., 2006). If this is indeed the case, the expression of these behaviors seems likely to be under the control of hard-wired motivational systems, perhaps similar to those that control other basic motivated behaviors such as eating, sex, and parental behavior. Indeed, one could go further to hypothesize that obsessions and intrusive thoughts are the cognitive interpretation (or *mis*interpretation), of the subjective experience of basic drive systems being activated. In this light, two fundamental processes seem relevant: how particular motivational systems are activated and how the expression of the motivated behavior quenches, or satiates, the subjective experience of motivation. Thus, compulsive behavior might involve the inability to achieve a sense of "satisfaction" (are hedonic processes involved?), or the inability to inhibit an ongoing motivated behavior (a NuAcc shell – lateral hypothalamus interaction?), or a dysfunction in the activational component of "wanting" (involving the control of dopamine release in the NuAcc?). There is a large body of literature on neural substrates of maternal, sexual, and pair-bonding motivation, and satiety mechanisms have been extensively studied with respect to male sexual motivation (e.g., Hull et al., 1999; Stolzenberg and Numan, 2011; Love, 2013). As already discussed, brain activity associated with maternal responses and pair-bonding is similar to that associated with obsessive compulsive symptoms (Leckman and Mayes, 1999; Fisher et al., 2006), and male sexual motivation and satiety involve dopamine and serotonin neurotransmission (Hull et al., 1999). Lorenz observed that the performance of certain species-typical behaviors seems to be intrinsically rewarding or pleasurable (Lorenz, 1938); an understanding why and how the performance of species-typical behaviors in itself is satisfying or rewarding seems important to understanding the roots of compulsions.

Motivational models for classifying OCD, in contrast to those based on overt symptom expression (e.g., those models that distinguish between "washers", "checkers", and ordering/symmetry), emphasize the subject's subjective experience of why they are performing the compulsion. Specifically, the OCD dimensions of "harm avoidant" and "incompleteness" have been proposed (Summerfeldt, 2004; Pietrefesa and Coles, 2008, 2009; Ecker and Gönner, 2008). Under this classification system, checking and repeating compulsions could be motivated by an anxiety or dread that something horrible will happen (e.g., *"The house will burn down if I left the stove on!"*) or by a vague feeling that the action was not performed completely or in exactly the correct way (e.g., "*I did not turn off the stove*

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in exactly the right way"). The latter type is associated with feelings of incompleteness and sensations that something is not quite right ("Not Just Right Experiences, or NJREs).

Szechtman and Woody (2004) and Woody and Szechtman (2011) have elaborated a neurobiological model related to the harm avoidant dimension of OCD that is based on a proposed innate "security motivation system", which is activated during certain circumstances when the presence of danger is uncertain, and which promotes apprehension and increased vigilance. In such circumstances, it is argued, in the absence of objective cues that would signal safety and thereby turn off the security motivation system, an alternative, behavior-based mechanism is necessary. The hypothetical behavior-based mechanism comprises the performance of certain species-typical precautionary behaviors, such as checking, which activate brainstem-mediated negative feedback that serves to de-activate the security motivation system. In OCD, they argue, there is a dysfunction in the mechanism by which precautionary behaviors normally mediate the negative feedback onto the security motivation system: the motivational system is turned on and promotes precautionary behavior, but the performance of these behaviors no longer results in the quenching of security motivation, resulting in the prolonged repetition of precautionary behavior. Hinds et al. (2012) recently presented evidence consistent with this hypothesis. These authors used changes in the Respiratory Sinus Arrythmia (RSA) amplitude as an objective measure of the autonomic nervous system's response to potentially threatening stimuli in the environment; in such circumstances, parasympathetic tone is reduced so that the sympathetic nervous system can be more quickly engaged. In the study by Hinds and colleagues, in response to a disgusting, apparently contaminated stimulus, the change in RSA was equivalent for both OCD washers and controls (indicating equivalent autonomic responses to the stimulus), but the amount of hand-washing time required to feel satisfied (and normalize the RSA amplitude) was much greater for OCD washers than for controls. Interestingly, in this same test, the RSA and behavioral measures did not differ between controls and OCD checkers.

One could propose that similar deficits in motivational control might underlie the incompleteness dimension of OCD. However, a possibly important distinction between circumstances in which the security motivation and the "incompleteness motivation" systems would be activated, is that the latter would be associated with objectively verifiable cues that could have the potential to signal goal completion. Thus, one's senses can verify that one's desk is sufficiently well-ordered, or that the stove is completely turned off in the correct manner. The persistent inability to quench incompleteness motivation might be experienced as not just right sensations, or cognitively as perfectionism (things not being good *enough*). The neural substrates responsible for persistent feelings of incompleteness and NJREs might be those that normally generate error signals when the actual and expected outcomes of goal-directed action are incongruent; possibly, these error-generating circuits are hyperactive in the incompleteness dimension of obsessive compulsive symptoms (Hajcak and Simons, 2002; Summerfeldt 2004; Maltby et al., 2005).

Conclusion

That obsessive-compulsive symptoms are particularly prominent during specific life stages suggests the involvement of certain neural processes. Subtle deficits in executive

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function would seem to promote the expression of obsessive-compulsive symptoms, and perhaps certain cognitive beliefs such as magical thinking and TAF, as well. Life event-specific hormonal changes, such as elevated progesterone levels during pregnancy and the luteal phase of the menstrual cycle, as well as neurotransmitter and neuropeptide systems involved in affiliative and motivated behavior, might also play a role in promoting obsessive-compulsive symptoms. Hyperactivity or hypersensitivity of neural substrates that underlie error or disgust processing, such as the anterior cingulate and insular cortices, might underlie sensations of incompleteness, NJREs, and contamination/washing obsessions and compulsions. The occurrence of clinical OCD symptoms might be due to the interaction of many of these, and more, factors.

Certain normal animal behaviors and behavioral phenomena seem particularly relevant to understanding OCD. Ritualized behavior can be observed in animals, in the form of behavioral routines and species-typical stereotyped behaviors, often within a particular spatiotemporal context. These behaviors often involve components that are repeated in a consistent manner and are maintained for a certain duration, after which the behavior is terminated, allowing the animal to switch its attention to a different behavior. What are the mechanisms that maintain and terminate the expression of these behaviors? This question is relevant to OCD in the sense that individuals with this disorder appear to have difficulty switching attention from one behavior to another, and compulsive behavior is in fact often quite stereotyped and repetitive.

Although the function of displacement behavior is not completely clear, it would seem to serve some sort of anxiety-reducing function. However, the "anxiety" associated with displacement behavior seems less related to fear, and more related to inner conflict, uncertainty, and the inability to carry out or obtain a desired behavior or goal. Are the neural systems that underlie this type of anxiety similar to those associated with the fear response, and how exactly does the performance of a displacement behavior negatively feedback onto the activity of these systems? This issue is potentially important for understanding why compulsions are performed as an attempt to neutralize obsessional anxiety and distress.

It is interesting to note that both acute displacement behavior and chronic pathological stereotypy can be induced by thwarting normal, species-typical motivation, such as the sex drive, in the case of MPOA-lesioned male rats, or foraging-associated tongue manipulation, in the case of captive giraffes (Hansen and af Hagelsrum, 1984; Fernandez et al., 2008). What are the processes by which the inability to carry out a motivated species-typical behavior leads to the expression of irrelevant, excessive, or repetitive behavioral patterns?

The suggestion by Lorenz (1938), that expressing certain species-typical behavioral patterns is in itself rewarding seems relevant to understanding why certain behavioral and thought patterns, such as those related to washing, checking, ordering, are particularly common obsessive-compulsive symptoms. What neural substrates and mechanisms underlie the feeling of satisfaction obtained when carrying out a species-typical behavior, and could these be dysfunctional in OCD, consequently leaving the individual with the sensation of dissatisfaction and lack of closure?

It is hoped that the present discussion will encourage OCD researchers to consider obsessive-compulsive symptoms within the context of normal behavioral processes and their underlying neural control mechanisms, in that way formulating new hypotheses on the pathophysiology of this disorder.

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Chapter III

OCD in Children: Saying Goodbye to Private Butt Nugget and Professor Just Right

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Abstract

Pediatric Obsessive-Compulsive Disorder (OCD) is a prevalent and debilitating condition for many young patients. OCD dictates compulsive behavior and rituals that complicate young patients' emotional, familial, academic, and social lives. This chapter reviews the literature on symptoms as well as offering both neurobiological and psychosocial explanatory models. Additionally, the extant treatment outcome research is briefly summarized and a modular approach to cognitive behavioral therapy (CBT) is explained. Finally, two cases are presented to illustrate the modular approach to CBT.

Pediatric obsessive compulsive disorder (OCD) is a common psychological problem challenging young patients, their families, and clinicians. The symptoms are often perplexing, embarrassing, and disabling contributing to patients' distress. Fortunately, modern clinical science offers solid conceptualization, assessment, and intervention strategies. This chapter aspires to build clinicians' knowledge about the phenomenology of childhood OCD and its treatment via cognitive behavior therapy (CBT).

The chapter begins with an overview of symptoms associated with pediatric OCD. Neuropsychiatric vicissitudes are described and a psychological formulation is provided. The treatment outcome literature is briefly summarized and readers are introduced to modular

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CBT. Finally, the chapter concludes with two confabulated cases which combine prototypical elements from multiple patients in order to illustrate salient points.

Symptoms and Phenomenology

The Diagnostic and Statistical Manual of Mental Disorders (5th edition) (DSM-5; APA, 2013) offers a diagnostic outline for OCD replete with multiple criteria. The first criterion addresses the presence of obsessions, compulsions, or both obsessions and compulsions. Obsessions are defined are repetitive enduring thoughts, urges, and/or images that are subjectively experienced as unwanted, intrusive, and anxiety producing. Additionally, individuals try to rid themselves of these unwanted mental intrusions by suppressing and/or neutralizing them. Compulsions are specified as recurring actions that individuals are compelled to perform by obsessive compulsive rules or demands. The compulsive behaviors are driven to reduce distress, or prevent harm yet they are illogically linked to what is prevented. DSM-5 requires that the obsessions and compulsions exhaust at least one hour per day as well as result in social and occupational (school) impairment.

OCD is quite impairing contributing to children's problems in finishing chores, completing assignments, bathing, sleeping, participating in typical routines (bedtime, going to school, family meals, etc.) as well as social discomfort. Children diagnosed with OCD frequently demonstrate passive avoidance (Franklin, Ryan, Foa, & March, 2004). Franklin et al explained that children will avoid situations that trigger their OCD symptoms (e.g doing school work, restaurants, etc).

In general, OCD is a multi-dimensional disorder and symptoms fall into multiple domains (Ivarsson & Valderburg, 2006; McKay & Piacentini, Greisberg, Graae, Jaffer, & Miller, 2006). More specifically, obsessive domains include contamination, aggression, sexual, hoarding, magical thinking, health/body, mortality, religious and miscellaneous categories (Merlo & Storch, 2006). In children, complaints of hearing recurring song lyrics or rhymes may occur (Mash & Wolfe, 2010).

Stewart et al (2008) found that OCD in children and adolescents is characterized by four factors including aggressive/sexual/religious/somatic/checking, contamination/cleaning, symmetry/ordering/counting/repeating and hoarding. The common obsessive symptoms in youth are aggressive thoughts/images and fear of contamination (Bernstein, Victor, Nelson, & Lee, 2013). Ivarrson and Valderburg (2006) noted that somatic concerns were associated with obsessions focused on illness and disease. Contamination fears also may include worries about substances such as household cleaners and sticky substances (Merlo & Storch, 2006). Merlo and Storch also explained that aggressive obsession may not only reflect fears of harming/hurting oneself or others but also blurting out inappropriate remarks as well as stealing or breaking objects. The content of the feared consequences may be relatively unusual (Franklin et al., 2004). For instance, Emily a 15 year old female patient diagnosed with OCD worried that if she came in contact with classmates who were not in the honors class and did not neutralize or wash, she would lose IQ points.

Other common obsessions include sexual, religious, superstitious, somatic, symmetry, scrupulosity, and fear of humiliation ruminations (Piacentini, Peris, March, & Franklin, 2012). Magical thinking, disaster oriented thoughts, and repetition rituals distinguished the

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superstitious obsessions. Magical thinking often shows up as possessing lucky numbers or colors. McKay et al. (2006) concluded that obsessions in young pediatric patients diagnosed with OCD closely resemble magical thinking and may not match up well to overt symptoms. Additionally, just right or symmetry obsessions and compulsions appear related to perfectionism (Storch et al., 2007a).

According to Merlo and Storch (2006), compulsions also assume various forms. Piacentini and colleagues noted that typical compulsive behaviors include washing, repeating, arranging, reassurance seeking, praying, confessing, and ordering. List making, touching, tapping, blinking/staring are characteristic compulsions in pediatric OCD (Butwinka & Gmitrowicz, 2010). Piacentini et al. (2012) explained that compulsive/ritualistic behavior may relieve feelings of distress, discomfort, incompleteness, and an uneasy sense that something is just not right.

Hoarding is an interesting subtype or point on the OCD continuum. Storch et al. (2007b) found that 26 percent of children diagnosed with OCD demonstrated hoarding symptoms. They also discovered a high prevalence of more externalizing symptoms among hoarders. Further, ordering and arranging symptoms were linked to hoarding. Similar to other forms of OCD, hoarding was associated with high levels of perfectionism. Additionally, hoarding is related to significant elevations in depressed moods (Bernstein et al., 2013; Stasik et al., 2012). Neziroglu, Bubrick, and Yaryura-Tobias (2004) cogently distinguished between collecting and hoarding. They noted collectors experience pride in their possessions, organize them in special places, display objects without embarrassment or shame, and enjoy talking about their collections. Indeed, collectors enjoy talking about their possessions since many peers collect the same things (e.g baseball cards, action figures, snow globes, etc.) (Franklin et al. 2004). Hoarders, on the other hand, feel embarrassed by their collections, live in clutter, and are reluctant to talk about or show their possessions to others. Their collections are often quite peculiar and idiosyncratic (chicken bones, scraps of plywood from Home Depot, etc.). Hoarders suffer an earlier age of onset than patients with OCD (Storch et al. 2007b).

Co-morbidity is common in pediatric OCD (Merlo & Storch, 2006). Recent estimates revealed that Major Depressive Disorder was co-morbid in 39 percent of children and 62 percent of adolescents diagnosed with OCD. Additionally, ADHD was co-morbid with OCD in 51 percent of children and 36 percent of adolescents. Finally, oppositional defiant disorder walked around with OCD in 51 percent of children and 47 percent of adolescents.

There are various age related differences in OCD symptom expression. Kircanski, Peris, and Piacentini (2011) affirmed that the average age of onset for OCD in children is between 8 and 11 years. Younger children demonstrate more magical thinking than older youngsters (If I don't tap the balcony three taps, my family will get cancer) (Franklin et al 2004). Further, adolescents report more sexual obsessions (Bernstein et al. 2013).

Storch et al. (2012) examined rage attacks in pediatric OCD. Rage attacks are conceptualized as extreme emotional reactions marked by anger and aggression that are inappropriate to the circumstances. Children subjectively report little perceived control over their reactions. Verbal attacks and threats are common. Storch and colleagues found that rage attacks were usually targeted at parents and siblings. They occurred at home and appeared elicited by parental limit setting, changes in routines, things having to be perfect, feeling contaminated, and inability to maintain the just right feeling. Storch et al. concluded that rage attacks appear maintained by negative reinforcement (e.g. increased calm following a rage episode as a function of escaping OCD stimuli or reduction of inner tension).

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Rage attacks are linked to levels of family accommodation. Family accommodation involves family routines, rules, activities, and behaviors to reduce symptom severity (Storch et al., 2012). Flessner et al. (2011) found that a majority of parents (63.5%) engaged in reassurance seeking with their children diagnosed with OCD. Approximately one-third of parents participated in their children's rituals and avoidance behavior daily. In addition to symptom severity, parental anxiety predicted increased accommodation.

Neuropsychiatric Issues

OCD is seen as a neuropsychiatric disorder (March & Mulle, 1998; March, Franklin, Leonard, & Foa, 2004; Stein & Feinberg, 2007). March et al. (2004) cited family, genetic, neuroimaging, and neuroendocrine/neurotransmitter studies as foundations for this conceptual understanding. Understanding the neuropsychological/neuropsychiatric basis is important in both conceptualization as well as treatment. The classic "boss back strategy" (March & Mulle, 1998) is founded on this neuropsychiatric formulation.

Contemporary affective neuroscience locates OCD in the frontal-thalamo-striatal circuitry which involves the frontal lobe, dorsolateral prefrontal cortex, orbitofrontal cortex, basal ganglia, and the anterior cingulate gyrus (Brand, Reid, & McKay, 2013). The anterior cingulate cortex is implicated in cognitive and emotional modulation (Hajcak, Franklin, Foa, & Simons, 2008). Szeszko, et al. (2004) also discovered that children diagnosed with OCD possessed a larger amount of anterior cingulate gyrus gray matter and smaller globus pallidus volumes than their counterparts. Accordingly, Hajcak and colleagues concluded that when these brain regions are activated, self-doubt, perfectionism, and the need for repeated actions to make things just right erupt. However, it is currently unclear whether this association reflects a state or a trait-like phenomenon.

Ornstein, Arnold, Manassis, Mendlowitz and Schachar (2010) posited that executive control and attentional difficulties in children diagnosed with OCD are due to compromises in the frontal-striatal pathways. They found that children diagnosed with OCD demonstrated slower letter sequencing, were highly distractible in trail making tasks, and recalled words that were never presented during the California Verbal Learning Test free recall task. Additionally, serotonin and dopamine neurotransmitter receptors are pivotal sites. Children diagnosed with OCD perform less competently on tests of memory, processing speed, and executive functioning (Andres et al., 2007). Flessner et al. (2010) reported that children diagnosed with OCD exhibited poorer performance on the processing speed factor on the WISC-III.

Pediatric Autoimmune Neuropsychiatric Disorders (PANDAS)

Pediatric Autoimmune Neuropsychiatric Disorders are associated with streptococcal infections (PANDAS) (Mell, Davis, & Owens, 2005). March and Mulle (1998) explained that for some youngsters OCD symptoms may erupt in the context of a Group A hemolytic streptococcal infection (GABHS) which is commonly referred to as a strep throat. Symptoms erupt whenever children suffer a streptococcal infection (Helbing & Ficca, 2009). In addition to prototypical OCD markers, PANDAS symptoms typically reflect ADHD-like behaviors,

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separation anxiety, mood fluctuations, sleep disturbances, nighttime bedwetting, increased daytime urinary frequency as well as fine and gross motor changes (Helbing & Ficca, 2009). Recently, Bernstein et al. (2013) noted that there were no differences in symptom presentation in children presenting with PANDAS OCD and non-PANDAS OCD.

Psychological Formulation

The obsessive compulsive behavioral cycle is explained well from an operant conditioning model (Kircanski et al., 2011; Piacentini & Langley, 2004). Simply, obsessions prompt an aversive internal emotional state such as anxiety, fear, worry, or dread. Compulsions are then enacted which diminish the aversive emotional state. Since these repetitive behaviors work to provide relief or escape, they are maintained by negative reinforcement.

In addition to learning theory paradigms, cognitive models explain the initiation, maintenance, and exacerbation of symptoms. Concepts such as thought-action fusion (TAF), inflated responsibility, overly valued notions of control, and intolerance for doubt are well-developed constructs. In these paradigms, OCD is marked by assuming excessive responsibility for harm to self or others (Reynolds & Reeves, 2008). Individuals diagnosed with OCD execute neutralizing mental rituals aimed to prevent harm.

TAF refers to patients believing there is no difference between a thought and action (Salkovskis, 1999). There are two types of TAF: morality and likelihood. In morality themed TAF, thinking about an action is the same as actually doing it (e.g. Thinking about stabbing someone in the eye with a knife is the same as really stabbing). There are multiple assumptions embedded within thought action fusion (Salkovskis, 1999). Patients diagnosed with OCD hold rigid rules such as failing to prevent harm is the moral equivalent of actually creating harm. Additionally, youth diagnosed with OCD suffer inflated responsibility believing that one's responsibility for preventing harm is not attenuated by extenuating circumstances. Patients believe that failing to ritualize to decrease harm is identical to the intention to cause harm. Finally, control is pivotal in patients diagnosed with OCD. Doubt and ambiguity is associated with danger (Foa & Kozak, 1995) whereas certainty is linked with perception of safety. Accordingly, youth diagnosed with OCD hold the expectation that they must exercise perfect, complete, and absolute control over their thoughts.

Reynolds and Reeves (2008) studied whether Rachman's (1993) and Salkovskis (1999) model of OCD applied to children. Reynolds and Reeves concluded from their review of non-clinical and correlational designs (n=7) that excessive responsibility appraisals and TAF punctuate pediatric OCD. Therefore, distorted cognitive processes may be rounded up as usual suspects in pediatric OCD and consequently targeted for treatment.

Rituals and routines enjoy a functional value. Repetitive actions contribute to a greater sense of perceived control (Purdon & Clark, 2005). When perceived control is threatened or thwarted, increased anxiety results. Individuals diagnosed with OCD then react by redoubling their control efforts (e.g. thought suppression, avoidance, etc) to restore emotional order. However, countercontrol measures actually create increased anxiety to decrease perceptions of control. Hence, this is often referred to as the paradox of mental control (Purdon & Clark, 2005).

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Treatment

CBT for Pediatric OCD is a well-established approach (Barrett et al., 2008; Franklin et al., 2010, 2011; March, 2011; POTS, 2004). Kircanski et al. (2011) concluded that meta-analyses, controlled trials, open clinical trials, and single case studies recommend CBT for pediatric OCD.

The Pediatric OCD Treatment Study (POTS, 2004) found that a treatment regimen combining CBT and sertraline was more effective than either CBT or sertraline alone. However, remission rate data demonstrated that the success rate children treated with CBT alone was no different than the combined protocol and was better than the sertraline alone condition. Accordingly, this study validated CBT as a first line treatment for pediatric OCD. There are several excellent treatment manuals available to clinicians. These manuals include *OCD in children and adolescents* (March & Mulle, 1998), *Family-based treatment for young children with OCD* (Freeman & Garcia, 2008), and *Cognitive behavioral therapy of childhood OCD* (Piacentini, Langley, & Roblek, 2007).

Several recent meta-analyses endorse CBT as the treatment of choice for pediatric OCD (Abramowitz, Whiteside, & Deacon, 2005). Abramowitz et al. found that the weighted mean effect size for CBT was .99. Watson and Rees (2006) showed similar strength for CBT in their review finding an effect size equaling 1.45. Moreover, Watson and Rees noted that CBT boasted greater effectiveness than pharmacotherapy.

Family accommodation decreases treatment effectiveness (Caparino et al., 2012; Storch et al., 2012). In a structural equation model, Caparino et al. (2012) found that high levels of family accommodation were related to increased symptom severity and functional impairment. Family based CBT which directly targeted family accommodation produced favorable outcomes (Merlo, Lehmkuhl, Geffken & Storch, 2009; Peris & Piacentini, 2013). Outcome data revealed that this directed treatment decreased family accommodation, blame, familial conflict, and symptom severity.

Modular Approach

Modular CBT is earning favor with researcher and clinicians (Chorpita, Daleiden, & Weisz, 2005; Friedberg & Brelsford, 2011; March, 2011; Weisz, 2013). Modular approaches take the empirically based procedures and group them into conceptually similar categories. Modules are formed based on practice elements that share similar content (Chorpita et al., 2005). Each module is related to the other modules by cognitive behavioral theory. Typical modules include psychoeducation, self-monitoring, basic behavioral tasks, cognitive restructuring, and exposure/experiments.

Psychoeducation typically readies patients for treatment by orienting patients to the symptoms, treatment options and course of treatment (Ong & Caron, 2008). Psychoeducation (PE) empowers patients with information and hope demystifying their distress as well as the treatment process (Cuipers, Munoz, Clarke, & Munoz, 2009; Goldfried & Davila, 2005; Wesseley, Bryant, Greenberg, Earnshaw, Sharpley, & Hughes, 2008; Wills, 2009). Self-monitoring alerts patients' attention to their thoughts, feelings, behaviors, and physiological arousal as well as challenging them to track these processes. Basic behavioral tasks (BBT)

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emphasize employing learning theory principles in order to change maladaptive overt actions. BBT include procedures such as contingency contracting, pleasant activity scheduling. relaxation, and social skills training. Cognitive Restructuring (CR) focuses on techniques that target private mental events such as maladaptive internal dialogues, judgments, conclusions, appraisals, and images. CR includes problem solving, self-instruction, and rational analysis. Experiments and exposures provide young patients with experiential learning opportunities that change action patterns (Barlow, Allen, & Choate, 2004).

Psychoeducation

There are a number of PE resources that are quite handy for parents and children. Good on-line resources are plentiful and include reputable sites such as the National Institute of Mental Health (www.nimh.org), American Academy of Child and Adolescent Psychiatry (www.aacap.org), the New York University Child Study Center (www.aboutourkids.org), Anxiety Disorders of America Association (www.adaa.org), Association for Behavioral and Cognitive Therapy (www.abct.org), and Division 53 of the American Psychological Association (www.clinicalchildpsych.org). Additionally, parenting books such as *Freeing Your Child* from OCD (Chansky, 2000) and *Talking Back to OCD* (March, 2007) provide parents with robust information and parenting tips. Finally, self-help books/workbooks for children such as *Up and Down the Worry Hill* (Wagner, 2000), What to do when your brain gets stuck (Huebner, 2007) and *It's the Thought that Counts* (Kant, Franklin, & Andrews, 2008) empower youth with coping strategies and knowledge. I (RDF) often use a miniature brain model to show young patients and their families the part of the brain where OCD is an "uninvited guest." Additionally, pictures from March and Mulle's (1998) manual showing various brain scans facilitates psychoeducation.

Psychoeducation is delivered actively with children and adolescents. Collaboration is essential. Children and parents are invited to review the psychoeducation and highlight points of agreement as well as disagreement. Clinicians typically work assiduously to individualize generalize information.

Assessment/Self-Monitoring

Assessment of OCD encompasses an array of approaches including diagnostic interviews, clinician-rated measures, and self-report methods (Grabill, Merlo, Duke, Hartford, Keeley, Geffken, & Storch, 2008). Accordingly, there are multiple assessment options. Common structured diagnostic interviews include the child version of the Anxiety Disorders Interview Scale (ADIS-C; Silverman & Albano, 1996), the Schedule for Affective Disorders and Schizophrenia for School-Age children (K-SADS;Kaufman, Birmaher, Brent, & Rao, 1997), and the Structured Clinical Interview for Diagnosis (SCID; First, Gibbon, Spitzer, & Williams, 1996). However, all of these clinician administered measures require training and are time consuming (Merlo & Storch, 2006).

The Children's Yale-Brown Obsessive Compulsive Scale (CY-BOCS; Scahill, Riddle, McSwiggin-Hardin, & Ort, 1997) is considered the gold-standard for assessing pediatric OCD (Lewin & Piacentini, 2010). The CY-BOCS is a semi-structured interview and assesses the

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presence as well a severity of OCD symptoms (Merlo & Storch, 2006). The CY-BOCS enjoys good psychometrics and is readily completed in about 15 minutes. The self-report version of the CY-BOCS is a viable alternative to or augmentation of the clinician administered CY-BOCS (Conelea, Schmidt, Leonard, Freeman, & Cahill, 2012). Conelea et al. noted a positive correlation between the clinician and patient version. Finally, the self - report CY-BOCS is a sound choice in time limited settings and as a way to track options.

Self-monitoring and scaling are referred to as ways to "map" the OCD (March & Mulle, 1998). Essentially, mapping demarcates the parameters of the obsessive-compulsive symptoms by identifying contextual information as well as the thought/feelings/behaviors/ physiological changes/interpersonal patterns accompanying the disorder. The mapping process yields a hierarchy. Hierarchies represent rank ordering of symptoms from the least distressing to the most distressing situations. The rank is determined by patients' subjective reports of distress (SUDS).

Friedberg, McClure, and Garcia (2009) described the use of "My World" creative mapping procedure for OCD. My World is a game-like intervention where children create their own game space on poster board. Spaces are labeled as "safe" zones are colored green. On the other hand, danger zones (e.g. sinks, basement, etc) are titled and colored red. When the game board is finished, play begins. Players roll the dice and land on either safe or danger zones. The child then scales their anxiety and reports their thoughts, behaviors, interpersonal patterns, and physiological sensations.

Basic Behavioral Tasks

Basic behavioral principles are recommended for handling reassurance seeking. Remember reassurance seeking is a form of checking. In her excellent book, Chansky (2000) offered several suggestions for handling requests for reassurance such as allocating reassurance coupons to the child, awarding points on a contingency chart when children do not seek reassurance, and earmarking blocks of time for seeking reassurance and not checking. Additionally, parents may instruct children to ask for reassurance in different ways such as singing it or asking the question incorrectly. When children successfully change their action, they receive contingent rewards.

Cognitive Restructuring

Cognitive behavioral therapists often refer to the cognitive restructuring procedures as facilitiative strategies (Silverman & Kurtines, 1996). Simply, they facilitate progress toward exposure. One of the first ways to cognitively battle the OCD is to objectify or distance the disorder from the child. Objectification of the OCD is directly related to the material presented in the psychoeducational module. Simply, children and parents must understand the disorder in order to gain distance from the symptoms.

March and Mulle (1998) presented a cogent rationale for distancing the child from the OCD. Objectifying the OCD by detaching the symptoms from the child enables "the child, family members, and therapist to become allies against OCD rather than battling with each

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other (March & Mulle, 1998, p.63).” Giving the OCD a name is a common initial step in the cognitive restructuring module (e.g.” Mr. I told you so”, “Captain Ridiculous”, etc).

Bossing back or talking back to the OCD is a classic intervention (March & Mulle, 1998). The boss back strategy is theoretically linked to the naming process. Once the OCD is named, children engage in a conversation where they argue, debate, or otherwise talk back to the OCD (e.g Pound sand, Captain Ridiculous, I’m not checking the light switch!). The boss back strategy can take many forms. OCD statements could be written on one side of index cards and a reply which bosses back the OCD is recorded on the other side. A role play where voices are given to OCD and the talk back strategy is another approach. Puppet play is often fun with young children.

It’s Me, Not OCD is a specific technique based on the boss back strategy (Friedberg et al., 2009). The child selects or draws a picture of OCD that reflects the name (e.g. Mrs. U.B. Perfect). Next, the picture is pasted on one side of a wooden paint stirrer and labeled OCD. OCD rules and commands are written beneath the picture. On the other side, a school picture of the child is pasted and titled, “Me!” On this side, talk back statements are noted.

Using self-instructional cognitive restructuring interventions to decrease symptoms is also commonplace during the boss back module (Chansky, 2000; Purdon & Clark, 2005). Making absolute OCD rules more conditional than absolute is central. For example, replacing the OCD rule “I will yell a bad word in class unless I count 1-2-3” with the self-instruction, “I might yell out a bad word in class unless I count 1-2-3.”

Exposure/Experiments

Exposures and experiments are consistent with Nock’s (2008) contention that actions speak louder than words. Behavior is an authentic form of communication as well as an accurate depiction of reality. Nock (2008) explained, “I could tell you that I am in peak physical condition as evidenced by the fact that I can lift 500 pounds over my head. However saying that does not make it so (p. 161).” Exposures and experiment puts understanding and reappraisal into genuine action. In the experiment and exposure module, young patients try on new action patterns for size by behaving differently (Rouf, Fennell, Westbrook, Cooper, & Bennett-Levy, 2004). Exposures and experiments are arguably the most powerful intervention pod for treating OCD (March & Mulle, 1998).

Friedberg, Gorman, Wilt, Biuckians, and Murray (2011) delineated several fundamentals in conducting exposures and experiments. First, exposures emphasize present oriented experiences as well as direct young patients attention to the here and now. Second, exposures propel increased perceptions of control and create authentic learning opportunities where patients appreciate that they are in control of the OCD rather than being controlled by the disorder. Third, exposures are emotionally evocative encounters which build self-correcting behavior changes. Fourth, exposures and experiments alter dysfunctional attitudes by providing young patients with disconfirming data. Fifth, all experiments and exposures are collaborative rather than prescriptive. Patients remain in charge of the process. Sixth, the best exposures and experiments approximate real life circumstances young patients encounter. In this way, transfer of learning occurs. Experiments and exposures also include a debriefing process where the experience is explicitly analyzed and the youngsters derive their own inferences.

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Gillihan et al. (2012) delineated several key strategies to remember when delivering exposure. First, encouraging and rewarding any approach behavior is essential. In this way, exposure builds on basic behavioral tasks where parents and teachers deliver rewards based on a contingency contract. For instance, stickers, points, and small prizes may be given to the child for approaching their feared situations. Neglecting to include these significant others likely will compromise treatment effectiveness.

Gillihan and colleagues suggested that clinicians mindfully consider the form of exposure. They advocated imaginal exposure for patients who mentally anticipate untoward consequences if OCD's warnings are unheeded. Additionally, imaginal exposure is good for patients who engage in thought action fusion. Failing to address mental compulsions in exposure is another common mistake (Gillihan et al. 2012). They remarked that mental compulsions during exposure tasks represent subtle behavioral avoidance.

There are also creative ways to make exposures fun (Friedberg et al., 2009). We often refer to exposures as therapeutic adventures. Germ scavenger hunt and musical contaminants are examples. Abramowitz, Deacon, and Whiteside (2011) recommended that clinicians adopt a playful attitude when conducting exposures with children diagnosed with OCD. They noted, "attempts to interject humor or silliness can make the experience more tolerable for the patient, the parent, and therapist (p. 309)."

The literature is replete with creative exposures and experiments (Marien, Storch, Geffken, & Murphy, 2009; McKay, Storch, Nelson, Morales, & Moretz, 2006). McKay collaboratively developed exposures with a young female patient diagnosed with OCD who feared sexual stimuli. The exposures included the patient watching TV ads for Viagara and browsing magazines for sexually provocatively advertisements. Marien and colleagues worked with a 13 year old female patient diagnosed with OCD who feared she was a lesbian and was also triggered by sexual stimuli. They collaboratively designed exposures where the girl watched the movie Dirty Dancing and skimmed "taboo" magazines such as Victoria Secret catalogues.

Friedberg et al. (2009) described their use of a "Musical Contaminants" game with an 8 year old girl diagnosed with OCD who believed she would be "poisoned" by germs. Musical contaminants is a modified version of musical chairs. The game begins with the lowest item on the OCD hierarchy being passed around among family members when music is playing. However, unlike in musical chairs, the person who holds the object when the music stops is the WINNER. The game provides practice because contact with the object increases as the game goes on. After each round, the therapist processes the exposure and items ranked higher on the OCD map are introduced.

Two Cases*

Bon Voyage to Private Butt Nugget

Derek was a 16 year old Euro-American male who presented with multiple OCD symptoms. He repeatedly laundered his clothes, vacuumed his house multiple times (e.g. up

* These cases are confabulated examples reflecting a combination of various patients' prototypical symptoms. All patient information is redacted, sanitized, and de-identified.

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to five time daily), and assiduously dusted his house for fear that the dust mites contained serious diseases such as E-coli, HIV, and cancer. He was particularly worried about being contaminated by bodily fluids especially urine which generalized to anything colored yellow or gold. He sought reassurance from his parents that he was contamination free (e.g. "Did I wash enough? Is this a pee stain?"). Derek was a star student but was also plagued by debilitating perfectionism which resulted in serious procrastination. Most recently during his last grading period, his procrastination cost him his first C and D in his academic career. He admitted to erasing through papers and re-reading materials up to "the magic number three." He disclosed, "Perfection is my life goal. Doing anything less than perfectly is not worthwhile and is a fricking waste of time."

The CY-BOCS revealed excessive and ritualized cleaning, checking that things are clean, excessive concern about symmetry and exactness, concerns about germs/illnesses, worries about bodily secretions, concerns about becoming ill, and spreading contaminants.

Psychological Formulation

Derek was plagued by numerous obsessions warning him about potential contaminations. He believed that unless he cleansed himself of potential diseases carrying substances (e.g. urine, dust mites, food stains, perspiration stains), he risked serious infection (e.g. E-coli, HIV, etc). His cleansing compulsions were maintained and exacerbated by negative reinforcement due to anxiety relief.

Derek's perfectionism may be understood as a function of his overly valued notions of control and responsibility. Turning in an imperfect assignment branded him as an "irresponsible, careless, not serious" student. In sum, Derek held a set of beliefs reflecting the sense "I must be constantly on guard against serious illness and doing irresponsible things because I am so vulnerable to getting sick in a world that is unpredictable, filled with danger, and out of order where other people carry disease, are reckless and careless."

Modular CBT

Psychoeducation

Psychoeducation with Derek included reading the autobiography written by a young patient diagnosed with OCD (Kant et al., 2008). Derek was invited to read and highlight personally salient parts of chapters. Both he and his parents were given verbal and written descriptions of CBT. Additionally, the therapist showed Derek a miniature model of the human brain to illustrate the parts of the brain affected by OCD.

Self-Monitoring

Derek particularly enjoyed naming the OCD. He readily referred to OCD as a "Butt Nugget" but after some reflection added the rank of "Private." He ranked the OCD as private

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for two compelling reasons. First, he chose private because he thought it was a low rank and disempowered OCD. Second, "private" worked because he wanted to keep his symptoms on the "down low."

Mapping Private Butt Nugget was somewhat more difficult. Due to his fears of negative evaluation and embarrassment, Derek was initially reluctant to disclose specific information about his OCD. He was ambivalent about other people knowing the particulars about his OCD and believed that disclosure could result in loss of control. However, the therapist worked with Derek to test out his predictions about loss of control. Subsequently, Derek became increasingly fluent with recording his thoughts, feelings, behaviors, interpersonal patterns, and physiological reactions. He reported beliefs such as, "If I don't wash, I'll die," "I am going to get a gross disease, suffer, and die," "If I get sick, it's because I wasn't careful enough," "The world is filled with horrible germs and deadly diseases," "I must always be perfect and in control," "All mistakes are avoidance," and "Any mistake could be fatal."

Basic Behavioral Tasks

The therapist elected to intervene via the cognitive restructuring module since contingency contracting and relaxation training were not indicated.

Cognitive Restructuring

The Cognitive Restructuring began by examining and Socratically processing Private Butt Nugget's orders. Derek identified several obsessional rules including " I am vulnerable to all germs," "Germs are bad", and "Germs carry bad things like plagues and if I let them get me, it means I am irresponsible."

The therapist designed a cognitive restructuring game called "Catch the germ." In Catch the Germ," the therapist initially presented Derek with a crash course in germ theory (Friedberg et al., 2009). Simply, Derek learned there are germs that will make you sick, some will make you stronger, and others are neutral. Additionally, while germs are omnipresent, contamination is an active process; germs have to be contracted. Next, Derek was given slips of paper and instructed to draw a happy face symbolizing good germs, a sad face representing the bad germs, and leave a third set blank signifying the neutral germs. Once the slips of paper were finished, the therapist brought out a Styrofoam cup labeled "Derek." The therapist told Derek that there were going to play a game where Derek would take the folded slips of paper and try to toss them into the cup placed about two feet away from him.

After the game, they would count the number and types of germs "caught" by Derek. Derek played the game and caught several germs. Two good, two bad and one neutral germ. Derek and the therapist processed these results. Following a Socratic dialogue, Derek was able to conclude "There are germs all around me but some are good and it takes a lot for a germ to get me sick."

Derek also completed a decisional balance analysis on his belief that "doing anything less than perfectly is a fricking waste of time." Essentially, the decisional balance processes involved examining the costs and benefits of holding onto this belief.

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Exposure

Graduated exposure was implemented with Derek. Derek's first set of exposures was to clothes with suspicious yellow or gold stain on them. The therapist worked with Derek's mother to co-engineer some of these experiments. For instance, his mother spilled some yellow coloring on his shirt. Derek held the shirt for a prolonged time (approximately 20 minutes), tolerated the anxiety, and washed his hands once. He then went home, washed the shirt, only once in the washing machine. However, Derek insisted on washing the shirt separately so it could not contaminate the clothes. Accordingly, for the next step on graduated hierarchy, Derek's mother repeated the staining process and Derek was challenged to wash the contaminated shirt with other clothes.

A second set of exposures was incrementally more difficult for Derek. In these sets of exposure, Derek faced down his fear of dust and dirt. Derek successively confronted dirt collection bins from the vacuum cleaner. Initially, they were placed two feet from him then they were placed gradually closer until he touched dirt/dust both tolerating increasing anxiety levels without escape or compulsive cleaning.

The third set of exposures focused on Derek's procrastination. The first graduated task to test Derek's perfection involved sending an email to his teachers explaining his late assignments. Derek was encouraged to make 2 errors in the email and then form predictions about what would happen after he sent the message to his teacher. He predicted that the teachers would respond critically and punishing. Of course, Derek's prediction was disconfirmed. When the teachers actually responded in an understanding way without even mentioning the errors, Derek's obsessional rules were disconfirmed.

The subsequent experiments were more challenging for Derek. Derek completed an assignment in session and omitted a minor detail (e.g. date). Next, Derek completed an assignment in session, agreed to proof read it once, and then mail the work to his teacher. Finally, Derek completed the task of proofing his assignment once and personally handing his work to his teacher.

Farewell to Professor Just Right

Samantha was a 10 year old female child who described herself as a "Neat freak" and an "Order Geek." She proudly stated, "I like things just so, Dr. Bob." Samantha presented with OCD symptoms such as fears of loss of control, need to have everything in a proper place, spending up to two hours changing her attire until the clothes fit perfectly and matched, taking up to 45 minutes to fix her hair so there was no hair out of place and her hair band was exactly placed on her head. Her room needed to be in the "proper" order with lip glosses arranged by date of purchase, her stuffed animals lined up exactly from tallest to smallest, and her clothes hung precisely hung in correct sections (pants, dresses, shirts) and organized by color. She explained that she placed the family DVD's in alphabetical order and arranged book on the bookshelves in alphabetical order by author's name. Interestingly, during the course of the initial clinical interview, she volunteered to arrange the books on my (RDF) bookshelf and straighten a wall hanging that had become askew.

On the CY-BOCS, areas of concern included fears of doing something embarrassing, worries about being responsible for something terrible happening, fears of losing things, need for symmetry, excessive list making, and repetitive checking. Samantha was also influenced by lucky colors (e.g. Pink is my special color. I always wear pink especially if there is

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something scary or difficult to do. It's good because I am a girl so nobody really knows about my pink habit.").

Formulation

Samantha's OCD rules dictated that she keep everything in order. Whenever she perceived disorder, her straightening and arranging neutralized her anxiety. Accordingly, her compulsions were negatively reinforced by anxiety relief. Doubt, uncertainty, and unpredictability were dangerous for Samantha. Order and symmetry ensured her sense of safety. Similar to Derek, Samantha held a set of beliefs characterized by the theme, "I must be absolutely sure everything is just right and in its proper place or else I am completely responsible for something bad happening to me or people I love in a world that is out of order and threatening where people have to rely on me to keep things straight, neat, and correct."

Psychoeducation

Written materials and experiential activities characterized PE with Samantha. Chansky's (2000) wonderful parenting book, *Freeing Your Child from OCD*, was recommended to her parents. Additionally, Samantha and her therapist reviewed *Up and Down the Worry Hill* (Wagner, 2000) and *What to do when your Brain Gets Stuck* (Huebner, 2007) in order to introduce key concepts.

Self-monitoring

Samantha selected the name "Professor Just Right" for her OCD. The Just Right title aptly captured her need for precision, symmetry, organization, and exactness. She chose "Professor" because she felt like OCD was lecturing her and constantly telling her what to do. Additionally, Samantha drew a stern looking professor figure wearing a mortar board cap and tassel.

Samantha captured several obsessional beliefs dispensed by Professor Just Right including "Messes mean you are out of control," "Control means you are safe," "If you keep things in order, nothing bad will happen," "You must control everything," "You must always be responsible," and "Being out of control means you are irresponsible."

Basic Behavioral Tasks

Parent management techniques were taught to Samantha so they could establish contingency contracts. Her parents were familiarized with ways to specifically set goals and deliver rewards. Samantha was oriented to the contingency contract and became aware of the salient contingencies.

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Cognitive Restructuring

Samantha's therapist elected to use "Pop up Monkeys (Friedberg et al., 2009) as a way to attenuate her overly valued notion of control. Pop-up Monkeys refer to small toys usually monkeys but they come in other animal shapes as well. They are plastic figures atop a spring and a rubber suction cup. Pop ups may be purchased at party stores or on-line.

The exercise involves teaching the child that the goal of the game is to have the pop-ups pushed down simultaneously. The child takes one toy at a time and pushes down the spring. After the first toy is down, she may continue to the second, and if two are remaining pushed down, she proceeds to the third pop up. If anyone springs up, she must start all over the beginning. After several trials, Samantha constructed several coping/self-instructional statements (e.g. "Sometimes I can try too hard to control things I am not in charge of. I can let things be sometimes," "Pushing too hard to control things I can't makes me feel worse," etc.).

Samantha and her therapist played the Chinese Finger Trap (Friedberg et al., 2009). A Chinese Finger Trap is a concrete, experiential procedure that illustrates the letting go metaphors used in ACT (Hayes et al., 1999; Heffner et al., 2002). The finger trap is a tube shaped toy made of woven straw. Players insert their fingers into the openings at each end of the straw tube. When fingers are inserted and then pulled outward, the tube tightens trapping the player. The trick to escape is to push in surrendering control to the trap which loosens its grip on the fingers.

Surrendering control for Samantha was difficult but when she learned that "letting go" is an active coping strategy rather passive resignation, she became more willing to experiment with less control.

Exposures

Graduated exposures and experiments with Samantha centered on four categories. The exposures concentrated on messes, disorder, imperfect appearance, and magical thinking about the color pink. Messes were the first target. In session, a hierarchically arranged series of exposures were established. One moderately ranked therapeutic adventure involved Samantha dumping a case of small beads on the floor and allowing them to scatter. Samantha tolerated the distress of seeing the mess without immediately searching for every piece. She waited for 25 minutes and then picked up the pieces without checking multiple times that she picked up all loose beads. Finger painting on a table was a higher ranked "mess." Samantha was able to tolerate covering her hands and table with globs of paint.

Order and arranging was next on the exposure agenda. As previously mentioned, Samantha assiduously categorized her toys and belongings. A moderate anxiety producing task for her to try in session involved bringing her Little Pet Shop and Polly Pocket toys to therapy. She stored her toys in clear plastic boxes that were precisely labeled by their contents. Her challenge was to play with the toys, put a Pet Shop toy in the Polly Pocket box, then store a Polly Pocket in the Pet Shop container, and leave them "disorganized." Homework assignments included putting her lip glosses and clothes in a different order. Her mother and father were coached to reward Samantha for her disorganization.

Completing experiments on Samantha's need to match her clothes and perfectly place her hair bands was more difficult. Typically, experiments were initiated in session where

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Samantha put on a "fashion show" where she modeled hair bands that were askew and outfits that were mismatched. The therapist and her family members were the audience members in the fashion show.

Decreasing Samantha's superstitious reliance on pink as a special color was the final step. A set of experiments were crafted where Samantha wore less pink attire in progressively more stressful situations. For example, early exposures involved Samantha wearing small amounts of pink attire (e.g. socks, bows) on less stressful days (e.g. half-days, no tests or assignments), and rating how well the day went. As time progressed, Samantha tested out her new action plan by wearing small pink accessories or completely different colors on days that were more anxiety producing (e.g. exam days) and noting the outcome. Her parents augmented the exposure with contingent rewards for approach behavior.

Conclusion

Pediatric OCD is a prevalent yet highly treatable disorder. Bringing bench/lab science to the direct care of young patients diagnosed with OCD propels better treatment. The recent advances in both the psychosocial and neurobiological understanding foster more comprehensive conceptualization and intervention. Modular CBT offers practitioners flexible yet theoretically sound ways to ameliorate these young patients' lives. Ideally, this chapter provides foundational knowledge that leads to improved patient care.

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Chapter IV

Cognitive Techniques for OCD Treatment

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Abstract

The exposure and response (ritual) prevention (ERP) was the first treatment revealed to be a truly effective therapy for the treatment of Obsessive Compulsive Disorder (OCD) symptoms, considered as one of the first-line treatments for such disorder back in the seventies. However, a significant percentage, circa 30% of the patients, didn't show any improvement. Many of them were presented with difficulties in joining the activities, and dropouts were very common. Misjudgments and dysfunctional beliefs that were observed almost as a rule in OCD patients not only could explain such difficulties, but have an important role in the appearance and maintenance of symptoms. Correcting them, therefore, could improve the treatment effectivity. Whether the addition of cognitive techniques indeed improves the results of the ERP therapy or not is still an open question, though the clinical impression is that they can be very useful and even relevant for the treatment of certain groups of OCD patients. It is believed that the cognitive approach, for those patients, clears the ground and makes the application of behavioral techniques possible. Though there could be some controversies on the theoretical field between the behavioral model, that proposes the exposition and response prevention, and the cognitive model, that proposes the correction of dysfunctional beliefs for the treatment of obsessive-compulsive symptoms (OC), in the clinical practice, the cognitive therapy is no longer seen as a competing model to the ERP therapy, and gradually cognitive techniques have been incorporated to behavioral techniques that are still thought to be the fundamental strategies for the treatment of OC symptoms. And indeed the designation of Cognitive-Behavioral Therapy (CBT) has been used more frequently. In the present

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chapter, we will describe the most used cognitive techniques for OCD disorder. We will also emphasize the patient's psychoeducation about the cognitive model of the disorder, the role of the thoughts and dysfunctional beliefs in the origin and maintenance of symptoms, their capacity to identify and record them and the use of cognitive techniques to correct them, with the aim to make the treatment more effective. We will mention situations in which the techniques could be useful and therefore referred, as well as situations in which those techniques could be either counterproductive or not useful.

Introduction

Although obsessive-compulsive disorder (OCD) is an important disease, little is still known about its etiology. Recent theories support an integrative hypothesis, where neurobiological (including genetic), psychological and environmental factors would be involved in the genesis of the disorder. Among the psychological factors, cognitive and behavioral theories try to explain the origin and maintenance of this disorder. [1]

Many researchers have been working on the development of cognitive models for OCD. [2, 3] In spite of being presented with different characteristics, there is a consensus regarding components that are essential to the cognitive model: (i) normality of intrusions: intrusive normal and pathological thoughts are part of a continuum; (ii) distortion when appraising intrusions: the core problem is the misappraisal of intrusive unwanted thoughts or obsessions; (iii) neutralization and avoidance: neutralization strategies increase the intensity of the obsessions and avoidances reinforce the misinterpretations of intrusive thoughts; (iv) dysfunctional beliefs: the tendency to make misappraisals of the obsessions derives from underlying dysfunctional beliefs, involving content of risk, danger, responsibility, uncertainty, need of control, perfectionism, etc. [4]

One hypothesis raised in the cognitive theory about the etiology for OCD is that obsessions are a variation of intrusive thoughts common to all individuals and are considered a natural phenomenon in human beings. What differentiates an obsession from a "normal" intrusive thought is the fact that the former seems to be more intense, insistent, time-consuming, and painful, while the later tends to be passing and with less emotional intensity. [5-7] By using the cognitive model to explain the maintenance of anxious symptoms, the unrealistic risk evaluation was highlighted as common in anxiety disorders carriers, when individuals appraise situations in an exaggerated manner in respect to the probability of its consequences to be negative. [8] Authors suggested that misappraisals could be critical to OCD, highlighting as relevant cognitions the dysfunctional perfectionism, exaggerated responsibility, and exaggerated risk evaluation. [9]

The cognitive-behavioral models have some points in common, as the following: a) emotions result from the appraisal of events, including the appraisal of thoughts; b) through the learning process, cognitive processes and structures (beliefs, schema etc.) pre-exist, affecting appraisals made by the individual; c) appraisals and emotional responses tend to interact reciprocally; d) consequences of the individual's behavior affect the appraisals and vice-versa. [10] The psycho-social approaches have been used in OCD with the aim to increase the improvement of the obsessive-compulsive (OC) symptoms. Among the psychotherapies, the behavioral therapy (BT) was introduced in a more systematic way in the seventies. BT, based on the learning theory, assumes that rituals or avoidances are strategies

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the individual "learns" to use to relief the subjective discomfort associated to the obsessions and, when being successful in having relieve, starts repeating such rituals while neutralizing, thus, temporarily their suffering and fears.

The therapy proposes a reversal of this condition through the use of two behavioral techniques combined which have been proved effective: the Exposure and Response (Ritual) Prevention (ERP). In this behavioral approach, the patient is encouraged to expose him or herself to the feared situations or objects and to refrain to perform the rituals that cause him relieve.

Although initially the discomfort is increased, if the individual tolerates for a sufficient time of exposure, the anxiety disappears spontaneously. Such phenomenon is called habituation: the gradual relief of the anxiety, with the exercises of exposure to the feared situations being repeated, until the anxiety fully disappears. [11]

The ERP therapy was tested in several clinical trials in the population proving to be effective in samples of adult patients, with success rates ranging from 75% to 85%, [12-15] especially when combined with pharmacotherapy. [16-18] The ERP therapy was considered as the first treatment proved to be a truly effective therapy for the treatment of OCD symptoms and one of the first-line treatments for such disorder back in the seventies.

However, 20%-30% of the patients do not adhere to the exercises or leave the therapy. [12, 19, 20] Misappraisals and dysfunctional beliefs, observed in OCD patients almost as a rule, could not only account for these difficulties as have an essential role in the emergence and maintenance of the symptoms. Therefore, correcting them could improve the treatment effectiveness.

Patients with predominance of obsessions or very strict convictions about their beliefs were little accessible to the ERP therapy, leading some authors to propose cognitive techniques associated or not to the ERP therapy. [3, 10, 21, 22]

The cognitive therapy have been proved effective for both patients with predominance of obsessions, considered refractory to the ERP therapy, [22] and patients with obsessions and compulsions. [23, 24]

Whether the addition of cognitive techniques indeed improves the results of the ERP therapy or not is still an open question, though the clinical impression is that they can be very useful and even relevant for the treatment of certain groups of OCD patients as the ones mentioned above. It is believed that the cognitive approach, for those patients, makes the application of behavioral techniques possible.

Though there could be some controversies on the theoretical field between the behavioral model, with the exposition and response prevention, and the cognitive model, with the correction of dysfunctional beliefs, for the treatment of obsessive-compulsive (OC) symptoms, in the clinical practice, the cognitive therapy is no longer seen as a competing model to the ERP therapy, and gradually cognitive techniques have been incorporated to behavioral techniques that are still thought to be the fundamental strategies for the treatment of OC symptoms. Indeed the designation of Cognitive-Behavioral Therapy (CBT) has been used more frequently.

In the present chapter, we will describe the most used cognitive techniques for OCD disorder. We will emphasize the patient’s psychoeducation about the cognitive model for the disorder, the role of the thoughts and dysfunctional beliefs in the origin and maintenance of symptoms, their capacity to identify and record them and the use of cognitive techniques to correct them, with the aim to make the treatment more effective. We will also mention

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situations in which these techniques could be useful, as well as situations in which those techniques could be either counterproductive.

The Cognitive Therapy for OCD Treatment

When researchers started to observe thoughts and dysfunctional beliefs as important factors in the origin and maintenance of OC symptoms, some of them tried to test the cognitive therapy as an alternative to the ERP therapy. To that end, clinical trials have been carried out using the emotional rational therapy (ERT) by Ellis [25] and the cognitive therapy based on the model proposed by Beck, [26] having initial favorable results for OCD patients.

The first study with 21 patients compared the efficiency of the ERT with ERP therapy having similar results. [27] On the other hand, a controlled clinical trial with 71 patients who attended 16 therapy sessions compared the ERP therapy with the cognitive therapy (Beck's model), where a higher number of patients treated with cognitive therapy were considered "recovered." [23] A randomized study with 75 patients identified similar efficiency of cognitive therapy compared to the behavioral therapy in reducing OC symptoms. [24] Two additional studies compared the efficiency of cognitive therapy with behavioral therapy and with fluvoxamine, with the results being similar both for short and long-term treatments. [28, 29] Even with these favorable results, some hypotheses were raised.

It was questioned whether the clinical trials with cognitive therapy used cognitive interventions only, or even indirectly, could be using ERP when proposing, for example, behavioral experiments to the patients.[30] It is also questioned if the changes in thought (obsessions) would be responsible for the changes in behavior (compulsions) or if the behavioral changes determine the changes in beliefs.

A study identified that the change in compulsions, more than in obsessions, was a better predictor of all treatment effects, suggesting that the change in compulsions is the critical factor to the symptoms improvement both in behavioral and cognitive therapy for OCD. [31]

In order to facilitate the progress of the therapy sessions, the introduction of cognitive techniques follows these steps: (i) psychoeducation about the cognitive model for OCD; (ii) training in identifying and recording obsessive thoughts and distorted appraisals; (iii) use of cognitive techniques and integration with ERP exercises. Before we discuss a little bit more about these steps, we will start with the relevance of the initial interview.

The Relevance of the Initial Interview

The objective of the initial interview is to motivate the patient to appraise his/her ability to tolerate anxiety and make sure about his/her effective adherence to the treatment, since it is common for patients to have questions doubts or feel ambivalent. Therefore it is critical to solve these issues before the therapy begins, minimizing the risk of dropouts and withdrawals. It is also important to assess the patient's available time to attend these sessions, do the exercises and, in case they need to pay for the treatment, afford these costs.

The first interview is a very important moment, because the first contact is critical to establish a bond with the therapist and strengthen the work alliance. Even this initial occasion

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strongly depending on the patient's personal aspects, the establishment of a bond also depends on the therapist's abilities of empathy, kindness, genuine interest in helping, and connecting to other people.

It is imperative the patient to feel he/she can trust on the therapist's professional competence, so that he/she can later accept an invitation that will probably imply an immediate increase of his/her anguish and distressing. [11]

Psychoeducation

Psychoeducation tries to motivate the patient for the treatment and the first step is to provide him/her basic information on OCD: what it is, what obsessions, compulsions and avoidances are, what their possible causes, course and prognostic are; if it interferes with work, interpersonal relationships, and family members, available treatments, and what the grounds of the cognitive therapy are.

Teaching the patient about the importance of becoming acquainted with the cognitive model for OCD must include an introduction to the concept of "misappraisals" of obsessions. In this phase, the therapist must introduce the expression "appraisal of thoughts" or "giving importance to the obsession," to explain why obsessions persist and to help the patient to realize that these appraisals are wrong. However, the patient should not be convinced that his/her appraisals are wrong, but to use the guided discovery to test the realistic or distorted basis of his/her appraisals. [4] This is only possible if the patient accepts that the cause of the highly persistent and distressing obsessions is his/her interpretation of something as a significant threat. OCD patients must adopt the view that compulsive rituals and neutralization strategies are useless attempts that only intensify the obsessions over time. [32]

It must be explained how the therapy sessions work: schedules, symptoms monitoring, revision of weekly tasks, records, ERP exercises and correction of thoughts and dysfunctional beliefs, during the sessions and as homework; execution of homework tasks in between sessions, time required to perform the exercises, records, therapy duration, etc.

Such aspects must be informed to the patient and, occasionally, with his/her relatives, aiming to motivate him/her for the treatment, since many patients entered the therapy with deeply rooted beliefs that their OCD is a disease deriving from a "chemical imbalance" or a "genetic defect." Others still struggle to distinguish a "normal" behavior related to health and personal hygiene, how we should care for our personal safety, from the excessive rituals and avoidances of OCD. [11] The therapist must create positive expectations of change, because patients with associated depression, for example, are often skeptical about the possibility of changes due to the disease duration, previous unsuccessful treatments, or have beliefs which are heavily intense and crystallized about the obsessions, or yet struggle to believe in the possibility of changing behavior patterns. Negative expectations regarding their own ability to tolerate the increase of anguish deriving from ERP exercises must be identified and addressed. Initially the therapist will be able to address results from research or other treatments, answer questions that no longer will exist when the patient performs the activities and is successful, once he/she discovers that can put up with the associated discomfort. It is important to highlight being successful depends essentially on adhering to the exercises: frequency and time devoted to them. All items mentioned during the educational phase da

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CBT can take much longer than one or two sessions as described in treatment handbooks. To have a better explanation regarding the issues cited so far, we will start by discussing the intrusive thoughts and their potential distorted appraisals. [11]

Cognitive Model for OCD

For the patient to become acquainted with the cognitive model for OCD, the clinician must help the patient to identify intrusive thoughts activated in different situations (objects, places, and people) and expressed as automatic catastrophic thoughts and misappraisals and misinterpretations inherent to these intrusions (obsessions).

It is well known that sensitive individuals, when facing normal intrusive thoughts, would interpret in a catastrophic manner the thought content as threatening, dangerous, relevant (I can get contaminated, someone will die).

Another negative interpretation is related to the occurrence of intrusive thoughts (having these thoughts means that I can be a homicide, a child abuser). These interpretations generate anxiety, fear, discomfort, and the individual feels compelled to do something to remove the threat (for example: performing rituals, avoidance behavior, and removing unwanted thoughts). This at first provides a temporary relief, but eventually causes the perpetuation of OCD and maladaptive dysfunctional beliefs (risk and responsibility exaggeration, giving magical power to thoughts). [5, 21, 33-36]

The negative interpretation of an occurrence of a thought or its meaning can be easily identified in patients with unacceptable thoughts of inappropriate violent, profane, or sexual content (“If I have this thought that means I’m a bad person...”), and patients with obsessive doubts, need to be sure (“If I’m not totally sure about this, it means that...”) and to check, with excess of responsibility and perfectionists (“If I make a mistake I can be... or it means that...”). Therefore, correcting such misappraisals and misinterpretations would have a significant role in reducing these patients’ anxiety and hence their OC symptoms. [11]

In the cognitive model, compulsive rituals, avoidance behavior, hypervigilance, and neutralizations are considered as consequences of obsessive thoughts, which are behaviors adopted by the patient with the aim to remove intrusions and perceived risks. [37] These behaviors are responsible for evidence and persistence of obsessive thoughts for two reasons: 1) they are reinforced by the immediate relief after the discomfort, temporary removal of the obsession and the sensation that the threat was withdrawn; and 2) they refrain the individual from learning that his/her appraisals are not realistic (for example, having an unwanted thought of violent content does not indicate the risk to commit it). [4, 10, 21]

Identifying and Recording Thoughts and Dysfunctional Beliefs

The way the patient will internalize the cognitive model is through the identification of these cognitions and their consequences, so that he/she could change them while putting into practice the relationship between thoughts, emotions, and behaviors. The patient will receive

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training in identifying thoughts (of catastrophic nature) that come to his/her mind when gets in touch with certain objects, places, people or situations that activate his/her obsessions.

In order to do this exercise, the therapist will ask the patient to remember an occasion in which he/she felt intensive fear and thereby felt compelled to perform a ritual or avoid contact (not touching some object, trying to remove him/herself from the scene, having to wash his/her hands, being taken by a doubt, checking), asking himself/herself the following questions:

1 What was the activating situation (object, place, and people)?
2 What came through your mind in that moment (what was the thought)? Or what do you imagine you could have thought, in case you don't remember?
3 What did you feel at that moment (fear, anxiety, physical discomfort, disgust)?
4 What did you feel compelled to do (or not to do) to relief anxiety or fear (ritual, repeating some acts to be sure, avoiding contact, performing a mental ritual, trying to remove the thought, asking other people, etc.)?

The therapist must encourage the patient to write down situations/thoughts into the Dysfunctional Thought Record (DTR) having three or more records during the week and ask him/her to bring them in the following session (see Table 1). These exercises can be performed some times in the clinic until the patient masters the exercise before doing part of the homework. Some authors use daily self-monitoring exercises, to help the patients to identify increasingly better the external (places, situations) and internal (thoughts and dysfunctional beliefs, fears, anxiety) factors, avoidance behavior, seeking for reassurance, that indeed reinforce the misappraisal processes. [38] Some patients struggle to identify their intrusive thoughts and the interpretations they lead to.

Table 1. Dysfunctional Thought Record Form

Date	a. Activating situation	b. Automatic thoughts	c. Consequences	
			What I felt	What I did
22/1	When leaving my house	My house can catch on fire	Fear, anguish	Checked all the sockets and the gas cylinder
23/1	When handing over a work project	It must have many mistakes	Doubt, anguish	Reread and rewrote the project several times

It is easier to identify symptoms related to contamination, obsessive doubts, checking, and unwanted thoughts, and more difficult to be identified in hoarders, aligners or patients with compulsions related to disgust. [11] To help the patient in identifying his/her thoughts and beliefs, we will show an example of DTR and then discuss a little bit about the relevance of the cognitive restructuring.

Guidelines for Filling the DTR

Identify date, schedule, place or situation where/when your mind was invaded by an obsession or when you felt compelled to perform a ritual. Answer the following questions, writing down your answers on the form.

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A Activating situation: in what situation, place or moment my mind was invaded by some unpleasant, inappropriate fear, worry, impulse, doubt, thought or image? Write down the situation on the table.
B Automatic thoughts: what went through my mind at that moment? Answers to this question are your automatic thoughts.
C Consequences: What did you feel at that moment (fear, anguish, guilt, disgust)? What were you lead to do to get rid of the discomfort or fear (ritual, avoidance, mental ritual)? If you had given in to the impulse to perform a ritual or avoid getting in touch, what did you feel after doing it?

Cognitive Restructuring

CBT is a collaborative process characterized as a guided discovery, where the patient with the help of the therapist is oriented to cognitively challenge his/her misappraisals and dysfunctional beliefs which shapes his/her obsessions to accept an alternative, more realistic explanation for these obsessions, hence helping the patient to learn to deal with them in a more adaptive way. [11] It is important the patient to understand that trying to suppress or remove a thought increases even more its frequency and intensity, as well as watching objects, places or people increases the frequency of the obsessions.

Many authors [3, 4, 22, 35, 39-43] proposed cognitive strategies to correct them and to be used in association with ERP therapy. Now, we are going to explain each technique used in the treatment for OCD.

Cognitive Techniques Used in OCD Treatment

The cognitive techniques proposed for the treatment of OCD symptoms are, in general, adapted from the ones initially described by Aaron Beck, for the treatment of depression, and by David Clark, for the treatment of anxiety disorders. [26, 44] They were suggested and adopted by many authors. [3, 35, 39, 41] We selected the techniques we believe to be the most useful and easiest to use in the therapeutic approach of obsessions and compulsions:

- Socratic Questioning
- The two-alternative technique
- Responsibility pizza technique
- Descending arrow
- Behavioral experiments or tests
- Correction of excessive risk evaluation
- Calculation of advantages and disadvantages
- Reminders

Using cognitive techniques is an ability only developed with practice. They demand the therapist to help the patient develop the habit to take a little look at his/her own mind, to

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realize and write down his/her automatic thoughts and beliefs so that he/she can later do the exercises.

The aim is to teach the patient to use them automatically whenever he/she realizes that his/her mind was invaded by an obsession or when feels compelled to perform a ritual. It is expected the therapist to do some practical exercises in order to certify that the patient has developed such abilities and is able to use the cognitive techniques by him/herself. The therapist must suggest the more appropriate techniques for each symptom or difficulty the patient presents. [11] We will examine each technique, with practical examples of how they must be used. Let's start with the Socratic Questioning.

Socratic Questioning

Probably the Socratic Questioning is the most important and the most applied cognitive technique in the treatment of OCD symptoms. Its name refers to the Greek philosopher Socrates, known for questioning concepts and beliefs of his fellow countrymen.

The objective of the Socratic Questioning is to make the therapist help the patient to replace his/her non rational way of thinking for a logical reasoning, to develop a scientific attitude while casting doubts over his/her convictions and by trying to confirm them in order to finally decide if he/she should keep or reject them, mainly when these are contrary to what is observed in daily life, as the behavior of other people.

When doing an exercise, the patient will be asked to face his/her convictions (manifested through automatic thoughts) not as solid true, but as mere hypotheses, which will be maintained in case he/she has evidence in favor or rejected in case evidence proves to be wrong, having the patient realize that OCD leads him/her to systematic misappraisals of risk and responsibility. Once the patient realizes this condition, he/she must be aware of these possibilities and ask him/herself whether his/her fears are indeed consistently evident or totally ungrounded. [11]

How to Do the Socratic Questioning

Following the identification of a catastrophic thought on the patient's DTR or during a session (through a report), the therapist must ask the patient to point out evidence in favor and against he/she is stating (his/her beliefs) and to see if there isn't the possibility of alternative explanations. [11]

The exercise can be done while the following situation is given as an example: a patient, before going to sleep, would not let her slippers misaligned and much less turned upside down, because she had the automatic thought or belief that, if she let her slippers misaligned, her mother could get sick or die and the guilt would be all mine.

The therapist can help the patient to make the questioning as follows:

- What evidence do I have that the thought that went through my mind at that moment or that the fears I felt are grounded? Is there any proof (or evidence) that slippers turned upside down can cause disasters?
- What pieces of evidence are contrary to what I thought? (In many occasions, the slippers remained upside down and yet no one got sick.)

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- Are there alternative explanations? (That I have such fears because I have OCD, for example, and not because slippers turned upside down indeed cause disasters.) Write at least one of them.
- Are my fears based in some real proof or do they occur because I suffer from OCD? What is more likely to be true?
- What is my imagination and what indeed will happen, in case I don't perform the ritual? (Test it: let the slippers turned upside down at night and see if any disaster happens.)
- What proofs do I have that really what I imagined at that moment is really going to happen?
- What would people who don't have OCD say about such fears and how would they behave in identical situations? How would most of the individuals behave in similar situations? (Does anyone get scared when notice a slipper upside down or most of them don't care?)
- What is the misbelief behind this ritual? (Risk or responsibility exaggeration or power of thoughts; magical thinking.)

Search for an example representing the patient's reality. Remind the patient that whenever he/she does Socratic Questioning, the answers must be written down for her to read them in other instances or discuss them with his/her therapist.

Two-Theory or Two-Alternative Technique (A and B)

This technique aims to test an alternative explanation that is at the same time logical, believable, and coherent with the facts, that is, a simple way to question thoughts and distorted beliefs or change his/her misinterpretation, with this being called as a 'two-theory technique.' [41]

The exercise suggests the patient to choose between two alternative explanations and its main objective is to improve patient's insight about his/her symptoms, distinguishing more clearly the "normal," adaptive, and healthy behaviors from the excessive and maladaptive ones.

Practical Exercise of the Two-Theory or Two-Alternative Technique

We use as an example for this exercise a patient, female, 32 years-old, who works with general and cleaning services in a store. Main symptoms: obsession for cleaning, fear of being contaminated; she performs compulsions through intense hand washing. From this example, the therapist can help the patient by presenting two alternative theories to explain what is happening to him/her:

- Theory A: you are, indeed, contaminated and need to wash yourself, because you can contaminate your family and be responsible for diseases and, maybe, for the death of some relatives.

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– Theory B: you are a person very sensitive to the fear of getting contaminated and react to this fear in a way that compromises your life with frequent and excessive cleaning.

Questioning

- Which of these two-alternative techniques is more likely to be true: A or B?
- Have you ever tried to deal with this problem according to the second hypothesis, that is, as if it were a problem about excessive worry or fear, or an OCD symptom, and not a real possibility? [41]

This very exercise can also be used to help the patient to improve his/her insight regarding many other symptoms, as it is in the case of obsessions of inappropriate (aggressive, sexual, profane) content, in which thinking about something is mistaken by committing something (fusion of thought and action), or when excessive rituals (checking, cleaning, alignment, hoarding, etc.) are performed. [11]

Technique of Reassigning Responsibilities or Responsibility Pizza

This exercise can be useful mainly for correcting distorted beliefs involving excess of responsibility, which are common in patients with obsessions of doubts, followed by checking rituals. The technique is named as so due to its rounded shape, where each slice represents the percentage of responsibility assigned to the patient him/herself and other factors. In the clinical practice, it is very common patients to assign 100% of the responsibility on themselves ("I am responsible for that…") when something bad happens. As a result, redrawing the pizza shape while assigning the percentage more likely to correspond to the responsibility for what had happened might help the patient to realize other factors that in the initial appraisal he/she didn't realize, due to his/her way of seeing him/herself as the sole responsible for misfortunes happening. [11] A drawing of a pizza is made according to the relative weight of each direct and indirect probable factor that could compete for a disaster (disease, accident, fire, burglary) to happen, including the percentage the patient assigned to him/herself. The exercise permits reassigning the responsibility in a more realistic manner. [3]

Practical Exercise of the Technique of Reassigning Responsibilities or Responsibility Pizza

A patient, female, 28 years of age, had given her mother a beautiful knife as a gift. When she gave her the present, her mother mentioned that in their culture, giving a knife as a gift would bring bad luck to the family, unless a coin was also given in exchange. As in that occasion the patient didn't have a coin to give her, the mother said that wouldn't accept the present. The patient didn't care for what her mother had said and insisted for her to accept the present. Three months later, her mother died and the patient started to feel responsible for her death. She believed the responsibility was all hers, because she gave her the knife even though she knew something bad could happen (cultural belief).

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At first, the therapist asks the patient to draw a pizza and to assign the percentage of the responsibility for what happened – the death of her mother. This patient gave herself 100% of responsibility for what had happened. Thereof, the therapist must help the patient as follows:

1 List the possible factors that could contribute for an event like that to happen: a) her mother didn't do a follow-up with a cardiologist; b) was hypertensive and didn't take the medication; c) was already in an advanced age (85 years-old); d) didn't eat well (diet high in fat and sugar); e) was a smoker; f) the patient having given her mother the knife as a gift; etc.
2 Assign equivalent percentages to the weights that, according to the patient's understanding, these different factors could have for this to happen, including in the list as the last item the percentage of the responsibility is assigned to the patient for having given her mother a knife.
3 Divide the pizza, writing in every slice one factor that, according to the patient, could have collaborated for the disaster. Set the size of the slices according to the percentage of the responsibility (weight) the patient believes each factor could have (10% of the responsibility is equal to slices of one tenth of the whole pizza; 25%, to slices of one quarter of the size; 50%, to the half, and so forth).
4 When finish this exercise, observe the percentage of the responsibility assigned to the patient and the percentage assigned to other factors involved.
5 Compare the percentage the patient assigned to herself with the weight she had previously assigned to herself (100%) (see Figure 1).

Descending Arrow

This technique permits the examination of imagined consequences both in short- and long-term periods through guided discovery. The exercise begins by identifying a catastrophic thought (associated to an OCD symptom), related to a possibility of a mistake or a disaster (accident, fire) to happen, and by explaining the consequences imagined by the patient connected through arrows.

Arrows remind intermediate beliefs and involve risks and responsibilities, causes and consequences. The therapist might help the patient by questioning: "If that happens, what does it mean to you?," "So?," "What can happen next?," "So what?," "What is the worst thing that could happen?," and so forth, until the patient describes the worst consequence possible according to him/her. [11] Misappraisals and underlying beliefs can be addressed using other techniques (correction of estimates of probabilities, evidence examination, behavioral tests, two hypotheses, etc.). [39]

Below there is an example of the technique used with a patient whose main symptom was the fear of being contaminated with diseases. The patient was swimming in the sea when he saw someone with a bleeding injury and soon thought: "He might have AIDS!"

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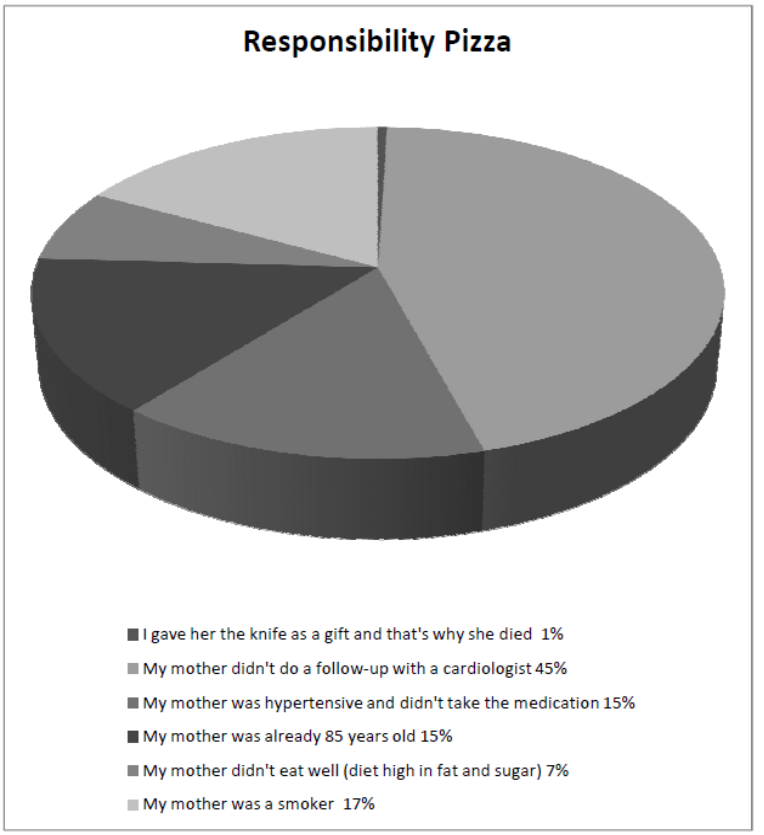


Figure 1. Responsibility pizza.

What if it is true that he has AIDS?
↓
The blood will contaminate the water
↓
So? The blood contaminates the water and the water touches my body
↓
So what? If the contaminated water touches my body, I can have a cut in my skin
↓
What if you have a cut in your skin? What do you mean? His blood will penetrate into my bloodstream
↓
What if the blood penetrates into your bloodstream? What is the worst thing that could happen? I'll get contaminated with the virus
↓
What if indeed you are contaminated with the virus? So, I'll contaminate my partner
↓
What if you contaminate your partner, what does this mean to you? That I am irresponsible, reckless and a very bad person

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In this exercise, the patient is able to notice his/her core belief about exaggerating the risk and responsibility. Even if the therapist already knows the core belief of his/her patient, it is important to do this exercise so that the patient has an insight about the distorted way he/she uses to appraise a situation.

Behavioral Experiments

Behavioral experiments are yet a way of exposure and ritual prevention in which the therapist helps the patient to behave as a scientist towards symptoms and consider his/her catastrophic thoughts or beliefs as mere hypotheses, testing evidence in favor or against them, so that in the last part of the experiment he/she can reject the ones that were not confirmed. This experiment is an effective way to test misappraisals, dysfunctional beliefs or obsessive catastrophic thoughts, and to treat superstitious symptoms where Socratic Questioning or evidence examination is not enough. [11]

Examples of behavioral experiments:

The therapist can suggest the patient some experiments:

1 Break the rule of performing a ritual in a certain number of times and see if something happens.
2 Revise only once (at most twice) e-mails, college essays, text posted on Facebook or Twitter and observe what happens (whether the consequence is tragic or not).
3 Leave a burning cigarette in the ashtray for 20 minutes and observe what happens.
4 Wear black or red (if he/she believes that it brings bad luck) or go to a wake without changing clothes when arriving there as homework and see if someone in your family gets sick or dies.
5 Leave the tap dripping for two hours and check if the kitchen got flooded.

Correction of Excessive Risk Evaluation

A very common distortion is to overestimate the probabilities that a disaster could happen. This overvaluation increases the need to perform rituals, especially repetitions and checking rituals. One way to correct this distortion when calculating probabilities is to do an exercise proposed by Van Oppen and Arntz. [3]

Practical Exercise for Excessive Risk Evaluation

The therapist must help the patient to recalculate the probabilities of a disaster to happen as follows:

1 Estimate the probability you imagine for a disaster to occur, in case you make a mistake (e.g., forgetting a burning cigarette in the ashtray, leaving the iron turned on, closing the gas cylinder, etc.) using percentages (%);
2 Set the sequence of necessary steps for a disaster to happen;
3 Calculate risks or probabilities of every step to occur;

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4 Estimate the cumulative odds (added odds). If there is 1 chance in 10, for example, two times is equal to 1 chance in 100 times (cumulative or added odd);
5 Compare the result with the degree of responsibility you have calculated before performing the exercise.

To check if the patient understands the exercise, the therapist can use the steps described earlier using the following example: what are the odds of your house catching on fire if you forget a burning cigarette in the ashtray before going to bed? Let's say that, without doing the exercise, you have calculated the probability of 10% (see Table 2).

Based on this example, we can conclude that one would have to smoke 10 million cigarettes for the chance to set the house on fire (odd calculated initially as 10%, or one in 10, versus one in 10 million, in a more realistic calculation).

In addition, the therapist can suggest the patient to talk to specialists to know if a certain ritual protects against or indeed increases the risk of accidents. For example, if the patient strongly tightens faucet handles and water meters several times, the odds that these would break or be dripping would increase.

Table 2. Adapted from Van Oppen and Arntz [3] by Cordioli [11]

Steps	Odds	Added Odds
I don't put out a cigarette	1/10	1/10
A tiny spark falls on the carpet	1/10	1/100
The carpet catches on fire	1/10	1/1.000
The carpet catches on fire and I don't realize immediately	1/100	1/100.000
It is so late when I realize the fire that I can't do anything else	1/100	1/10.000.000

Patients with fear of being contaminated could be informed about the way some diseases are transmitted or contaminated, as in the case of HIV virus, H1N1, risk factors for cancer and also what does not represent any risk (e.g., shaking hands with someone with HIV, or visiting a patient with cancer in the hospital). Consulting a specialist might help the patients to realize that their belief or catastrophic thought are distorted or ungrounded. [11]

How to Live with Uncertainty

Patients with beliefs related to uncertainty need to acquire some cognitive strategies to learn to live better with uncertainty and accept it as impossible to avoid, since the absolute sure is an attainable state in many situations. Thus it is necessary the patient to understand that: a) monitoring does not prevent catastrophic events from happening, because there is no way to eliminate human beings' vulnerability nor predict if people are subjected to misfortunes, coincidences, accidents. Even the permanent monitoring won't stop those events from happening, such as getting involved in a car accident or avoiding a hurricane to destroy your house. It is necessary to learn to live with them; b) checking rituals could increase the probabilities of disasters, because it is common for patients with checking symptoms to damage water faucets by strongly tightening them, cable of electrical devices by plugging and unplugging them, and gas meters by repeatedly closing and opening them; c) help the patient

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to live with uncertainty, because he/she must expose him/herself (in imagination) to the possibility that something bad happens (making mistakes, death of a relative, someone in the family get sick) in order to identify his/her own resources and to learn to deal with it, since the aim of the therapy is to learn to live with a possibility that they occur; d) The therapist can assume temporarily the patient's responsibility of taking decisions subject to the possibility of making mistakes (for example: seal an envelope containing a document the patient signed but didn't check; sitting aside the patient who fears running over pedestrians while driving). [11]

Technique of the Advantages and Disadvantages

This exercise helps reducing the symptoms of repetitions, obsessive ruminations and perfectionism. Two columns are drawn up. On the left column, list the advantages, and on the right one, the disadvantages of performing a ritual, for example, checking repeatedly the gas cylinder before going to bed, or if the car door is closed or not. The two lists must be appraised in order to check if the advantages outweigh the disadvantages. Example: Advantages and disadvantages of not checking the stove (see Table 3).

Table 3. Advantages and Disadvantages

ADVANTAGES	DISADVANTAGES
● I fight back against OCD ● I learn to live with the doubt ● The odds of having the need to perform the ritual are reduced ● I learn to share the responsibilities ● I feel like a winner	● Arguing with my relatives, because I go back so many times that we end up arriving late in engagements ● I lose too much time ● Checking the stove leads me to do more checking ● I still have the OCD symptom

Reminders

Reminders can help the patient to identify the intrusive thoughts and stop motor or mental rituals during the ERP tasks. The reminders can be written in a notebook or in short cards that can be carried in wallets, purses, shirt pockets or, as the patient wishes, in a poster displayed in his/her bedroom or workplace. Short sentences can be written together with the therapist, who will help the patient based on beliefs and symptoms, for example:

- No matter how hard we try, risks are part of life and there is no way to avoid them!
- Do these thoughts come through my mind because they're real or because I have OCD?
- That's OCD!
- Look at the OCD!

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Conclusion

The use of the cognitive therapy as an alternative strategy to the ERP, although effective, is not yet confirmed possibly due to some practical difficulties. Its application is more complex, since it demands a stronger basis from the therapist into the theory and the cognitive models and a more specific training.

The use of cognitive techniques also demands some conditions from the patient, such as having some ability to be introspective and identification of automatic catastrophic thoughts (before rituals): since they can be of little help for the treatment of rituals non preceded by obsessions, although being effective to treat the sensory phenomena, disgust or "non just right" sensations are common in OCD or in patients with compulsions for hoarding.

It is also difficult to use cognitive techniques based on logical reasoning, as evidence examination, when the cognitive phenomena are based on magical thinking, aspect observed in several manifestations for OCD, such as the superstitious type or even in obsessions and compulsions of contamination. [11]

Research did not prove so far, for example, if CBT is more effective than ERP therapy for treating OCD patients both in short- and long-term periods. [45-47]

It was also not yet analyzed if the adherence to the exercises is higher when the adopted approach is CBT and if it would be more effective in reducing the intensity of overvalued ideas than ERP therapy. A seemingly consistent difference is that dropout rates in CBT are smaller than the ones observed in patients who undergo ERP therapy. [47]

It is well known that psychotherapeutic treatment for OCD begins with ERP techniques, especially when symptoms are mild or moderate and compulsions predominate, the patient has a good *insight and* adheres to the exercises. Cognitive techniques are introduced, in general, in sessions 3 or 4, when ERP exercises have already started, with this being a good occasion to make the patient familiar with the cognitive model and train him/her in the use of cognitive techniques.

As an exception to this sequence in the treatment, we have patients with predominance of obsessions, many times presenting with no rituals (for example, ones who are presented with unwanted thoughts of inappropriate violent, profane, or sexual content), patients with endless doubts, mental rumination related to the need to be sure, guilt and excessive responsibility, considered as difficult to be approached with techniques behavioral only. The same happens to patients with overvalued ideas (many times with no *insight* about the nature of their symptoms), with no motivation for treatment and adherence to ERP tasks. Approaching and correcting such beliefs and convictions can represent a chance to include these patients in the therapy. In these situations, the initial sessions are used to approach beliefs and misappraisals with the aim to improve *insight* and motivation for the treatment. [11] The use of cognitive techniques may pave the way for ERP exercises, while guaranteeing a more complete treatment to OCD patients.

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Chapter V

The Dimensionality of Obsessive-Compulsive Symptoms: A Review of the Literature and an Update of Results in Clinical Settings

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Abstract

Obsessive-compulsive disorder (OCD) has a heterogeneous symptom presentation and a dimensional nature, according to current cognitive formulations. However, although numerous factorial analyses have examined the disorder's content dimensions, there is no agreement about the dimensions that characterize OCD symptoms, due in part to the measures used. Recently, different instruments have been developed that offer a promising dimensional alternative (i.e., DY-BOCS, DOCS, INPIOS, ITTIS) to previous self-reports, and alternative structural proposals. The present paper reviews existing data about OCD heterogeneity, emphasizing results provided by dimensional instruments, and it updates data about the OCD dimensional structure and severity using the Obsessional Intrusive Thoughts Inventory (INPIOS). The INPIOS was specifically designed to assess the content and frequency of 48 obsessional intrusive thoughts, as well as the unpleasantness, difficulty to control and interference of the most upsetting intrusions/ obsessions experienced. Responses of 79 OCD participants are analyzed and compared to responses of a matched non-clinical group. Although both OCD and community subjects experienced intrusions representative of all types of obsessional contents, differences

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appeared regarding the frequency, unpleasantness, difficulty to control, interference and prevalence of some dimensions.

Keywords: Obsessive-compulsive disorder, intrusions, dimensions, severity

1. Introduction

Standard diagnostic classifications (DSM-VI-TR, American Psychiatric Association [APA], 2000; ICD-10, World Health Organization [WHO], 1992) have described obsessive-compulsive disorder (OCD) as a unitary nosological entity, and its diagnostic definition is based on the presence of clinically significant obsessions or compulsions described in a general way. However, in recent decades there has been an increasing interest in considering OCD as a heterogeneous condition, and several studies have described the existence of different OCD subtypes, taking into consideration diverse criteria such as age of onset, family history of OCD, response to pharmacological treatment, neuropsychological features, or the clinical manifestations of obsessions and compulsions (e.g., Abramowitz, Franklin, Schwartz, & Furr, 2003; Harris & Dinn, 2003; Mataix-Cols, Marks, Greist, Kobak, & Baer, 2002). A significant number of theoretical and empirical studies have focused on identifying specific subtypes of OCD based on symptom themes or contents. However, most studies, congruently with the OCD cognitive model (e.g., Rachman, 1997), which considers obsessions and compulsions as the pathological end of a dimension, have adopted a different perspective when analyzing obsession and compulsion themes (e.g., Bloch, Landeros-Weisenberger, Rosario, Pittenger, & Leckman, 2008; García-Soriano, Belloch, & Morillo, 2008; McKay et al., 2004). This dimensional perspective has received support from a growing body of genetic and neurobiological data, and also from comorbidity and treatment response studies (Leckman et al., 2010). Also supporting a dimensional view of OCD, epidemiological data indicate that most OCD patients (81% according to the National Comorbidity Survey Replication) endorse symptoms in multiple content areas (Ruscio, Stein, Chiu, & Kessler, 2010). Moreover, although most patients can be categorized in a particular symptom subgroup, some patients (26% according to Matsunaga, Hayashida, Kiriike, Maebayashi, & Stein, 2010) do not fit into one main category, as they present more than one relevant obsessional content, or their symptom content is highly idiosyncratic (Matsunaga et al., 2010). This dimensional view of OCD has been recently incorporated in the last DSM version (APA, 2013).

Obsessive-compulsive disorder cognitive models describe obsessions as extreme variants of unwanted obsessional intrusive thoughts (OIT) experienced by the general population. Clinical obsessions and OITs share similar or identical contents, as supported by a vast number of studies (e.g., Belloch, Morillo, & García-Soriano, 2007; Morillo, Belloch, & García-Soriano, 2007; Purdon & Clark, 1994; Purdon & Clark, 1993; Rachman & de Silva, 1978), but they differ on the frequency, distress and uncontrollability of thoughts (Rachman & Hodgson, 1980). Therefore, when describing obsessive-compulsive (OC) symptom dimensionality, different variables should be examined, such as the frequency or time occupied, the distress, or the uncontrollability experienced when having the symptoms.

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2. Obsessive-Compulsive Symptoms Heterogeneity

As mentioned above, although some studies have analyzed OCD heterogeneity from a categorical perspective, empirically identifying subgroups of patients present in OCD samples (cluster analysis), most of the studies have used a dimensional approach by examining either the factorial structure of clinical OC manifestations (e.g., Leckman et al., 1997) or the structure of non-clinical OIT content (e.g., Lee & Kwon, 2003).

2.1. Proposals Analyzing Clinical Manifestations of Obsessions and Compulsions: Results from the Symptom Checklist of the Yale-Brown Obsessive-Compulsive Scale

Most of the studies analyzing the clinical manifestations of obsessions and compulsions have used the same instrument to explore the OCD structure: the *Symptom Checklist of the Yale-Brown Obsessive-Compulsive Scale* (Y-BOCS-SC; Goodman et al., 1989a, b), which includes the largest pool of obsessive and compulsive contents. The Y-BOCS-SC includes a checklist of over 50 obsessions and compulsions, organized, following clinical criteria, into 15 categories: eight categories of obsessions (aggressive, sexual, religious, contamination, hoarding, symmetry, somatic and miscellaneous) and seven categories of compulsions (washing, hoarding, ordering, checking, repeating, counting, and miscellaneous). From a categorical perspective, efforts have been made to classify OCD patients into distinct symptom-based subgroups in which each individual is assigned to one OC category. For example, Abramowitz et al. (2003) identified five different OC clusters labelled: harming obsessions and checking; contamination obsessions and washing; hoarding; symmetry obsessions and ordering; unacceptable thoughts and mental rituals. Calamari et al. (2004) identified 7 symptom-based clusters, including contamination, harming, hoarding, obsessionals, symmetry, certainty, and contamination/harming.

From a dimensional approach, Baer (1994) was the first to conduct an exploratory factor analysis of the 13 major symptom categories from the Y-BOCS (excluding miscellaneous obsessions and compulsions), identifying three factors: symmetry and hoarding; contamination and checking; and pure obsessions. Following a similar methodology, subsequent studies found different structures, although most of the studies (e.g., Cullen et al., 2007; Hasler, Kazuba, & Murphy, 2006; Leckman et al., 1997) reported a 4-factor structure: (1) symmetry obsessions and ordering, repeating, and counting compulsions; (2) aggression, sexual, and religious obsessions; (3) contamination obsessions and cleaning compulsions; and (4) hoarding obsessions and compulsions (see Bloch et al., 2008, for a meta-analytic review). This four-factor structure was not exactly replicated by subsequent studies that factor analyzed the Y-BOCS at the item (*versus* categorical) level, or through confirmatory analysis (e.g., Denys, de Geus, van Megen, & Westneberg, 2004; Feinstein, Fallon, Petkova, & Liebowitz, 2003; Wu, Watson, & Clark, 2007). These studies posed some questions that remain unanswered, especially those about aggressive obsessions and checking compulsions and hoarding obsessions and compulsions. On the one hand, aggressive and sexual obsessions and checking compulsions are reported as a single factor in some studies (Cavanilli, DiBella,

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Siliprandi, Malchiodi, & Bellodi, 2001; Hasler et al., 2005; Mataix-Cols et al., 2002) and as independent factors in others (aggressive and sexual obsessions *versus* checking compulsions; e.g., Denys et al. 2004; Pinto et al., 2007), while aggressive obsessions, checking and contamination compulsions constitute a dimension in still others (e.g., Feinstein et al., 2003). On the other hand, hoarding has been proposed as an independent OCD dimension in most studies (e.g., Cavanilli et al., 2002; Feinstein et al., 2003; Leckman et al., 1997), but it has also been associated with symmetry/ order (Baer, 1994) or with checking compulsions (Tek & Ulug, 2001). Furthermore, recent proposals suggest that hoarding could be both a separate syndrome and an OCD symptom, as proposed for (Mataix-Cols et al., 2010) and described in DSM-5 (APA, 2013).

Confusion about the Y-BOCS factor structure, both from categorical and dimensional approaches, could be a consequence of using only some of the rationally derived Y-BOCS-SC checklist symptom categories (Denys et al., 2004; Feinstein et al., 2003; Pinto et al., 2008), thus excluding the miscellaneous category, which includes OC symptoms related to superstitious and mental rituals, or including heterogeneous categories such as checking (e.g., Bloch et al., 2008; Summerfeldt, Richter, Antony, & Swinson, 1999). Moreover, studies differ in the strategy used to classify OCD symptoms and analyze the Y-BOCS factor structure. Whereas some studies analyze the number of symptoms endorsed by the patient within each theoretical category, others examine the presence or absence of symptoms in each theoretical category, and still others examine symptom "severity" (absent, present, or prominent) (Bloch et al., 2008). The first two methods especially do not seem to be the best way to analyze OCD symptom dimensionality, as they use a categorical response scale (i.e., presence or absence of each symptom) *versus* a dimensional scale (e.g., frequency of each symptom) (Mataix-Cols, Rosario-Campos, & Leckman, 2005; Wu et al., 2007).

2.2. Proposals Analyzing Obsessional Intrusive Thoughts: Results from the Revised Obsessional Intrusions Inventory

Studies analyzing heterogeneity from the OIT perspective have mainly used the first part of the *Revised Obsessional Intrusions Inventory* (ROII; Purdon & Clark, 1993, 1994). This is a self-report questionnaire designed to assess the presence and frequency of unwanted intrusive thoughts, images and impulses with egodystonic content, analogous to clinical obsessions (ROII-first part), as well as the appraisals and control strategies associated with the most upsetting intrusive thought referred to by participants (ROII-second part). Different authors have consistently found two main factors (Belloch et al., 2007; Belloch, Morillo, Lucero, Cabedo, & Carrió, 2004; Lee & Kwon, 2003; Moulding, Kyrios, & Doron, 2007; Purdon & Clark, 1993), labeled by Lee and Kwon (2003) as autogenous and reactive intrusions. The former correspond to the Y-BOCS factor that includes aggressive, sexual and religious obsessions, while reactive intrusions correspond to symmetry/ ordering and contamination/ cleaning factors. However, even though the ROII reflects the dimensionality assumption better than the Y-BOCS, it has some important shortcomings that limit the extension of these results to the study of OCD heterogeneity (e.g., limited representation of some clinically relevant obsessional themes, such as religion, order or symmetry) (García-Soriano, Belloch, Morillo, & Clark, 2011).

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Thus, although most studies using the Y-BOCS suggest a four-factor structure, and those using the ROII suggest a two-factor structure, there are relevant inconsistencies among the studies, in part due to limitations of the instruments, which do not include a representation of all OC relevant contents and do not usually take OC dimensionality into account. These limitations suggest the need to develop new assessment tools which, by better appraising OCD heterogeneity and dimensionality, would make it possible to confirm the dimensional structures of OC symptoms. Responding to that need, in the past decade different evaluative alternatives have been offered, and they will be reviewed in the next section (Table 1).

3. Further Efforts to Encompass OC Symptom Heterogeneity and Dimensionality

From the first perspective (i.e., to analyze the clinical manifestations of obsessions and compulsions), Rosario-Campos et al. (2006) developed the *Dimensional Yale-Brown Obsessive-Compulsive Scale* (DY-BOCS). This is a dimensional version of the Y-BOCS that allows researchers and clinicians to evaluate the presence and severity of various obsessive-compulsive symptom dimensions. It includes an 88-item self-report checklist of obsessions and compulsions grouped in 6 sections about contamination obsessions and cleaning compulsions; hoarding obsessions and compulsions; symmetry, "just right" perception, ordering, counting and arranging obsessions and compulsions; obsessions about harm due to aggression/ injury/ violence/ natural disasters and aggressive compulsions; sexual/ moral/ religious obsessions and compulsions; and miscellaneous obsessions and compulsions that include somatic concerns and superstitions, among others. The DY-BOCS also contains a series of questions about the severity of the symptoms in each dimension, through a self-report and clinician administered scales. The latter consists of a series of 3 items about the severity of the obsessions and compulsions of each content dimension experienced during the *last week*: (a) frequency (time occupied by the obsessions and related compulsions), (b) amount of distress caused by them, and (c) degree of interference with family life, friendship, ability to perform at work or school (including avoidance). Additionally, it contains a Global symptom severity scale and an Impairment score.

From a different perspective, Abramowitz et al. (2010) developed *The Dimensional Obsessive-Compulsive Scale (DOCS)*, with the objective of appraising the OCD dimensional severity of 4 symptom dimensions. This is a 20-item self-report instrument with four OC symptom subscales (contamination; responsibility for harm, injury, or bad luck; symmetry, completeness, and exactness; and unacceptable obsessional thoughts). Each of the subscales includes a general description of the symptom dimension, with examples of contents representative of obsessions, compulsions, and avoidance behaviors, but no items assessing specific obsessions or compulsions. Then, it includes 5 items assessing (from to 0 to 4) different severity dimensions *during the past month*: (a) time occupied by obsessions and rituals, (b) avoidance behavior, (c) associated distress or anxiety, (d) interference with daily routine, and (e) difficulty disregarding the obsessions and refraining from the associated compulsions. Interestingly, checking and ordering compulsions are assessed in two different dimensions, depending on their function: order is conceptualized as a response to the "incompleteness feeling" (symmetry dimension) or to prevent harm (responsibility for harm

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and mistakes dimension); and checking is conceptualized as a response to prevent either harm (responsibility for harm and mistakes dimension) or contamination (contamination dimension).

Finally, two new instruments have recently been developed to analyze OITs in clinical and nonclinical samples: the Obsessional Intrusive Thoughts Inventory and the International Intrusive Thoughts Interview Schedule Version 7. The *Obsessional Intrusive Thoughts Inventory* (INPIOS; García-Soriano & Belloch, 2013; García-Soriano et al., 2011) was developed based on the ROII to overcome its limitations. It is a self-report questionnaire designed to assess the frequency of unwanted obsessional intrusive thoughts, images and impulses, as well as the appraisals and control strategies associated with each participant's most upsetting intrusive thought. The first part consists of a list of 48 items grouped in six first-order factors: aggressive; sexual, religious, and immoral; contamination; doubts, mistakes, and necessity to check; symmetry and order; and superstition/ magical thinking intrusive thoughts. These dimensions are collapsed into a second-order structure composed of two dimensions: "Type I moral based" and "Type II non-moral based". Type I includes two first-order factors, aggressive intrusions, and sexual, religious and immoral intrusions; and type II includes the other four dimensions. Respondents rate each statement from 0 (*I have never had this intrusion*) to 6 (*I have this intrusion frequently during the day*). The second part asks participants to choose from the 48 items the most upsetting intrusion they had experienced during the past three months. Next, keeping this intrusion in mind, the person is asked to rate from 0 (*not at all*) to 4 (*extreme*) the emotional reactions linked to the intrusion (unpleasantness, anxiety, sadness, guilt, and shame); difficulty in controlling the intrusion, the interference the intrusion produces (3 items: difficulty of controlling it, success controlling/suppressing, and interference); and 8 dysfunctional appraisals associated with the intrusion (importance of the thought, thought-action fusion (TAF)-moral, personal significance, TAF-likelihood, responsibility, importance of control, over-estimation of threat, and intolerance to uncertainty). Then, subjects are asked to record how often they use a set of control strategies to get rid of the intrusion (5 factors: general strategies to control anxiety, covert thought control strategies, distraction, compulsions, and doing nothing). Frequent and upsetting OITs are considered obsessions in clinical samples, as tested in a group of OCD patients (García-Soriano et al., 2011).

The *International Intrusive Thoughts Interview Schedule Version 7* (IITIS; Clark et al., 2013; Radomsky et al., 2013) is a 101-item structured interview developed by the Research Consortium on Intrusive Fear to collect quantitative and qualitative data about individuals' experiences of unwanted intrusive thoughts and the associated appraisals and control strategies.

First, a detailed description of unwanted intrusive thoughts is introduced, emphasizing the distinction between OIT and other forms of cognition (i.e., worries or rumination), and participants are asked if they have experienced this type of intrusion *in the past three months*.

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Table 1. Symptom dimensions of the reviewed instruments

	DY-BOCS[1]	DOCS[1]	INPIOS[2]	ITTIS[2]
Symptom Dimensions	sexual, moral, religious obsessions and related compulsions	aggressive, violent, sexual, and religious obsessions with mental rituals and other forms of neutralizing (Unacceptable obsessional thoughts)	sexual, religious, and immoral intrusions	sexual intrusions
				religious/ immoral intrusions
			aggressive, harm intrusions	harm/ injury/ aggression intrusions
	harm due to aggression/ injury/ violence/ natural disasters/ unintentionally harming others obsessions and related compulsions (e.g., checking that one has not harmed other people)	obsessions about causing harm by various means, and checking, reassurance seeking, and related compulsions (Responsibility for harm/injury/bad luck)	doubts, mistakes, and necessity to check intrusions (including doubts about having caused harm)	doubt intrusions
	contamination obsessions and cleaning compulsions	contamination obsessions and washing and cleaning compulsions (Contamination)	contamination intrusions	contamination intrusions
	symmetry, "just right" perception, ordering, counting and arranging obsessions and compulsions to count, or order/ arrange	"just right" obsessions and compulsions involving ordering and repeating (Symmetry/ completeness/ exactness)	symmetry, order intrusions	(see Other)
	miscellaneous: include somatic concerns and superstition obsessions and compulsions	(see Responsibility for harm)	superstition/ magical thinking	'other': include order, symmetry, melodies, superstitious intrusions
	hoarding obsessions and compulsions			

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Table 1. Continued

	DY-BOCS[1]	DOCS[1]	INPIOS[2]	ITTIS[2]
				victimization intrusions
Severity appraisals	Frequency (time)	Time occupied	Frequency	Frequency
	Distress	Distress/ Anxiety	Anxiety	Distress or interference
	Interference	Interference	Interference	
		Difficult to disregard	Control difficulty	Difficulty removing
		Avoidance	Unpleasantness	Importance

Note. [1]= refer to obsessions and compulsions; [2]=refer to unwanted obsessional intrusive thoughts; DY-BOCS=Dimensional Yale-Brown Obsessive-Compulsive Scale; DOCS= Dimensional Obsessive-Compulsive Scale; INPIOS=Obsessional Intrusive Thoughts Inventory; ITTIS=International Intrusive Thoughts Interview Schedule Version 7.

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Second, there is a section divided into 7 content areas (i.e., contamination, harm/ injury/ aggression, doubt, religious/ immoral, sexual, victimization, and 'other') that includes, first, a description of the contents, with examples of typical OIT, and questions about the form of the intrusion (i.e., a thought, image, impulse, or feeling), as well as the perceived anticipated consequences of having such a thought. Participants also report on the frequency, distress/ interference, importance, and difficulty in removing the relevant OIT type, using a six-point rating scale from 0 (*never/ not*) to 5 (*frequent/ extremely*). Participants are then asked to indicate which type of intrusion was most distressing, and answer questions about this most distressing thought on a 6-point scale: appraisals about the meaning and importance of the most distressing unwanted IT (i.e., overestimation of threat, importance of thought, intolerance of anxiety/distress, need to control, responsibility, intolerance of uncertainty, perfectionism, thought-action fusion and ego-dystonicity); control strategies to cope with the intrusion (i.e., distraction, thought replacement, thought stopping, self-reassurance, reassurance seeking, ritualizing, neutralization, rationalization, avoidance); and finally, questions about control failures (i.e., misinterpretation of control significance, thought-action fusion, possibility inference, unrealistic control expectations, inflated responsibility, and faulty inference of control).

The last two instruments share a similar structure that follows the OCD cognitive model, since they are mainly focused on the analysis of intrusions rather than compulsions and/or rituals, which are considered control or neutralizing strategies motivated by the intrusions. Moreover, these instruments appraise not only the frequency, interference, and difficulty of controlling the intrusions, but also the associated *appraisals and control strategies.* The two instruments differ on the way heterogeneity is appraised and on the instrument's format (self-report *versus* face-to face interview).

3.1. Answers to Unsolved Questions from Recent Assessment Tools

This new generation of instruments (i.e., DY-BOCS, DOCS, INPIOS, ITTIS) was developed to answer previously unresolved issues, adopting a dimensional perspective that emphasizes OCD heterogeneity. They were developed to appraise the *dimensionality* of OC symptom contents, going beyond the idea of dimensionality according to the frequency or duration of the symptom contents. Instead, they appraise the dimensionality of the distress or anxiety associated with different symptom contents, as well as the interference and avoidance that these contents produce in the individual. Thus, these instruments consider OC symptom severity as a multidimensional construct responding to former OIT definitions (e.g., Rachman & Hogdson, 1980), empirically-based parameters of severity (Deacon & Abramowitz, 2005), and the severity parameters described by the DSM (APA, 2000). Interestingly, and unlike previous instruments, they ask responders to appraise the severity of all the contents experienced (*vs* global obsessions and compulsion severity). This is especially true for the DY-BOCS, DOCS and ITTIS, which explore the frequency or duration, distress and interference of *all* the experienced obsessions/ intrusions during the past week, month or 3 months, respectively.

Regarding the *content dimensional structure,* only the INPIOS includes an empirically derived structure to date. This structure was derived from the exploratory and confirmatory analyses of the list of intrusions included in the first part in a non-clinical sample (García-

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Soriano et al., 2011). The first-order structure is similar to the solutions reported in most of the studies using OC symptom instruments (e.g., Leckman et al., 1997; Summerfeldt et al., 1999) and the second-order structure is analogous to the two-factor models derived from the study of unwanted intrusive thoughts (e.g., Belloch et al., 2004; Lee & Kwon, 2003; Purdon & Clark, 1993, 1994). The remaining instruments do not include data about factor analysis; their structures are theoretically driven after examining research on the OC symptom structure (See Table 1). One of the main differences among the instruments lies in the consideration of hoarding symptoms. Hoarding constitutes an independent factor in the DY-BOCS, but not in the other instruments, which consider hoarding as a distinct phenomenon, in agreement with recent proposals (e.g., APA, 2013; Rachman, Elliott, Shafran, & Radomsky, 2009). In this sense, efforts should be made to ensure that hoarding symptoms evaluated by the DY-BOCS are not being confused with hoarding disorder (Pertusa, Fernández de la Cruz, Alonso, Menchón, & Mataix-Cols, 2012). Another difference has to do with the consideration of *aggressive* obsessions. On the DY-BOCS, obsessions about violence, aggression, and unintentionally harming others are placed with checking compulsions, and not with sexual and religious thoughts. By contrast, on the DOCS, obsessions about aggression or violence are placed together with sexual and religious obsessions, and obsessions about unintentional aggression are placed with checking compulsions. In the same vein, on the INPIOS, intrusions about aggression or violence constitute a factor that, together with sex/religious in Type I-moral based intrusions and doubts about having caused a tragedy or unintentional harm (e.g., having left something dangerous in a place where children have access to others), is included with other doubts, such as having left something unlocked at home or having made a mistake at work (Doubt OIT). Thus, while the DOCS and INPIOS coincide with recent proposals that suggest separating aggressive obsessions related to accidental harm (pathological doubt) from those concerning acts of impulsive aggression (unacceptable thoughts) (Pinto et al., 2008), the DY-BOCS adopts a different perspective. Finally, another difference between the instruments is the inclusion of a separate dimension for *superstitious* OIT in the INPIOS; the remaining instruments also collect those contents, but they are included in the others, miscellaneous, or responsibility for harm/ injury/ bad luck dimensions of the ITTIS, DY-BOCS and DOCS, respectively. In this sense, as recently suggested, the miscellaneous dimension of the DY-BOCS needs an item-level analysis in order to further delimit it (Pertusa et al., 2012), a practice that could be extended to the ITTIS Others category.

All the reviewed instruments focus on obsessions and, in line with the standard definition of compulsions (e.g., APA, 2000, 2013), consider *compulsions* as a way to neutralize the anxiety or distress triggered by obsessions or prevent a feared event. Unlike the former Y-BOCS or other widely used instruments (e.g., Obsessive-Compulsive Inventory-revised; Foa et al., 2002), the new instruments allow compulsions to be associated with different obsessional contents, and they try to assess OCD symptoms based on function rather than form (Wheaton, Abramowitz, Berman, Riemann, & Hale, 2010). However, only the INPIOS and ITTIS focus on the appraisal of the cognitive phenomena (i.e., intrusions) and, moreover, appraise compulsions in a different section, together with a list of other control strategies that also include avoidance and covert compulsions and rituals.

Promising results will come from these new proposals that have been developed to overcome the limitations of previous instruments and better capture the OCD dimensionality and heterogeneity. Unlike previous instruments, all the new instruments allow clinicians and

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researchers to evaluate severity from a multidimensional perspective associated with specific OC contents. Promising results are expected from the examination of the structure of the DY-BOCS and INPIOS symptom/ OIT checklists.

4. Updated Data from the Obsessional Intrusive Thoughts Inventory (INPIOS) Proposal

Finally, in order to further support the multi-dimensionality of obsessions' severity, taking their content into account, some new data will be provided. With this objective, the responses to the INPIOS provided by a group of OCD patients and community counterparts will be analyzed, focusing on the distribution of the OIT contents, and the frequency, unpleasantness, interruption and control difficulty of these intrusions.

4.1. Method

Two groups of subjects participated in the study. The first group was recruited from outpatient mental-health clinics in the Spanish National Health System and the OCD Research Unit of the Universidad de Valencia (Spain). It was composed of 79 patients who met DSM-IV-TR (APA, 2000) criteria for OCD. Of them, 52% were women, and the mean age (*SD*) was 34.75 (11.99) years. On average, the patients had a severe disorder: Yale-Brown Obsessive Compulsive Scale (Y-BOCS)-total score (Goodman al., 1989a, b): *M* (*SD)* = 25.66 (6.71); and the duration of their disorder was 9.36 (8.87) years. All the patients had OCD as their main Axis I diagnosis, and up to 22% of the patients had a comorbid diagnosis. Regarding Axis I comorbidity, patients presented the following diagnoses: 5 major depression, 2 dysthymia, 1 bipolar, 3 panic disorder, 3 generalized anxiety disorder, and 1 social phobia. Regarding Axis II comorbidity, patients presented these diagnoses: 1 Borderline and 2 Schizotypal. The second group, recruited by advanced Psychology students who received course credit for their recruitment efforts, included 79 non-clinical community adults (51.9% men, mean age of 36.33 (9.84) years) extracted from a larger community sample, matching their basic socio-demographic features (i.e., gender, age, studies level, socio-economic level) with those of the clinical participants.

Both groups of participants completed the Obsessional Intrusive Thoughts Inventory (INPIOS; García-Soriano et al., 2011; García-Soriano & Belloch, 2013) and the Obsessive-Compulsive Inventory-Revised (OCI–R; Belloch et al., 2013; Foa et al., 2002), along with other questionnaires. An individualized, face-to-face administration of the INPIOS (previously described) was carried out. The OCI-R was completed as a self-report. It is a widely used questionnaire containing 18 items that assess the distress associated with various obsessive-compulsive symptoms (washing, checking, ordering, obsessing, hoarding and neutralizing). The OCI-R total score was used.

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4.2. Results and Discussion

4.2.1. Preliminary Analysis

As expected, no differences were found between the OCD group and the community group on age (t (156)=0.203, p=.840), gender (χ^2 (1)=0.228, p=.633), marital status (χ^2 (3)=0.859, p=.835), socio-economic level (χ^2 (4)=6.195, p=.185), or level of studies (χ^2 (2)=3.118, p=.210). Significant differences were found on obsessive-compulsive symptoms (OCI-R total) in the expected direction, with OCD participants scoring higher (M =28.27, SD= 12.69) than community individuals (M=18.15, SD= 13.54; t(154)=4.81, $p \leq .001$).

4.2.2. Content Dimensionality

Participants were distributed according to the content of their most upsetting OIT from the INPIOS 1st part, and the percentages of subjects who chose each content are displayed in Table 2. OCD patients most frequently chose doubts, mistakes, checking, contamination, superstition, and aggression as their most upsetting OIT. Next in frequency were the sexual, religious, immoral, and symmetry/ order dimensions. In the non-clinical sample, the most common dimension was also doubts, followed by the superstition and aggressive dimensions, sexual, contamination and symmetry/ order. The data suggest, first, that both OCD and community subjects experienced intrusions that were representative of all types of obsessional contents. Moreover, chi-square analysis revealed no differences between groups in the proportion of clinical and non-clinical participants who selected aggressive, sexual, religious or immoral, order and superstition contents as their most disturbing intrusion. Nevertheless, the two groups differed on the frequency with which they chose doubts and contamination as their most disturbing intrusion. A higher proportion of community participants selected doubt as their most disturbing intrusion, whereas a higher proportion of patients selected contamination. These results coincide with previously reported results, except for the similar percentage of subjects who chose superstitious thoughts, which in previous studies were higher in the OCD sample (García-Soriano et al., 2011).

Table 2. Percentage of participants who chose their most upsetting unwanted intrusive thought in each content dimension

INPIOS most upsetting UIT content:	**OCD sample % (n [a])**	**Community sample % (n)**	**X^2 (df=1)**
Aggressive	16.5% (13)	12.7% (10)	0.458
Sexual/religious/immoral	10.1% (8)	11.4% (9)	0.066
Symmetry & order	1.3% (1)	5.1% (4)	1.859
Doubts, mistakes, checking	26.6% (21)	49.4% (39)	8.706**
Contamination	22.8% (18)	8.9% (7)	5.750*
Superstition/ magical	19.0% (15)	12.7% (10)	1.188

Note. % (n); *$p \leq .05$; ** $p \leq .01$; [a] three patients were not included, as their unwanted intrusive thought was highly idiosyncratic and could not be categorized in any of the categories.

INPIOS = Obsessional Intrusive Thoughts Inventory; OIT=obsessional intrusive thought.

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The data suggest that OCD patients and community samples cannot be distinguished based on the content of their most upsetting intrusions, which in the OCD group correspond to their most upsetting obsession. Thus, the results contradict the Rassin and Muris (2007) proposal of obsessions being characterized by abnormal contents, and they support the dimensionality of the contents appraised by the INPIOS (i.e., aggression, sexual/religious, symmetry/order, contamination, superstitious).

4.2.3. Severity Dimensionality: Frequency, Anxiety, Unpleasantness, Interruption, and Control Difficulty

Next, OCD and community participants were compared on four variables appraising the "severity" of their OITs. OCD participants scored higher on all the severity dimensions (i.e., frequency of the OIT (t(129.63)=9.73; p<.001); anxiety (t(156)=11.46; p<.001); unpleasantness (t(156)=11.95; p<.001); interruption (t(156)= 10.13; p<.001); and control difficulty (t(156)=12.64; p<.001)). To better examine the dimensionality of these variables, community participants were divided into two groups according to their obsessive-compulsive symptom distress (community subjects scoring above and below OCI-R percentile 50). The results, reported in Table 3, further support the dimensionality of intrusions: large significant differences appeared when comparing the three groups on the study measures. OCD patients scored higher than community participants, and those non-clinical participants scoring above percentile 50 (OCI-R>13) on the OCI scored higher on the severity measures than those participants who scored below percentile 50. There were no significant differences between the two groups of community participants, except on the concentration interruption produced by the OIT.

Table 3. Differences between groups (OCD sample, community participants scoring above and below percentile 50 on OCI-R total score) in the frequency, anxiety, unpleasantness, interruption and control difficulty of their most upsetting unwanted intrusive thought

		Community sample			
INPIOS most upsetting UIT:	**OCD sample (n=79)**	**OCI-R < P_{50} (n=32)**	**OCI-R ≥ P_{50} (n=47)**	**F (2, 155)**	***Partial eta squared***
Frequency	5.22 (0.87)a	3.94 (1.43)b	3.02 (1.29)c	56.99***	.424
Anxiety	3.09 (0.96)a	1.66 (1.07)c	1.19 (0.74)c	69.68***	.473
Unpleasantness	3.59 (0.69)a	2.41 (0.95)b	1.74 (0.87)c	83.36***	.518
Interruption	2.82 (1.12)a	1.41 (1.07)b	0.98 (0.83)b	53.69***	.409
Control difficulty	3.13 (0.94)a	1.63 (1.21)b	0.74 (0.79)c	95.69***	.553

Note. *M* (*SD*); *P*= percentile; *** p≤ .001

INPIOS = Obsessional Intrusive Thoughts Inventory; OCI-R = Obsessive-Compulsive Inventory Total Score; OIT=obsessional intrusive thought.

Taken together, the results showed that OCD and community participants' most upsetting OITs have similar contents, and those contents are experienced mostly in the same proportion (except in the contamination and order dimensions). However, there is a continuum of the frequency and disturbance with which those OIT are experienced. At one extreme lie the most

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upsetting OIT the OCD participants experience, which constitute their main obsession, and at the other extreme lie the OIT experienced by community individuals with low obsessive-compulsive symptoms. On the one hand, clinical OIT (OCD sample) were experienced between daily and frequently during the day, patients felt very anxious, obsessions were very-extremely unpleasant, they interrupted their concentration very much, and it was very difficult to control them. On the other hand, non-clinical OIT (community samples) were experienced between once or twice a week (OCI≥*13*) and once or twice a month (OCI<*13*), and they were between unpleasant (OCI≥*13*) and somewhat unpleasant (OCI<*13*); participants felt between anxious and somewhat anxious, the intrusion interrupted concentration a little, and it was between difficult (OCI≥*13*) and somewhat difficult (OCI<*13*) to control.

Conclusion

Obsessive-compulsive disorder is a heterogeneous condition; nevertheless, although substantial efforts have been made to analyze the structure of obsession and compulsion symptom dimensions, results have been inconsistent in part due to instrument limitations (Bloch et al., 2008; Leckman et al., 2010; McKay et al., 2004). Most of the studies have factor analyzed the categories of obsessions and compulsions included in the Y-BOCS symptom checklist in OCD samples; however, there is an interesting alternative to analyzing OCD content dimensions that is highly consistent with cognitive proposals. It consists of factor analyzing unwanted intrusive thoughts experienced by clinical and non-clinical individuals. Results from the two approaches are different, but could be integrated, as autogenous intrusions map onto the unacceptable/ taboo thoughts dimension from the Y-BOCS factor analyses, and reactive intrusions may map onto the other OC dimensions. In recent years, a new generation of instruments has been developed from both perspectives (i.e., measuring obsessions and compulsions *versus* OIT) in an attempt to improve previous deficiencies. These instruments (DY-BOCS, DOCS, INPIOS, ITTIS) take into account the heterogeneity and dimensionality of OC symptoms, a dimensionality that is defined at different levels (frequency, distress, interference). Although promising results are expected regarding the description of OCD heterogeneity, at the moment only the INPIOS factor structure has been empirically analyzed (in a non-clinical sample). Additional work is needed to factor analyze the DY-BOCS at the item level, and replicate the INPIOS factor analysis in clinical samples. The new instruments coincide in the relevance given to obsessions as the key element in OCD. Moreover, all of them include the dimensions empirically supported by former factor analysis (i.e., symmetry/ ordering; sexual/ religious; contamination/ cleaning compulsions), although differences are found in the solution given to these symptom dimensions, which are not always confirmed (i.e., harm, checking and hoarding symptoms) or even appraised (i.e., superstition). Finally, results from the analysis of the INPIOS support the dimensionality of the six different obsessional contents appraised, taking into account not only the frequency, but also the unpleasantness, interference and control difficulty of these dimensions. Although efforts are still needed to factor analyze the generation of new instruments and obtain a more well-defined psychopathological description of OC symptoms, in the meantime we can continue to work with the proposed structures when developing OCD patients' treatments, as experimental work (Abramowitz et al., 2003; Coehlo & Whittal,

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2001) investigating specific symptom categories has found support for the relevance of taking OC manifestations into account.

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Chapter VI

Associations Between Hoarding and Eating Disorders in an Italian Clinical Population

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Abstract

From a phenomenological standpoint, eating disorders (ED) and hoarding disorder (HD) have several features in common, and both conditions are closely associated with impulse control disorder (ICD) and obsessive-compulsive disorder (OCD). So far, no studies seem to have used specific measures for assessing HD to investigate the presence of hoarding symptoms in patients with ED. There is also a paucity of research on HD in Italy, and even less attention has been paid to its association with ED.

The aim of the present study was to identify features of hoarding in a sample of patients with ED, comparing them with a group of self-identified HD individuals. Judging from the literature, the two groups could be expected to have similar levels of compulsive acquisition and obsession with order and symmetry, and similar anxiety and depression symptoms.

Our study sample of patients with ED included 15.15% who scored higher than the cut-off for hoarding psychopathology in the Saving Inventory-Revised (SI-R). On the subscales of the SI-R, 12% of the ED patients exceeded the clinical cut-off in the Clutter and Acquisition domains, and 9% in the Discarding domain. A comparison between our ED and HD groups showed, however, that the self-identified hoarders scored higher than the ED patients both on the subscales of the SI-R, and on the scales designed to measure their obsessive-compulsive behavior, while the ED patients scored higher for anxiety and depression. Our initial hypotheses were therefore not confirmed, even though the patients with ED showed some features of hoarding.

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Further research is needed, focusing on specific types of ED (e.g. bulimia and binge eating), and analyzing in more depth those domains related to impulse control that seem to characterize both ED and HD.

1. Introduction

The term 'hoarding' first appeared in the scientific literature to describe a food acquiring and storing behavior considered typical of rodents (Herberg & Blundell, 1967). The relationship between humans' and animals' hoarding behavior would undeniably be interesting, but it has been studied very little and is beyond the scope of our study. In Italy, the first report on this type of behavior was published by Mingazzini (1893), reporting on a population in a psychiatric institution in Rome. He described a number of behavioral features that he defined as *polycollectionism* and *polykleptocollectionism*, both involving the indiscriminate collection and accumulation of objects of little value and food, which might also be stolen in the case of polykleptocollectionism. With time, various articles described this psychopathological phenomenon, mainly in anecdotal accounts (e.g. Bolman & Katz, 1966; Gutheil, 1959), but it is thanks to the systematic work done by Randy Frost that hoarding disorder (HD) really came to the attention of the scientific community.

Frost's first studies date from the 1990s (Frost & Gross, 1993; Frost & Hartl, 1996), when a definition of hoarding was proposed that is still generally applicable today: 1) the acquisition and consequent incapacity to discard large quantities of apparently useless objects of limited economic value; 2) the cluttering of living space in the home to such a degree that this prevents the activities for which they were intended; 3) significant distress and interference with social and occupational functioning. The reasons why hoarders acquire and accumulate objects are much the same as the motives of non-hoarders, i.e. for their affective value, because they are expected to be useful in future, and to avoid unnecessary waste (Steketee, Frost & Kyrios, 2003). A similarity with collectors would seem appropriate but, unlike hoarders, collectors seek objects as a hobby. The objects they acquire have a symbolic or affective and instrumental value. They are chosen in a given thematic context by means of planned searches, and they are often put to normal use or carefully arranged in spaces set aside for the collection. In most cases, collectors experience no distress or social impairment, and their collecting activities do not interfere with their private or social lives (Nordsletten & Mataix-Cols, 2012).

Another, no less important feature of HD concerns a shortage of insight: hoarders rarely acknowledge that they have a problem. It is often a member of the family who raises the issue, or a neighbor who call the authorities because of a practical problem posed by the hoarder's clutter becoming a public health or fire hazard. Given the egosyntonic nature of the disorder, hoarders have strong emotional reactions, becoming angry, aggressive and distressed when someone touches their things or when they are faced with having to dispose of them. These characteristics make their treatment very difficult, and it is usually scarcely successful (Mataix-Cols et al., 2010; Tolin, Frost, & Steketee, 2007).

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1.1. Epidemiology, Onset and Course

According to a study by Grisham, Frost, Steketee, Kim and Hood (2006), conducted on a sample of 51 individuals with hoarding problems, 60% of the sample developed symptoms early, at around 12 years of age, and 80% were symptomatic by the age of 18. Only a small percentage (14%) reportedly had any remission of their symptoms once they had developed. Samuels et al. (2008) examined accumulation behavior in a sample of 742 adults from 34 and 94 years of age taking part in the *Hopkins Epidemiology of Personality Disorder Study* with a view to estimating the prevalence of hoarding in relation to their sociodemographic characteristics. Their results showed that the prevalence of HD increased with the participants' increasing age, ranging from 2.3% for the youngest group to 6.2% among the most elderly.

After its onset, HD generally follows a chronic course, with symptoms worsening over the years; spontaneous remissions are rare (Gilliam & Tolin, 2010). In the Italian general population, the prevalence of self-identified hoarding reaches as high as 6% (Bulli et al., 2013).

The prevalence of female gender is a controversial issue in the literature. Frost, Steketee and Tolin (2011) found the percentage of women as high as 78.7% in a sample of hoarders, whereas Samuels et al. (2008) calculated the presence of hoarding in the general population as amounting to 5.6% for men and 2.6% for women. No gender-related differences were identified in Italian studies on nonclinical populations (Bulli et al., 2013; Bottesi & Novara, 2012). In the clinical population, the severity of the hoarders' symptoms (when compared with patients who had obsessive-compulsive disorder [OCD], but were not hoarders) seemed to depend on the female component of the sample, i.e. the female hoarders had more severe psychiatric symptoms, which included eating disorders (ED), than the non-hoarder OCD patients (Wheaton, Timpano, Holland LaSalle-Ricci, & Murphy, 2008).

1.2. Comorbidity with Anxiety and Depression

Few studies have assessed this comorbidity with HD considered as a separate disorder. In fact, HD has only recently been included in the DSM-5 as one of the disorders belonging to the obsessive-compulsive spectrum (Mataix-Cols et al., 2010; Pertusa et al., 2008). Because hoarding symptoms were initially seen as a subtype of OCD, most studies have focused on the characteristics that the two disorders have in common, or that distinguish between them (for a review, see Pertusa et al., 2008).

Hoarding has stronger comorbid associations with other anxiety disorders, and particularly with generalized anxiety disorder (GAD) and social phobia. A recent study (Tolin, Meunier, Frost, & Steketee, 2011) showed that 28% of the patients in a sample recruited at a clinic for treating anxiety disorders also had hoarding symptoms. These findings were confirmed by Frost et al. (2011) in a population diagnosed as hoarders (n=217), in which the prevalence of depression was as high as 50.7%, followed by GAD (24.4%), social phobia (23.5%), OCD (18%), and specific phobia (14.3%).

In an interesting study on a pediatric and adolescent population with OCD, Storch et al. (2007) demonstrated that 26.25% of the patients had hoarding symptoms and were also obsessed with symmetry and order.

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One of the main limitations of the majority of studies about the topic of HD concerns sample selection: the subjects considered are usually in the care of clinics for treating anxiety disorders, where the proportion of individuals with OCD and hoarding symptoms is particularly high. Moreover, cases are sometimes recruited by means of invitations circulated in Internet, and the individuals who respond usually acknowledge their problem to some degree, and may therefore be scarcely representative of the population of hoarders.

1.3. The Relationship Between Hoarding and Impulse Control Disorders

The egosyntonic nature of HD and the fact that sufferers' behavior includes the acquisition of both paid and unpaid objects makes the link between hoarding and impulse control disorders (ICD) evident. Mueller, Mitchell, Crosby, Glaesmer and de Zwaan (2009) found in a German study population that approximately 2 out of 3 participants had strong correlated scores for hoarding measures and compulsive shopping measures. Frost et al. (2011) also showed that hoarders report compulsive buying in 60.8% of cases, episodes of kleptomania in 9.9%, gambling in 5.7%, and trichotillomania in 4%.

Judging from the above-outlined research, hoarders differ from patients with OCD in that they have: a) an early onset of symptoms; b) more severe anxiety, depression and impulse control symptoms; and c) in the clinical setting, women have more severe psychiatric symptoms, including ED.

1.4. The Relationship Between Eating Disorders and Hoarding

Studies investigating hoarding in populations with ED have focused mainly on analyzing how they correlate with symptoms of OCD. The link between ED and OCD has been well known for many years (e.g. Bulik et al., 2003; Milos, Spindler, Ruggero, Klaghofer, & Schnyder, 2002). A large study by Kaye, Bulik, Thornton, Barbarich and Masters (2004) demonstrated a 41% comorbidity with OCD in a population with an ED, irrespective of the subtype (anorexia, anorexia with bulimia, or bulimia alone). The first study considering hoarding symptoms in patients with ED, (Halmi et al., 2003) used the Yale-Brown Obsessive Compulsive Scale (Y-BOCS; Goodman et al., 1989) and estimated the lifetime presence of obsessive-compulsive symptoms between 68% and 79%; the patients did not differ significantly from the controls with OCD in terms of hoarding symptoms (16.5% vs 13.8%).

A similarity has also emerged in the content of the obsessions of anorexia and OCD patients after excluding those related to food and slimness (Solyom, Freeman, & Miles, 1982). The most frequently encountered types of obsessions in a group of ED patients has to do with symmetry and order (identified in 68.8-72% of cases) (Bastiani et al., 1996; Matsunaga et al., 1999). On the other hand, considering patients with OCD, some studies have identified a history of ED (26% of anorexia, 16% of bulimia and anorexia; Tamburrino, Kaufman, & Hertzer, 1994). Rubenstein, Pigott, L'Heureux, Hill and Murphy (1992) found that 13% of patients diagnosed with OCD had a diagnosis of ED as well. Fontenelle, Mendlowicz, Soares and Versiani (2004) were the first to look for features of ED in a sample of hoarders recruited among a population with OCD: they reported that 26.7% of the HD patients had ED too (bulimia nervosa in 13.3% and binge eating in the remainder).

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Conversely, Frost et al. (2011) in a population of hoarders, found that only 1.4% of patients showed ED, whereas 0% of patients with OCD had ED as well. Like hoarders, ED patients also have a high likelihood of comorbidity with ICD. In a large sample of ED patients, Vandereycken and Van Houdenhov (1996) found that 47.1% also had a history of kleptomania. In a more recent and extensive study on patients with a history of ED (N=709), the lifetime prevalence of ICD was 16.6%, and the most commonly reported syndromes related to compulsive buying and kleptomania (Fernández-Aranda et al., 2008).

Like hoarders, patients with ED:

a. tend to develop their disorder around the age of puberty (APA, 1994);
b. the symptoms are egosyntonic in nature (APA, 1994);
c. the symptom content associated with OCD seems to relate with symmetry and ordering obsessions;
d. patients have a comorbidity with ICD and especially with compulsive buying and kleptomania.

While the gender distribution amongst hoarders is unclear, the prevalence of female gender in ED nears 90% (APA, 1994). To the best of our knowledge, no studies have ever been conducted on hoarding symptoms in patients with ED with the aid of instruments specifically designed for the assessment of HD. The present study involved two groups of individuals, i.e. a group of patients hospitalized for ED, and a group of women with HD who were enrolled after responding to an invitation published in daily newspapers, in Internet or on advertisements posted in public places. The aim of this study was to look for any evidence of hoarding features in a group of patients suffering from ED. To be more specific, the patients with ED were expected to return scores similar to those of the self-identified hoarders on the scale designed to measure acquisition, while the people with HD should have higher scores in the scales assessing cluttering and difficulty with discarding things. Patients with ED and people with HD were also expected to obtain similar scores on scales measuring order and symmetry, and to experience similar levels of anxiety and depression.

2. Method

2.1 Participants

The study involved 33 patients with ED: 15 had a diagnosis of anorexia nervosa, 7 had bulimia nervosa, 1 had binge eating disorder, and 10 not meeting the criteria for a specific disorder were diagnosed as having eating disorders not otherwise specified (EDNOS). All patients were hospitalized at the Developmental Neuropsychiatry Department of the Sant'Orsola Hospital in Bologna (Italy). The study also involved 31 self-identified female hoarders from the general population of northern Italy. The patients with ED were younger (M=19.36 years, SD=2.76) than the HD subjects (M=26.26, SD=11.43; $F_{(1,63)}$=11.32, p=.001). As for their occupation, 90.9% of the sample with ED and 77.4% of the HD cases were students ($\chi^2_{(3)}$=2.89; p=.41); most of the patients with ED had a middle-school or high-

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school diploma (90.9%), while 93.6% of the HD sample had a high-school diploma or university degree ($\chi^2_{(2)}$=14.22; p=.001).

2.2. Instruments

Saving Inventory - Revised (SI-R; Frost, Steketee & Grisham, 2004; Italian version edited by Melli, Chiorri, Smurra, & Frost, 2013): this questionnaire assesses the main characteristics of a compulsive hoarding behavior by means of three items on a 5-point Likert scale; respondents' answers concern the previous week. In addition to the total score, there are also three subscales for clutter, difficulty with discarding objects, and acquisition. In both the original version and the Italian translation, the total and subscale scores show a good internal consistency and test-retest reliability. Scores in excess of 40 indicate a disposophobia.

Obsessive Compulsive Inventory – Revised (OCI-R; Foa et al., 2002; Italian version edited by Sica et al., 2009): this questionnaire assesses the symptoms characteristic of OCD, grouped into six categories: *washing*, *checking*, *ordering*, *obsessional thoughts*, *hoarding* and *mental neutralizing*. The OCI-R has shown a good construct validity, internal consistency and test-retest reliability, both in patients with OCD and in controls.

Beck Anxiety Inventory (BAI; Beck, Epstein, Brown & Steer, 1988; Italian version edited by Sica, Coradeschi, Ghisi & Sanavio, 2006): this test measures the severity of the physiological signs of anxiety. It has a good internal consistency and test-retest reliability at one week. Good psychometric properties have been confirmed in the Italian version too.

Beck Depression Inventory – II (BDI-II; Beck, Steer & Brown, 1996; Italian version edited by Ghisi, Flebus, Montano, Sanavio & Sica, 2006): this questionnaire identifies the presence and severity of affective, cognitive, motivational, psychomotor and vegetative signs of depression according to the DSM-IV criteria. The BDI-II has shown a good internal consistency (both in groups of students and in patients), a good test-retest reliability at one week, and a good convergent and discriminant validity. The Italian version of the BDI-II has confirmed these good reliability and validity characteristics.

2.3. Procedure

All ED participants were inpatients whereas the hoarders were recruited from the Italian general population. The latter answered to advertisements in public places and published newspapers, and then they were invited to complete a questionnaire online.103 people completed the online questionnaire and 54 of them obtained scores higher than the clinical cut-off in the SI-R; among the latter, only the 31 females were chosen for the study in order to make the two groups comparable in terms of gender. All participants read and signed a form to give their informed consent to participation in the study. Then they completed a form providing socio-demographic details and answered the self-report questionnaires, which had been arranged into a battery in counterbalanced order to avoid any order effects.

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2.4. Statistical Analyses

Pearson's chi square analyses and univariate analysis of variance (ANOVA) were used to compare the two groups in terms of the socio-demographic variables. Chi square analysis was also used to test the differences between the groups as regards the clinical cut-offs exceeded on the subscales of the SI-R and OCI-R.

Univariate analyses of covariance (ANCOVA) were used to test the differences between the two groups' hoarding characteristics, obsessive-compulsive symptoms, anxiety and depression, inputting age as a covariate (given the significant age difference between the two groups).

3. Results

3.1. Hoarding and Obsessive-Compulsive Characteristics in the ED Group

In the group with ED, 5 patients (15.15%) returned a total SI-R score above the cut-off for psychopathological features of hoarding; these patients had a diagnosis of either bulimia (6%) or EDNOS (9%).

On the subscales of the SI-R, 12% of the ED patients obtained scores higher than the cut-off in the domains for clutter and acquisition, while 9% exceeded the threshold in the Discarding domain.

3.2. Differences Between the ED and HD Groups in Hoarding and Obsessive-Compulsive Traits

The HD group returned significantly higher scores than the ED patients on the SI-R subscales (see Table 1). The age covariate was never significant ($p>.05$).

The obsessive-compulsive characteristics of the self-identified hoarders were characterized by higher levels of checking, ordering, mental neutralizing and hoarding than in the patients with ED. The age covariate was only significant in the case of the checking trait ($F_{(1,63)}=10.75$; $p=.002$). The two groups showed no differences in terms of washing and obsessing (Table 2).

As for the numbers of individuals in each group that exceeded the cut-off for each subscale of the SI-R and OCI-R, the hoarders exceeded the cut-offs more frequently than the ED patients on all subscales except for the washing and obsessing domains (Table 3).

Table 1. Means and standard deviations for the two groups on the subscales of the SI-R

	ED (N = 33)	HD (N = 31)	$F_{(1,64)}$	p
SI-R Clutter	6.82 (±6.20)	19.03 (±5.21)	54.91	<.001
SI-R Discarding	8.85 (±5.21)	15.68 (±4.41)	28.22	<.001
SI-R Acquisition	9.97 (±5.92)	16.26 (±4.11)	17.79	<.001

Note: ED = eating disorders; HD = hoarding disorder; SI-R = Saving Inventory-Revised.

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Table 2. Means and standard deviations obtained by the two groups on the subscales of the OCI-R

	ED (N = 33)	HD (N = 31)	$F_{(1,64)}$	p
OCI-R washing	2.69 (±3.39)	4.42 (±1.78)	3.56	.06
OCI-R checking	2.69 (±3.09)	5.61 (±2.75)	7.01	.01
OCI-R ordering	4.09 (±3.83)	7.77 (±3.83)	9.81	.003
OCI-R obsessing	4.84 (±3.47)	5.84 (±2.54)	.79	.38
OCI-R mental neutralizing	1.34 (±2.64)	3.85 (±1.43)	17.14	<.001
OCI-R hoarding	3.56 (±3.12)	7.68 (±3.26)	24.58	<.001

Note: ED = eating disorders; HD = hoarding disorder; OCI-R = Obsessive Compulsive Inventory-Revised.

Table 3. Frequencies (percentages), by group, of individuals exceeding the cut-off for the SI-R and OCI-R subscales

	ED (N = 33) frequencies (%)	HD (N = 31) frequencies (%)	$\chi^2_{(2)}$	p
SI-R Clutter	4 (12.1)	24 (77.4)	27.69	<.001
SI-R Discarding	6 (18.2)	23 (74.2)	20.24	<.001
SI-R Acquisition	7 (21.2)	22 (71)	15.97	<.001
OCI-R washing	7 (21.2)	8 (25.8)	.13	.71
OCI-R checking	5 (15.1)	12 (38.7)	4.26	.04
OCI-R ordering	9 (28.1)	22 (71)	11.56	.001
OCI-R obsessing	9 (28.1)	7 (22.6)	.26	.61
OCI-R mental neutralizing	5 (15.1)	14 (45.2)	6.52	.001
OCI-R hoarding	6 (18.8)	20 (64.5)	13.61	<.001
OCI-R total	11 (34.4)	28 (90.3)	20.90	<.001

Note: ED = eating disorders; HD = hoarding disorder; SI-R = Saving Inventory-Revised; OCI-R = Obsessive Compulsive Inventory-Revised.

Table 4. Means and standard deviations obtained by the two groups for the measures of anxiety and depression

	ED (N = 33)	HD (N = 31)	F(1,64)	p
BAI	21.68 (±13.49)	12.10 (±7.87)	10.13	.002
BDI-II	22.47 (±11.56)	9.45 (±9.40)	25.20	<.001

Note: ED = Eating Disorders; HD = Hoarding Disorder; BAI = Beck Anxiety Inventory; BDI-II = Beck Depression Inventory-Second Edition.

Finally, for symptoms of anxiety and depression (as measured by the BAI and BDI-II), the ED group scored significantly higher than the HD group. When age was considered as a covariate, it did not prove significant (p>.05) (Table 4).

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4. Discussion

This is the first study to have explored hoarding features in a sample of patients with ED by comparing them specifically with a sample of individuals with HD. As many as 15.15% of our sample of patients with eating disorders also exceeded the clinical cut-off for HD; these were women diagnosed with bulimia (6%) or EDNOS (9%). When the single domains of hoarding behavior were investigated, most of these patients (12%) also seemed to have constructs relating to clutter and acquisition. Assuming a biunique correspondence between ED and HD, our results would partially confirm those reported by Fontenelle et al. (2004) in a population with HD, who found patients with bulimia and binge eating (in higher proportions than in our sample) and no cases of anorexia nervosa. On the other hand, our results appear to contrast with those of Frost et al. (2011), who found a lower proportion of eating disorders among individuals with HD.

When compared with HD subjects, our patients with ED obtained significantly lower scores on all subscales of the SI-R and OCI-R except for obsessing and washing. Consistently, when the percentages of women exceeding the cut-offs on each subscale were considered: 71% of the HD group exceeded the threshold on the acquisition subscale in the SI-R, as opposed to 21% of the ED group; and 71% of the HD group exceeded the cut-off on the subscale for order and symmetry, as opposed to 28% of the patients with ED. Our hypothesis that the scores obtained by the ED and HD would be similar was therefore not confirmed for the acquisition subscales in the SI-R or the order and symmetry subscale in the OCI-R. The women with HD had more severe obsessive-compulsive symptoms as well as their hoarding traits (90% of these HD individuals exceeded the cut-off for the total score in the OCI-R), whereas the patients with ED seemed to be characterized more by a picture of anxiety and depression (as measured by the BAI and BDI).

These findings should be considered with caution, because the present study had at least two important limitations:

1. our ED patients had been admitted to public hospitals due to problems relating to their eating behavior, and this could explain their high scores on the anxiety and depression scales, and their limited attention to other problems;
2. our sample of HD cases is not representative of the population of hoarders because it consisted of self-identified cases presenting in response to an invitation published in the press or in Internet.

Therefore, while we cannot rule out an association between eating and hoarding disorders, future research should concentrate on individuals with bulimia and binge drawn from an outpatient population, and could focus more particularly on the domains relating to impulse control, which seem to characterize both ED and HD.

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Chapter VII

Internet-Based Cognitive Behaviour Therapy for Obsessive Compulsive Disorder

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Abstract

Cognitive behaviour therapy is the psychological treatment of choice for obsessive compulsive disorder. Traditionally, cognitive behaviour therapy is delivered via face-to-face sessions with a specialist clinician. It often involves significant portions of therapist-assisted exposure. However, there are many barriers to accessing appropriate treatment and there is a clear need for more effective treatment dissemination. Can CBT be successfully delivered without seeing the patient? Can OCD be properly treated over the Internet? This body of research has been steadily increasing over the last decade, and this review provides the latest evidence for the efficacy, cost-effectiveness, and acceptability of online treatments for OCD. The outcomes of online programs will be compared to those found for traditional face-to-face cognitive behaviour therapy. The clinical advantages and disadvantages of Internet-based treatments will also be discussed and exemplified with examples drawn from the online clinic that we operate. This article concludes with recommendations for future research.

Introduction

Obsessive compulsive disorder (OCD) is characterised by persistent, intrusive, and ego-dystonic obsessions (recurrent thoughts, urges, or images) and/or compulsions (repetitive

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behaviours or mental acts performed to relieve distress) (American Psychiatric Association [APA], 2013). OCD has an estimated 12-month prevalence of 0.6- 1.8% and is associated with substantial psychiatric comorbidity, disability, and economic costs (APA, 2013; Andrews, Henderson, and Hall, 2001; Crino, Slade, and Andrews, 2005; DuPont, Rice, Shiraki, and Rowland, 1995). Evidence-based treatments, notably cognitive behaviour therapy and antidepressant medications, are established (Fineberg, Reghunandanan, Brown, and Pampaloni, 2013; National Institute for Health and Clinical Excellence, 2005; Olatunji, Davis, Powers, and Smits, 2013). However, many individuals with OCD remain untreated, receive inappropriate treatment, or delay seeking treatment for many years (Andrews et al., 2001; Marques, LeBlanc, Weingarden, Timpano, Jenike, and Wilhelm, 2010).

There are multiple barriers to effective treatment of OCD. People with symptoms of OCD frequently report that they feel ashamed of their symptoms and about needing help and fearful of being judged negatively if they seek assistance (Marques et al., 2010). In addition to uncertainty about who to consult and whether treatment is effective, accessing treatment is significantly complicated by difficulties associated with cost, transportation, and inconvenience (Marques et al.). One option for addressing barriers to treatment is internet-based cognitive behaviour therapy (iCBT). This approach aims to provide the same content of face-to-face therapy remotely via the Internet. CBT may be particularly well suited to an online application because it employs a structured skills-based approach to address specific symptoms (e.g., behaviour avoidance). Compared to traditional face-to-face CBT, it is particularly convenient, economical and private. But does it work? This review aims to examine the current evidence for the efficacy, acceptability, and clinical utility of online CBT for OCD.

Internet-based CBT for OCD

Although research regarding online treatments for OCD is in its infancy, the evidence supporting iCBT programs for anxiety and depressive disorders is substantial. In 2010, Andrews, Cuijpers, Craske, McEvoy and Titov conducted a meta-analysis of 22 randomised controlled trials (RCTs) of computerised CBT compared to a control condition for adults diagnosed with major depression, panic disorder, social phobia or generalized anxiety disorder. They found that the mean pre to post treatment effect size superiority of computerised CBT was 0.88 (Hedges *g*) on primary symptom measures, with a corresponding number needed to treat of 2.15. Gains were maintained at follow-up (based on 14 RCTs with median follow up = 26 weeks) and evident across all four diagnoses. Adherence and treatment satisfaction was good with a median of 80% of people completing their program and a median 86% reporting that they were either satisfied or very satisfied. These results appear comparable to traditional face-to face CBT programs (Butler, Chapman, Forman, and Beck, 2006). Could online CBT be just as effective for people with OCD?

OCD is often seen as a heterogeneous and complex disorder requiring specialist clinical treatment. Traditional face-to-face CBT is typically delivered on an individual or group basis over 10-20 sessions lasting 30-120 minutes each (Foa, 2010; Olatunji et al., 2013). It can involve considerable portions of therapist-assisted exposure whereby clinicians guide their patients through a variety of distressing tasks in order to challenge and overcome difficult

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thoughts, feelings, and behaviours. Avoiding these distressing exposure tasks is very tempting for most people with OCD and without support completing such tasks may seem impossible. Unfortunately, a comparatively small number of clinicians are adequately trained to deliver CBT, especially the large portions of exposure and response prevention (ERP) that are required. This has lead researchers to consider how computers could be useful in the treatment of OCD.

Baer and colleagues (1987; 1988) first reported using computers to assist in behaviour therapy in the 1980s; they subsequently developed *BT Steps*, the first comprehensive computerised behaviour therapy program for OCD (Baer and Greist, 1997). *BT Steps* is not an online program; it is delivered over the telephone via an automated interactive voice-response system. The course involves 9 steps; at each step people with OCD read a chapter from a standardised 190-page workbook based on ERP and then telephone the computerised system which provides consultation and reinforces treatment goals. The program involves (i) self-assessment (psycho-education, symptom identification, goal setting and involving others in therapy, steps 1-4), (ii) ERP (steps 5-8), and (iii) relapse prevention (step 9). Clinicians supervise patients through the course by sending regular progress reports and answering any questions left by patients on the computerised system. *BT Steps* is now called *OCFighter* and is available online. The content of the program is similar with patients receiving emails with treatment tips and activities worksheets at each step of the course. Support workers provide telephone contact during the course.

A number of trials have examined outcomes for *BT Steps* (i.e., not the online version of the program). The largest RCT (N = 218) found that BT Steps (within group Cohen's d = .84) was marginally less effective than clinician-guided behaviour therapy (CGBT, d = 1.22) but more effective than relaxation therapy (RT, d = .35) in improving symptoms of OCD (using the self-rated Yale Brown Obsessive Compulsive Scale, YBOCs) (Greist, Marks, Baer, Kobak, Wenzel, Hirsch, et al., 2002). Estimates of treatment responder status were 38%, 60%, and 14% for BT Steps, CGCBT and RT, respectively, on the Clinical Global Impressions scale. Gains were maintained over 4 months, but only 35% of BT Steps participants took part in 2 or more exposure homework sessions. This completion rate is problematic because it was clearly found that the patients who used the system longer and completed exposure steps achieved the best gains, comparable to the gains made in clinician-guided behaviour therapy (Greist et al., 2002). Smaller studies report similar results and found that brief, scheduled clinician contact throughout the program enhanced outcomes and that using the program shortened the length of subsequent clinician-guided ERP (Bachofen, Nakagawa, Marks, Park, Greist, Baer, et al., 1999; Kenwright, Marks, Graham, Franses and Mataiz-Cols, 2005; Marks, O'Dwyer, Meehan, Greist, Baer, and McGuire, 2000; Nakagawa, Marks, Park, Bachofen, Baer, Dottl et al., 2000). BT Steps' promising outcomes and the recent proliferation of online CBT for anxiety disorders prompted research teams to develop online programs specifically for OCD.

An evidence base has accumulated for three online OCD treatment programs, although there are numerous OCD-related web-based applications available (e.g., online OCD workbooks, exposure monitoring applications, or online self-help groups for compulsive hoarders such as Muroff, Steketee, Himle, and Frost, 2010). The first program, developed by Wootton and colleagues (2011)involves (i) eight online lessons (completed once per week), (ii) written resources (lesson summaries and homework assignments, as well as additional resources such as a handout on managing sleep problems), (iii) twice weekly therapist

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contact, (iv) an online forum, and (v) automated reminder emails. The program comprises psycho-education (lesson 1), cognitive restructuring for cognitive biases (e.g., over importance of thoughts, intolerance of uncertainty, lessons 2-8), ERP (lessons 3-8), and relapse prevention (lesson 8) based on the techniques described in Abramowitz (2006a; 2006b), Bennett-Levy et al. (2004), Kozak and Foa (1997), Rachman (2003), and St Clare, Menzies, and Jones (2008). This program has been subsequently shortened to 5 lessons (completed over 8 weeks) and has also been produced in book format (Wotton, Dear, Johnston, Terides, and Titov, 2013). The second iCBT program from Andersson and colleagues (2011) is completed over 15 weeks. Patients access a 100-page self-help manual based on the treatment protocol of Abramowitz (2009). The manual is divided into 15 modules (later versions give 10 modules available online and via downloadable mp3 files). After each module, patients submit homework exercises and correspond with a therapist over email (and telephone as required). Administrators also contact patients via short mobile text messages. Psycho-education and treatment rationale (modules 1-2) is standardised, but participants can elect to follow one or all four fictional characters who exemplify common subtypes or symptoms of OCD (washing, checking, symmetry, or violent thoughts). The remainder of the course comprises cognitive restructuring for metacognitions (e.g., thought-action fusion, module 3), daily ERP (modules 4-13), establishing values-based goals (module 13), and relapse prevention (14-15). This program was also subsequently shortened to 10 modules completed over 10 weeks (Andersson, Enander, Andrén, Hedman, Ljótsson, Hursti, et al., 2012) with therapists individualising the order of modules 5-10. The third program, *OCD-STOP!*, reported by Klein, Meyer, Austin and Kyrios (2011) is completed over 12 weeks. The program is clinician guided (via email) for those with moderate to severe symptoms or fully automated self-help for those with mild symptoms.*OCD STOP!* comprises CBT-based downloadable lessons, workbook exercises, video and audio files.

Treatment Outcomes

A number of RCTs and open trials have found promising evidence for the effectiveness of iCBT for OCD (see Table 1). Andersson et al. (2012) conducted the first RCT comparing their iCBT program to an attention control condition (online supportive therapy). In the control group patients received short mobile text messages, email and telephone calls from administrators and therapists to discuss the past week's experiences and any distressing events. Patients completed a mean of 7.28 (out of 10) iCBT modules, with six patients classed as non-completers because they did not begin ERP (there were no drop outs in the control condition). iCBT participants improved more than the controls with respect to OCD symptoms (as assessed by the clinician-rated YBOCS, post-treatment between groups d = 1.12). iCBT also led to a moderate pre-post treatment effect size superiority on secondary measures of OCD symptoms (d = .62) and global functioning (d = .57), with a small effect size superiority for reductions in depression symptoms (d = .19). Gains were maintained at 4 months post-treatment. Unfortunately, the sample did not include people reporting severe OCD symptoms (YBOCS > 31), no post-treatment diagnostic assessments were conducted, and the follow-up period was reasonably short. Nevertheless, the study's results were largely consistent with this team's previous small open trial which found that 43% of participants no

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longer met diagnostic criteria for OCD post-treatment (Andersson, Ljótsson, Hedman, Kaldo, Paxling, Andersson et al., 2011).

Table 1. Treatment outcomes of iCBT for adults with OCD

	Trial conditions	Mean clinician input (minutes)	YBOCS pre-post treatment ES (*d*)[a]	Patients achieving clinically significant change	Adherence	Follow-up outcomes
Andersson et al. (2011) Open trial	15 weeks N = 23	92	1.56	61%	87%	-
Andersson et al. (2012) RCT	iCBT vs. OST 10 weeks N = 101	iCBT = 129 OST = 17	iCBT = 1.54 OST = .48 iCBT> OST	iCBT= 60% OST = 6%	iCBT = 88% OST = 100%	Gains maintained at 4 months
Wotton et al. (2011) Open trial	8 weeks N = 22	86	1.53	29%	81%	Some loss of gains at 3 months
Wotton et al. (2013) RCT	iCBT vs. bCBT vs. WL 8 weeks N = 52	iCBT = 88.63 bCBT = 102.73	iCBT = 2.16 bCBT = 1.65 WL = .11 iCBT=bCBT> WL	iCBT = 47% bCBT = 40% WL = 0% iCBT = bCBT	iCBT = 94% bCBT = 85% WL = 95%	Some loss of gains at 3 months
Klein et al. (2011) Open trial	12 weeks N = 17	0	.83[b]	-	63%	-

*Note:*iCBT: internet-based cognitive behaviour therapy; OST = online supportive therapy (attention control condition); bCBT = bibliotherapy-administered cognitive behaviour therapy; WL = waitlist control.

[a]Within group, uncontrolled effect sizes (Cohen's *d*) of the clinician administered YBOCS. Intention-to-treat data were used.

[b]Based on e-PASS (online psychological assessment and referral system) clinical disorder severity rating (not the YBOCS).

A second RCT conducted by Wotton et al. (2013) compared iCBT against bibliotherapy (bCBT, the iCBT program in book format) and a waitlist control group in 52 adults diagnosed with OCD. Completion rates across the therapy conditions were the same with an average number of 4.33 and 4.30 lessons (out of 5) completed in the bCBT and iCBT groups, respectively. Outcomes were not significantly different in the CBT conditions, however, both therapies were associated with a moderate to large pre-post treatment effect size superiority over the waitlist group in terms of reducing OCD symptoms (*d* = .63- 1.57) with a small to moderate effect size superiority in improving symptoms of depression (*d* = .26-.56).There were some losses in gains following the relatively short 3 month follow-up period; participants meeting criteria for clinically significant change fell from 47% at post-treatment to 27% at follow-up for the iCBT group and from 40% to 20% for the bCBT group. These results were comparable to a previous open trial using the 8 lesson iCBT program

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(Wootton,Titov, Dear, Spence, Andrews, Johnston, et al., 2011). Interestingly, when the control group was treated (using iCBT) with less total amount of therapist contact during treatment (about 60% of the time in the previous iCBT and bCBT groups) outcomes remained satisfactory, albeit somewhat weaker (d = 1.11- 1.13 for OCD symptoms, 33% achieving clinically significant change). Like previous studies, this finding emphasises that satisfactory therapeutic outcomes can be achieved by a small amount of clinicians' time. Another interesting finding of the Wootton et al. trial is that follow-up gains, although not directly compared, appeared somewhat more stable in the treated waitlist group who received automated emails during the follow-up period compared to the therapy groups who received no such emails.

Lastly, an open trial by Klein, Meyer, Austin and Kyrios (2011) found initial evidence to suggest that the fully-automated self-help version of *OCD STOP!* was effective in reducing OCD symptoms (d = .83) and improving quality of life (d = .87) with 63% of patients completing the 12 week course. However, the study was very small (N =17) and uncontrolled, and the measures employed to assess patient progress were not widely used or psychometrically established which limits comparisons across other outcome studies.

The outcomes of iCBTappear to be comparable to those attained in traditional face-to-face CBT. In a recent meta-analysis of 16 RCTs (N = 756), Olatunji et al. (2013) found that the mean pre to post treatment effect size superiority of face-to-face CBT compared to control conditions (waitlist or placebo) was 1.39 (Hedges g) on OCD measures (all but one study used the YBOCs) and .51 on depression measures. CBT also outperformed control conditions at follow-up with respect to OCD symptoms (g = .43) (based on 3 RCTs, n = 111). However, direct comparisons between iCBT and face-to-face CBT are yet to be made and consequently conclusions remain preliminary. Nevertheless, it seems clear that iCBT requires much less therapist time to administer. Across RCTs, participants completing face-to-face CBT in the Olatunji et al. meta-analysis received 5-23 treatment sessions (each session lasting an hour on average), whereas iCBT programs required only 1-2 therapist hours per patient across the full course of treatment.

Acceptability and Cost Effectiveness

The treatment trials conducted by Wootton and colleagues evaluated patient satisfaction. In Wootton, Titov, Dear, Spence, Andrews, Johnston et al. (2011) all participants reported that they were either very satisfied or mostly satisfied with their iCBT program, and that they would recommend the program to a friend with OCD. Similarly in Wotton et al. (2013) 70% of participants were either extremely satisfied or satisfied, while 30% reported feeling neutral about their iCBT program. In an online survey-based study examining attitudes towards internet-based treatment for OCD, Wootton, Titov, Dear, Spence and Kemp (2011) found that most people with symptoms of OCD reported that they would be willing to try iCBT and expected that they would benefit from it. They saw the advantages of iCBT as convenience (no need to travel to appointments, time saving), reduced costs, and increased privacy. About a third of the sample also thought that iCBT would be less embarrassing than face-to-face treatment. Respondents did not report many disadvantages for iCBT, although some (7-10%) reported that they would prefer face-to-face therapy, or felt their problems were too severe for

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iCBT or that they would not be able to communicate their ideas online. This internet sample shared similar demographic features and severity of OCD symptoms to a hospital outpatient sample of adults diagnosed with OCD. Like studies examining iCBT for other disorders, these findings suggest that the majority of people with OCD find iCBT an acceptable form of treatment.

The cost-effectiveness of iCBT for OCD has not been well studied. Preliminary cost estimates of *BT Steps* suggested that it is likely to be a cost-effective alternative to clinician-guided behaviour therapy (McCrone, Marks, Greist, Baer, Kobak, Wenzel, et al., 2007), however, no such comparisons have been published for contemporary iCBT programs. It is possible to extrapolate cost estimates from the amount of therapist time involved in administering iCBT compared to traditional CBT (i.e., 1-2 hours vs. 5-23 hours, respectively), although this does not take into account the significant costs involved to developing and maintaining online programs. Further research is clearly required.

Clinical Advantages and Disadvantages

There are multiple advantages for patients and clinicians who use iCBT. The Clinical Research Unit for Anxiety and Depression, which is where the present authors work, operates a specialist anxiety clinic in a large metropolitan hospital. The Clinic receives 400 new patient referrals per year, which used to mean that the staff could not keep up with the demand for CBT and that there was always a long waiting list. However, three years ago we began to offer iCBT to patients as a routine first step of care. Presently, nineteen out of twenty patients elect to do iCBT at home rather than come into the clinic for face-to-face CBT. Adherence is above 70% with benefits comparable to face-to-face CBT for the majority of patients. Importantly, we no longer have a waiting list, and referring clinicians and their patients are pleased. Clinicians now have more time to spend with the complex and difficult-to-get-better patients, particularly those with OCD. For some patients iCBT is used as an adjunct to face-to-face treatment. The iCBT course provides standardised psycho-education and foundational CBT skills which allows clinicians to focus sessions on more challenging tasks like ERP. Clinicians also report that iCBT for OCD has improved treatment integrity, reduced therapist drift, and helped limit therapy-interfering behaviours such as excessive reassurance-seeking or prolonging unproductive therapy.Trainee clinicians, in particular, have benefited from adhering to the highly structured, time-limited approach employed in iCBT. Advantages to iCBT are also evident in patient feedback:

> ‘I am definitely feeling an improvement in some areas of my life already. I really wish this program had been around many, many years ago. Interestingly, some of Bella's [the iCBT character with OCD] issues are identical to mine! Thank you for including me in the program, I look forward to continuing and improving my symptoms.’ Jean, 43

> ‘The course material I have read has made me aware how caught up I was in the vicious cycle and I felt I was spiralling on a downwards path. It is a scary feeling. I feel I am now taking control and I'm equipping myself with awareness, knowledge and understanding to enable me to avoid slipping back. Thanks for your support, it means a lot to me to know that help is not far away.’ Paul, 36

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However, other patients have found it difficult to apply the iCBT content and struggled to relate to the depictions of OCD. This is understandable given the heterogeneity of OCD and the difficulties inherent in ERP. Some patients also report feeling disconnected from their clinician because they are not attending face-to-face appointments. In these cases, our clinic provides face-to-face treatment to supplement or replace iCBT.

Future Research

The body of literature concerning iCBT for OCD is small and there are consequently many avenues for future research and development. The longer-term (6-24 months) therapeutic benefits of existing iCBT programs need to be evaluated and maximised as some loss of gains during follow-up is evident in both iCBT and face-to-face CBT. RCTs directly comparing traditional CBT and iCBT are needed, as are RCTs evaluating combined treatments. Importantly, treatment programs need to be evaluated by teams of researchers who are independent to the development of such programs. Furthermore, large scale effectiveness studies, such as Mewton, Wong and Andrews (2012), are essential to examine if iCBT for OCD can be used as part of routine clinical care. It is yet to be determined if specialist clinicians are needed to administer iCBT programs or if technicians under supervision, nurses, or general physicians can adequately provide iCBT. Moreover, the use of iCBT need not be confined to adult populations; more research is required to examine its utility in pediatric and geriatric populations.

Conclusion

The evidence is mounting: iCBT for OCD is efficacious and acceptable. It helps overcome the multiple barriers clinicians and patients face in traditional treatment, and only requires a tenth of the time needed to administer face-to-face CBT. When patients receive regular consultation and support from clinicians and complete necessary ERP tasks, iCBTmay be as beneficial as face-to-face CBT.

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Chapter VIII

Obsessive-Compulsive Love: Phenomenology, Correlates and Psychosocial Impact

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Abstract

The construct of obsessive-compulsive love has not been clearly established despite its widespread presence in media outlets such as film, music, and literature. To help clarify this construct and how it affects individuals, this study investigates the phenomenology of obsessive-compulsive love as well as its association with various emotional states, impairments, and relationship conflict. Participants (N = 445) completed measures assessing obsessive-compulsive symptoms, emotional dysregulation, anger, stress, anxiety, depression, loneliness, psychosocial functioning, and relationship conflict with other-sex peers. Additionally, they completed a newly developed measure of obsessive-compulsive love. Results suggest obsessive-compulsive love is relatively common and functionally impairing. In addition, it is associated with a host of difficulties, such as obsessive-compulsive symptoms, emotional dysregulation, anger, stress, anxiety, depression, and loneliness. Lastly, obsessive-compulsive love was found to impact conflict in other-sex relationships indirectly through the influence of emotional dysregulation and anger. Collectively, these results suggest that obsessive-compulsive love is relatively pervasive, personally distressing, and problematic for relationships.

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Keywords: Obsessive-compulsive love, obsessive love, limerence, passionate love, emotional dysregulation, anger

1. Introduction

The drive to establish loving relationships can contribute to obsessive thoughts about individuals who potentially could reciprocate feelings of affection, especially when this drive is unsatisfied (Acevedo & Aron, 2009). A number of researchers and theorists have described these obsessions over the past 50 years. For example, Berscheid and Hatfield (1969) dichotomized love into "passionate" and "companionate" types in their pioneering work on love and romantic relationships.

They suggested that passionate love involves experiencing an intense state of infatuation or longing to connect with another individual that can result in obsessive thoughts, intolerance of uncertainty, and emotional dysregulation whereas companionate love involves passion infused with attachment, commitment, and intimacy.

Similarly, Lee (1977) devised six different types of love (two of which included obsessive qualities) while working to advance research on love and romantic relationships. According to Lee, "Eros" or romantic love involves an intense focus and desire for union with a beloved person without obsession whereas "Mania" involves feeling jealous, full of doubt about the partner's sincerity, unsure about their commitment, and a palpable affective reaction (e.g., experiencing physical symptoms such as inability to sleep or eat, acute excitement, depression). Thus, Eros and Mania in combination can result in passionate or obsessive love as previously described by Berscheid and Hatfield.

Consistent with previous research and scholarship on passionate love, Hatfield and Sprecher (1986) developed the Passionate Love Scale (PLS), which includes items that measure Eros, Mania, and overlapping aspects of obsessive love (e.g., "Sometimes I feel I can't control my thoughts; they are obsessively on my partner;" "I sometimes find it difficult to concentrate on work because thoughts of my partner occupy my mind"). Research with the PLS suggests that obsessive love is relatively distinct from other forms of love (e.g., compassionate, romantic) as well as independent of the influence of gender, length of relationship, and the type of statistical analysis used (principal components analysis, principal axis analysis) (Aron, Norman, Aron, McKenna, & Heyman, 2000).

As a third major contributor to scholarship on obsessive love, Tennov (1979) coined the term "limerence" to describe a phenomenon that involves obsessive love and unrequited desire for another person. According to Tennov, limerence is a multicomponent cognitive and emotional state of attachment to a partner or potential partner. First, in all limerence cases, strong interest develops for another person that becomes heightened if the other person (i.e., the "limerent object") reciprocates (or is perceived to reciprocate) this interest. Second, obsessive thoughts develop about being in a relationship with the limerent object if he or she does not reciprocate a similar desire. Third, the person who desires to be in a relationship may experience an emotional high from being near and/or possibly encountering the desired person; conversely, the affected person also may experience psychological suffering if he or she is prevented from seeing the desired person.

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Lastly, the person who desires to be in a relationship generally perceives the desired person to be near perfect and is blind to flaws he or she may display. Although interesting and well-articulated in case studies empirical research on limerance is relatively limited (Tennov, 1999).

1.1. Obsessive-Compulsive Love

Extant theories of passionate love, Eros, Mania, and limerence include an obsessive component. However, they are relatively silent on behaviors that individuals display in response to love-related obsessions or due to the negative emotional states associated with these intrusive thoughts. Individuals who obsess over a current or desired relationship may experience distressing emotions (e.g., anxiety, frustration, anger) if they perceive that there is a problem in the relationship or the other person does not reciprocate their feelings of affection. In response to these negative emotions, affected individuals may develop compulsive behaviors that temporarily mitigate the distressing emotional state. For example, a person who obsesses over what a desired partner thinks about him or her (e.g., "does he like me?") may perform compulsions that temporarily reduce his or her anxiety (e.g., repeatedly asking for reassurance about being liked). However, the effectiveness of these distress reducing compulsions is short lived and these behaviors ultimately may frustrate or upset the desired partner if they are performed in excess (Cupach & Spitzberg, 2004).

The relationship between obsessive thoughts (i.e., intrusive and distressful thoughts) about partners/relationships and the performance of associated compulsive acts (i.e., rituals performed to mitigate anxiety/distress) parallels a cognitive-behavioral process that occurs in individuals with obsessive-compulsive disorder (OCD) (Abramowitz et al., 2010). Obsessive-compulsive disorder is a relatively common psychiatric disorder that is characterized by the presence of intrusive and anxiety-provoking thoughts (obsessions) and performance of behaviors that temporarily reduce anxiety (compulsions) (American Psychiatric Association, 2000). Although different in symptom content from OCD, obsessions and compulsions associated with the desire to experience and/or maintain a loving relationship may be functionally similar to OCD-related obsessions and compulsions. However, although distinct in its own right, this phenomenon—the phenomenon of obsessive-compulsive love—also may cause significant distress and functional impairments in affected individuals.

In addition to being personally distressing, obsessive-compulsive love may be associated with a variety of cognitive and affective variables that impact psychosocial functioning. For example, individuals whose romantic gestures or compulsions are frustrated (i.e., not reciprocated or appreciated) may feel upset, angry, lonely, stressed, anxious, or depressed (Cupach & Spitzberg, 2004; La Greca & Harrison, 2005). Moreover, obsessive-compulsive love may negatively affect relationships because the potentially upsetting nature of these behaviors, their association with negative affective states (e.g., anger, emotional dysregulation), and how these states may affect potential partners (Aron, et al., 2000). However, despite the distress that obsessive-compulsive love may cause, its potentially deleterious effects on relationships, and its widespread presence in film, music, and literature, obsessive-compulsive love has received little empirical attention (Cupach & Spitzberg, 2004; Moore, 2006).

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1.2. The Current Study

This study is a preliminary investigation on the phenomenology of obsessive-compulsive love in young adults. Additionally, the relationships between obsessive-compulsive love and anger, emotional dysregulation, stress, anxiety, depression, loneliness, obsessive-compulsive symptomology, and functional impairment were examined. Lastly, because of the established relationship between anger and emotional dysregulation and relationship problems (Ellis & Malamuth, 2000; Kim, Pears, Capaldi, & Owen, 2009), we tested the influence of obsessive-compulsive love on relationship conflict through anger and emotional dysregulation. We anticipated finding positive relationships between obsessive-compulsive love and anger, emotional dysregulation, stress, anxiety, depression, loneliness, obsessive-compulsive symptomology, and functional impairment. We also expected that the direct and indirect effects of obsessive-compulsive love would impact relationship conflict through the influence of anger and emotional dysregulation. However, no specific hypotheses were specified because of the preliminary nature of this investigation.

2. Method

2.1. Participants

Participants (N = 445) were enrolled in undergraduate psychology courses at a large university located in the southern United States. The majority of study participants (78%; N = 348) were female. Participants ages ranged from 18 to 56 years (M = 21.63, SD = 4.99). Twenty-three percent of participants were college freshmen, 20% were sophomores, 23% were juniors, and 34% were seniors. Fifty-one percent of participants listed their ethnicity as White/Caucasian, 20% as Hispanic/Latino, 14% as Black/African American, 8% as Asian, and Mixed/Biracial.

2.2. Measures

2.2.1. Obsessive-Compulsive Love Scale (OCLS)

The OCLS was developed to assess obsessive-compulsive love. The OCLS includes seven items that were selected following content and principal component analysis with data obtained from an unpublished pilot study by Grabill, Storch, Clemans, and Graber (2010). Only items with high factor loadings (λ = .60) were retained when a one-factor solution was imposed. Participants rated the degree to which each item on the OCLS affected them on a 4-point scale ranging from "not at all" to "severely." Results of the current study support the psychometrics of the OCLS. The measure displayed an adequate single-factor solution, $\chi^2(1, N = 445) = 59.77$, $p < .01$, CFI = .96, SRMR = .04, and RMSEA = .09 (95% CI = .06 - .11), and moderately high factor loadings were found for all scale items (λ = .61 to .74).

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2.2.2. Sheehan Disability Scale (SDS; Sheehan, 1983)

The SDS is a 3-item self-report questionnaire that aims to assess impairment in academic/work, social life or leisure, and family/home functioning due to psychopathology or psychiatric symptoms. The SDS is rated on a 10-point scale that ranges from "little or no" impairment to "extremely" impaired. The SDS displays good internal consistency (α = .89), sensitivity, and specificity (Sheehan, Harnett-Sheehan, & Raj, 1996).

2.2.3. Obsessive Compulsive Inventory-Revised (OCI-R; Foa et al., 2002)

The OCI-R is an 18-item, self-report inventory of obsessive-compulsive symptoms. Items are rated on a 5-point scale and higher scores indicate greater levels of distress due to symptoms. Acceptable test-retest reliability (r = .67 - .70) and good internal consistency (α = .86 - .88) values have been found for the OCI-R (Foa et al., 2002).

2.2.4. Depression Anxiety Stress Scales-21 (DASS-21; Lovibond and Lovibond, 1995)

The DASS-21 is a 21-item self-report instrument designed to measure the three related negative emotional states of depression, anxiety, and tension/stress. Each of the three DASS-21 subscales contains 7 items. A three factor solution has been found for the DASS-21 and subscales display good to excellent internal consistency (α = .80 - .91) (Sinclair et al., 2011).

2.2.5. UCLA Loneliness Scale, Version 3 (UCLALS-3; Russell, 1996)

The UCLALS-3 is a 20-item measure of subjective feelings of loneliness or social isolation. The measure is rated on a 4-point scale with higher scores indicative of higher levels of loneliness. The UCLALS-3 displays good to excellent internal consistency (αs = .89 - .94) and good test-retest reliability (r = .73) (Russell, 1996).

2.2.6. Clinical Anger Scale (CAS; Snell, Gum, Shuck, Mosley, and Kite, 1995)

The CAS is a 21-item self-report measure of anger. The CAS is rated on a 4-point scale and elevated total scores indicate clinically significant anger. The CAS displays excellent internal consistency (α = .94) and a single-factor structure (Snell, et al., 1995).

2.2.7. Difficulties in Emotion Regulation Scale (DERS; Gratz and Roemer, 2004)

The DERS is a 36-item measure that is designed to assess multiple aspects of emotion dysregulation (e.g., lack of emotional awareness, poor emotional regulation, and lack of emotional clarity). Items are rated on a 5-point scale with higher scores suggesting higher levels of emotional dysregulation. The DERS displays excellent internal consistency (α = .93) and test-retest reliability (r = .88).

2.2.8. Network of Relationships Inventory, Revised (NRI-R; Furman & Buhrmester, 1985)

The NRI-R assess the quality of relationships with significant others. The measure includes 45 items that are rated on a 5-point scale. In lieu of including all NRI-R items, the current study included the 3 items that are subsumed under the NRI-R Conflict subscale,

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which assesses conflict in relationships with other-sex individuals. The NRI-R Conflict scale displays good internal consistency (αs = .76 - .89) (Furman & Buhrmester, 1985, 2009).

2.3. Procedure

Students taking psychology classes were offered the option to participate in this study to meet course requirements for research participation. All survey items and questionnaires were entered into Checkbox (an online survey program) through a secure participant access portal. Participants were provided with study details through the Checkbox platform and they indicated their consent through selecting a box indicating their agreement with all study procedures. A university institutional review board approved all study procedures.

2.4. Data Analysis

Descriptive statistics, single-order correlations, and linear regression analyses were calculated to examine the phenomenology of obsessive-compulsive love and assess relationships among study variables. Mplus v. 6.11 (Muthén & Muthén, 1998-2011) was used to conduct all structural regression analyses between observed variables. Maximum likelihood (ML) estimation was used to specify and test all models because endogenous variable distributions were multivariate normal and study variables had few missing data (< 3%). Model fit was assessed using the chi-square test of independence, the comparative fit index (CFI), the standardized root mean square residual (SRMR), and the root mean square error of approximation (RMSEA). Competing models were compared using the Akaike information criterion (AIC). Lastly, model trimming was guided by theoretical and statistical criteria (i.e., non-significant structural regression paths were constrained to zero) with a goal of finding a parsimonious and explicable model with adequate data fit.

3. Results

Table 1 presents means, standard deviations, and percentages of participants who endorsed each OCLS item. The majority (64%) of students reported that their life or relationships were at least “slightly” impacted by obsessive-compulsive love symptoms. Specifically, 36% reported no interference in functioning, 36% reported slight interference in functioning, 23% reported moderate interference, and 6% reported severe interference in functioning related to obsessive-compulsive symptoms.

Table 2 presents means, standard deviations, minimum and maximum values, reliability coefficients, and intercorrelations among study variables. Relationships between demographic variables and OCLS scores were investigated prior to testing associations among inferential variables. No significant associations were identified between OCLS scores and participants’ sex, race/ethnicity, grade level, relationship status, number of previous relationships, sexually active status, and number of previous sexual partners.

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Table 1. Means, standard deviations, and percentages of participants who endorsed each OCLS item

Items	*M*	*SD*	Not at all	Slightly	Moderately	Severely
1. I feel that I can't control my thoughts about someone with whom I desire to be in a relationship	2.00	.87	33%	38%	25%	4%
2. I call, text, or message someone who I desire to be in a relationship with to check on them	1.87	.87	42%	33%	23%	3%
3. I repeatedly think about my relationship/friendship with someone with whom I desire to be in a relationship	1.97	.88	36%	37%	23%	5%
4. I ask the person with whom I desire to be in a relationship about how they feel about me even when I know the answer	1.86	.89	43%	33%	19%	5%
5. I worry about how the person with whom I desire to be in a relationship feels about me, so I repeatedly tell them how I feel about them	1.80	.88	46%	32%	18%	4%
6. I feel jealous when the person I desire to be in a relationship with looks at other potential partners	2.25	.96	25%	38%	26%	12%
7. I think about the person with whom I desire to be in a relationship when I feel like I should be doing something else	2.09	.87	29%	39%	27%	5%

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Table 2. Means, standard deviations, minimum and maximum values, reliability coefficients, and intercorrelations among study variables

Measures	*M*	*SD*	Minimum	Maximum	α	*r*
OCLS Total	13.84	4.52	7	27	.85	--
DERS Total	93.59	17.08	53	161	.87	.41**
CAS Total	9.21	9.44	0	42	.93	.39**
UCLALS Total	44.33	11.08	20	79	.93	.27**
SDS Total	7.95	6.34	0	27	.84	.64**
DASS-21 Stress	10.85	8.40	0	34	.84	.32**
DASS-21 Anxiety	7.27	7.90	0	34	.84	.35**
DASS-21 Depression	8.82	9.19	0	42	.89	.36**
OCI-R Total	17.59	11.89	0	55	.91	.37**
NRI-R Conflict	5.37	2.55	3	15	.92	.25**

Note: OCLS = Obsessive-Compulsive Love Scale, DERS = Difficulties in Emotion Regulation Scale, CAS = Clinical Anger Scale, UCLALS = UCLA Loneliness Scale, SDS = Sheehan Disability Scale, DASS-21 = Depression Anxiety Stress Scales-21, OCI-R = Obsessive-Compulsive Inventory-Revised, NRI-R Conflict = Network of Relationships Inventory, Revised Other-Sex Conflict Subscale, α = Chronbach's alpha, *r* = Pearson correlations with OCLS total score, * $p < .05$, ** $p < .01$.

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Table 3. Effects decomposition for study variables

	Endogenous Variables							
	NRI-R Conflict						CAS	DERS
Exogenous Variables	Total	Direct	Total Indirect	Indirect through CAS	Indirect through DERS	Indirect through DERS and CAS	Total	Total
Model 1								
CAS Total	.41**	.41**	---	---	---	---	---	---
DERS Total	.03	.03	---	---	---	---	---	---
OCLS Total	.24**	.07	.17**	.16**	.01	---	.39**	.41**
Model 2								
CAS Total	.45**	.45**	---	---	---	---	---	---
OCLS Total	.10**	---	.10**	.05**	---	.05**	.21**	.41**

Note: OCLS = Obsessive-Compulsive Love Scale, DERS = Difficulties in Emotion Regulation Scale, CAS = Clinical Anger Scale, UCLALS = UCLA Loneliness Scale, SDS = Sheehan Disability Scale, DASS-21 = Depression Anxiety Stress Scales-21, OCI-R = Obsessive-Compulsive Inventory-Revised, NRI-R Conflict = Network of Relationships Inventory, Revised Other-Sex Conflict Subscale; the symbol "---" means the effect is not in the model; * $p < .05$, ** $p < .01$

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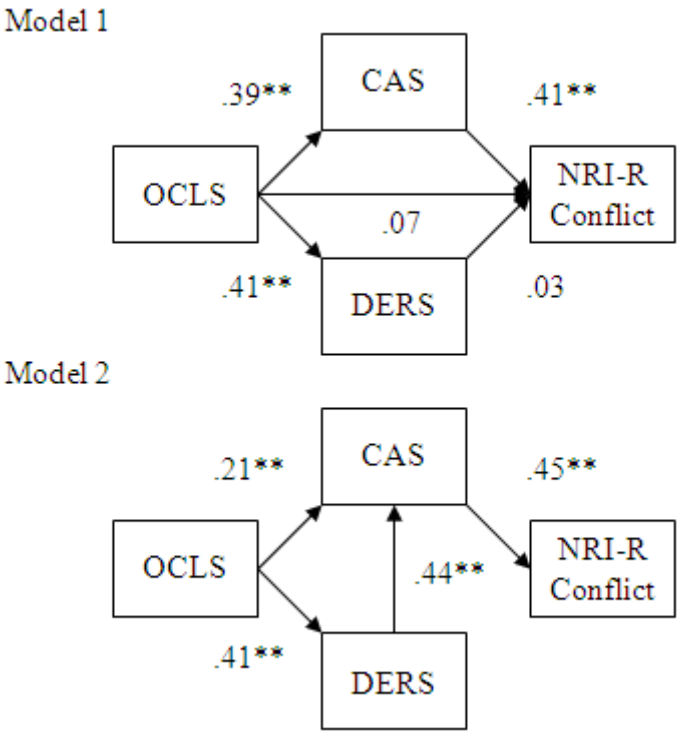


Figure 1. Structural relationships among variables of interest. *Note*: OCL= Obsessive-Compulsive Love Scale; CAS=Clinical Anger Scale; DERS=Difficulties in Emotion Regulation Scale; NRI-R=Network of Relationships Inventory, Revised Conflict Scale; *p<.05, ** p<.01.

3.1. Structural Regression Models

Table 3 presents total, direct, and indirect effects for study variables across different models. Figure 1 displays associations among OCLS, CAS, DERS, and NRI-R Conflict scores for each tested model. A structural regression model (Model 1) was specified to test the direct effects of OCLS scores on NRI-R Conflict scores as well as the indirect effects of OCLS scores on NRI-R Conflict scores through the influence of CAS and DERS scores. Overall, this model displayed poor data fit, $\chi^2(1, N = 439) = 90.95$, $p < .01$, CFI = .73, SRMR = .11, RMSEA = .45 (CI = .38 - .53), AIC = 8643.02.

Regarding structural relations, significant direct and indirect effects of CAS scores on NRI-R Conflict scores were observed as anticipated. However, contrary to expectations, no association was observed between DERS and NRI-R Conflict scores. Additionally, despite finding a significant single-order correlation between OCLS and NRI-R Conflict scores (r = .25), no direct relationship was observed between these measures.

A second structural regression model (Model 2) was tested to investigate a second indirect pathway between OCLS scores and NRI-R scores through the influence of DERS and CAS scores. Individuals with emotional regulation problems likely engage in angry/aggressive behaviors that can lead to conflict in social interactions (Cupach & Olson, 2006) and contribute to romantic relationship problems (Kim et al., 2009). Therefore, when

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under duress, individuals with emotion regulation difficulties may feel overwhelmed and express anger or threatening emotion, which in turn, may contribute to relationship conflict. The non-significant paths between OCLS and NRI-R Conflict scores as well as non-significant paths between DERS and NRI-R Conflict scores were constrained to zero. Model 2 displayed excellent data fit, $\chi^2(1, N = 439) = 2.62$, $p = .27$, CFI = .99, SRMR = .02, RMSEA = .03 (95% CI = .00 - .10), AIC = 8552.69. Thus, compared to Model 1, Model 2 displayed superior data fit as evidenced by a non-significant chi-square fit test value and a lower identified AIC value. In Model 2, significant indirect effects of OCLS scores on NRI-R Conflict scores were found through the influence of CAS scores as well as through the influence of DERS scores on CAS scores.

4. Discussion

Results of this study suggest that obsessive-compulsive love is relatively common in young adults regardless of their gender, race/ethnicity, relationship status, sexual behavior, number of previous relationships, or number of previous sexual partners. In this vein, almost two thirds of individuals reported experiencing at least slight interference in their life or relationships and almost a third reported experiencing moderate or severe interference in their life because of their obsessive-compulsive love symptoms. Therefore, obsessive-compulsive love appears to affect a range of young adults with diverse backgrounds and social experience.

In addition to being personally distressing, obsessive-compulsive love also co-occurs with symptoms of obsessive-compulsive symptomology, emotional dysregulation, anger, stress, anxiety, loneliness, and depression. Furthermore, it also appears to impact other-sex relationships. Although no direct association was identified between obsessive-compulsive love and relationship conflict in the tested structural models, these variables were related through the influence of emotional dysregulation and anger. Therefore, similar to an "emotional cascade" model of negative affect in which intense rumination contributes to emotional dysregulation (Selby, Anestis, & Joiner, 2008), obsessive-compulsive love may overwhelm individuals' emotional regulation capacities, which can then motivate them to find alternative outlets to vent their negative emotions. Thus, obsessive-compulsive love appears to contribute to relationship conflict through various pathways. It may directly imbue feelings of anger and/or contribute to emotional dysregulation that then compounds angry feelings that may be directed at the other person. Either way, the co-occurrence of obsessive-compulsive love and anger may be particularly deleterious to relationships. By itself, anger is highly caustic to relationships (Baron et al., 2007) and it also is predictive of aggression and relationship dissatisfaction in romantic partners (Ellis & Malamuth, 2000). Therefore, partners or potential partners may feel particularly threatened by individuals who fixate on them and display strong and potentially hostile negative emotions.

The relatively high numbers of young adults affected by obsessive-compulsive love in the current study may be surprising because of the deleterious effects it appears to exert on individuals and relationships. However, universal and adaptive pair-bonding processes may undergird this phenomenon (Graham, 2011). For example, although problematic for developing mutually satisfying long-term relationships, the mechanisms behind obsessive-

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compulsive love (e.g., a need for bonding, sexual desire) may help to foster relationships that ultimately lead to copulation (Hazan & Diamond, 2000). In support of this notion, a recent meta-analysis by Graham (2011) found romantic obsession to be related to experiencing short—yet not long-term relationships—and Acevedo and Aron (2009) found that passionate love (i.e., love with obsession) is associated with satisfaction in new relationships and dissatisfaction in long-term relationships. Therefore, obsessive-compulsive love may be adaptive in circumstances in which the need to foster progeny takes precedence over the development of long-term relationships. In this regard, obsessions about a desired partner may motivate affected individuals to endure adverse conditions and take extreme measures to bond with a desired partner. Recent findings from neuroimaging studies seem to support this notion as experiencing passionate and obsessive love has been associated with distinct activation in reward centers of the brain that correspond with highly motivational and goal-directed behavior. Specifically, experiencing passionate and obsessive love has been linked to hyperactivity in the right ventral tegmental area, right caudate nucleus, posterior cingulate, and posterior hippocampus (Acevedo, Aron, Fisher, & Brown, 2012; Aron et al., 2005).

Furthermore, in addition to orienting individuals toward potential partners, obsessive-compulsive love also may help preserve unstable or tenuous relationships. For example, the compulsive behaviors (e.g., checking, reassurance seeking) performed to neutralize distress associated with obsessive love may help to guarantee the desired partner's fidelity in a manner that is similar to the process of mate guarding, which involves fending off other potential mates and ensuring that a partner does not defect (Buss, 2002). Thus, obsessive-compulsive love may have its greatest utility when the need to procreate and prevent cuckoldry usurps the need to establish long-term companionship.

Results of the current study should be interpreted in the context of several limitations. First, female college students were oversampled and results many not generalize to dissimilar populations. Second, relationship conflict as assessed on the NRI-R Conflict scale measured conflict with other-sex individuals only. Therefore, results of this study may not apply to lesbian, gay, bisexual, transgender, or questioning students whose preference for partners and relationships may differ in meaningful ways. Third, a cross-sectional study design prevents the determination of temporal relationships among study variables. Fourth, the current study may have been affected by a monomethod bias. Fifth, although the use of observed variable structural regression modeling allows for model and relationship testing, alternative models may exist that better explain relationships among study variables. Sixth, because of the preliminary nature of this investigation, important variables related to obsessive-compulsive love may have been omitted in this study and may warrant inclusion in future research. Lastly, students' self-reports were assessed as opposed to their actual behaviors.

Despite these limitations, the results of this study provide preliminary support for the construct of obsessive-compulsive love and illustrate how it negatively influences individuals and relationships. Specifically, results of this study suggest that obsessive-compulsive love is a relatively common phenomenon that is associated with functional impairment and indirectly related to relationship conflict as well as a range of negative affective states and behavior problems.

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Chapter IX

The Importance of Quality of Life in Patients with OCD

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Abstract

Quality of life (QOL) assessment and improvement are an increasingly important aspect of health care, particularly in the field of mental health. QOL in those with OCD is significantly impaired when compared to that of the general population and of patients with other psychiatric and medical disorders. Therefore, it is important to not only address the relentless symptoms but also this populations' overall well-being. Likewise, QOL in OCD appears to be largely affected by comorbid conditions, suggesting that practitioners should take comorbid disorders into account during treatment planning. While QOL in OCD can improve with medications and with both individual and group psychotherapy, it does not seem to improve to the levels enjoyed by the community. Thus, QOL assessment in both clinical and research settings should be assessed at baseline and during follow up so that we can examine the disease burden, monitor treatment efficacy, and strive for true relief. Thus, treatment providers should endeavor to reach symptom remission while also striving for improvements in functioning and quality of life.

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1. Introduction

Symptoms of Obsessive-compulsive disorder can lead to great anguish for patients, particularly when it is an impediment to accomplishing occupational, social, and recreational aspirations. Symptom relief is the first step but quality of Life must also improve in order to afford patients the ability to enjoy work, love, and play. As argued in a review of QOL and depression [1], QOL can be considered the ultimate indicator that treatments have succeeded.

The World Health Organization (WHO) defines quality of life (QOL) as the "individuals' perception of their position in life in the context of the culture and value systems in which they live and in relation to their goals, expectations, standards and concerns [2]." Quality of life is further identified as one's satisfaction with health, and with occupational, social, and other activities according to self-report[1]. The most frequently cited QOL measures in the literature are the Medical Outcomes Study Short Form-36 (SF-36) [3, 4]; the World Health Organization Quality of Life Assessment (WHOQOL-100)[5], and its brief form: the World Health Organization Quality of Life Assessment- Abbreviated Version (WHOQOL-BREF)[6]; and the Quality of Life Enjoyment and Satisfaction Questionnaire (Q-LES-Q) [7].

QOL impairment in OCD has been shown to be higher than QOL impairment in panic disorder [8], social phobia [8], heroin dependence[9], and in patients on hemodialysis[9] and kidney transplant recipients [9]. It has been comparable with QOL impairment in schizophrenia and depressive disorders [8, 10]. Moritz et al. [11] compared QOL of OCD patients to QOL in a healthy sample of the population, and z-scores for QOL in OCD patients were 2-4 standard deviations lower than the z-scores for the healthy sub-sample.

2. Overview

2.1. QOL Domains and OCD Symptoms

QOL can be divided into several domains. Generally, deficits in OCD are more pronounced on the social, emotional, and mental domains of QOL measures than on the physical domains. In Bobes et al. [9] the QOL domains most significantly impaired were first social functioning, then role limitations due to emotional impairment (role-emotional), mental health, general health, and vitality. Physical functioning, role limitations due to physical impairments (role-physical), and bodily pain were not significantly different, nor were the physical health summary measures. Moritz et al.[11] found that the domains most affected in OCD participants were social functioning, mental health, vitality, and role limitations due to emotional problems. The scores were below the 25th percentile of the normative sample. However, Moritz et al. also found that QOL impairment due to OCD can also extend to somatic domains, particularly in those patients with compulsions and long illness duration [11].

The evidence is mixed regarding whether patients' symptom severity is predominantly obsessive versus compulsive differentially affect QOL. Stengler-Wenze [12] found compulsion severity to be associated with QOL impairment on WHOQOL-BREF on three subscales whereas *obsession* severity was not. Conversely, in Gururaj et al. [13], YBOCS total symptom severity and *obsession* severity were significantly associated with QOL

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impairment but compulsion severity was not. Other studies have found both compulsions and obsessions to be significantly related to several QOL subscales, but with a difference in which, compulsions or obsessions, had a stronger association; Moritz et al. [11] found a stronger association with compulsions in SF-36 subscales of QOL and Eisen et al. [14] found stronger associations with regards to obsessions in seven Q-LES-Q categories. The lack of consensus suggests additional studies are needed to better differentiate the effects of obsession and compulsion symptom severity on QOL, as this might influence the management of OCD symptoms.

Additional studies are also needed to integrate these results by investigating the differential effects of OCD symptoms on different QOL domains and vice versa in order to allow practitioners the ability to personalize a patient's treatment plan to their particular symptoms and thus QOL and vice versa.

Other modifiers of patients' symptoms and QOL domains should be studied further and considered by practitioners as well. Low social support is indicative of a low QOL [15]. Fewer years of education, higher HAM-D scores, higher YBOCS scores, and more contamination/washing symptoms can predict impaired physical QOL [16]. Higher YBOCS compulsion scores, the presence of a comorbid mood disorder, and higher anxiety scores can predict impaired QOL in the mental health domains. Ideally, starting at the initial contact with the patient, the provider should be considering such predictors. By identifying these predictors, physicians could target these predictors and thus tailor treatment to the patient's individual needs [17-19].

2.2. Impact of Comorbidity

It appears that among patients with OCD, those with comorbidities have greater QOL impairment than those without comorbidity [20] and the further QOL impairment can be attributed to comorbidity [13]. Unfortunately, this is pertinent to a large proportion of patients with OCD. Surveys indicate that at least 50% of OCD patients have at least one other psychiatric disorder, most commonly another anxiety disorder [21]. Over a third suffer from anxiety symptoms, over a third suffer from depressed mood, and about a third meet diagnostic criteria for major depressive disorder [22]. Other common comorbid conditions are body dysmorphic disorder [28] and impulse control disorders [23].

Two studies in particular speak to this impact of psychiatric comorbidity. Gururaj et al. [13] found correlations between social and family QOL with OCD symptoms dissolved after controlling for depression symptom severity using the BDI. Besiroglu et al. [24] found that in individuals suffering from OCD and subsequent MDD, the severity of depressive symptoms contributed more than the severity of OCD symptoms to impairment of QOL. In addition, the effects of medical comorbidity contributed to poorer QOL in Rodriguez-Salgado et al. [25] and Koran et al. [26]. Given the substantial impact of comorbid conditions and their prevalence, it is paramount that treatment strategies aim to achieve remission in these conditions as well as OCD.

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3. Treatment

Standardized measurements of QOL should be used before starting treatment and then monitored over the course of the treatment, whether the treatment is pharmacotherapy, psychotherapy, or a combination of the two. They should be used in research settings as well so that the effect of various treatments on QOL can be better understood and applied.

3.1. Pharmacotherapy

QOL improves in treatment studies using oral fluvoxamine [27], escitalopram [28, 29], paroxetine [30], sertraline [30], IV citalopram [31], and clomipramine [32, 33]. Research should continue to investigate how the timing and duration of pharmacotherapy treatments promote the intended improvement of QOL to apply such findings to patient care so that patients can be treated past mere symptom reduction and toward truly regaining satisfaction with their health. Koran et al. in a 54-week trial [30] showed little to no QOL improvement in studies checking after 12 weeks of treatment, however after 40 to 52 weeks, mental health and role emotional problems were improved [27]. This strongly indicates that it is important for practitioners to consider monitoring QOL over an extended period beyond symptom improvement. Practitioners should also keep in mind, when considering prolonged treatment courses, that adverse side effects themselves can have a negative effect on QOL [15] and treatment options should be discussed with the patient after careful consideration of risks and benefits.

Considering research has also shown that some patients do not experience improvement in QOL, even with improvement in OCD symptoms due to medication, there seem to be factors beyond OCD symptom severity that affect QOL [15]. Thus, more investigation into what those factors are is warranted, so that the goal of improved QOL can be targeted appropriately in clinical settings.

3.2. Psychotherapy

Psychotherapy can dramatically impact a patient's understanding of OCD, their coping, and their QOL. Exposure and response prevention (EX/RP) [34], is the first-line treatment for OCD and both individual and group CBT have a positive impact on QOL in OCD [11, 35]. Similar to SSRIs, CBT shows improvement in the four domains of the WHOQOL-BREF QOL [36].

Psychotherapy should include a directive approach to QOL improvement. Individual domains of QOL, specific symptom spectrums, and modifiers of adherence should be addressed. Considering that adherence may be influenced by a patient's baseline QOL and comorbidities as suggested in Maher et al. [37], QOL outcomes may improve if these factors are targeted by practitioners. In addition, utilization of motivational interviewing (MI), a client centered goal-oriented method designed to motivate patients to change, may improve adherence and QOL. While Simpson et al. [38] found that EX/RP both with and without motivational interviewing showed significant improvement in symptom reduction and QOL,

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findings from both Maltby et. al. [39] and Merlo et al.[40] suggest that motivational interviewing may yield more adherence and may accelerate treatment progress. Practitioners should also target individual problems such as victimization, safety, employment, and finances as suggested in Saxena et al. [36]. Overall, more research attention should focus on the specific aspects of psychotherapy that would be most useful for QOL improvement.

4. Discussion

OCD appears to hamper patients' ability to enjoy life and their quality of life just as strongly as other severe mental illnesses like major depression and schizophrenia and just as much as other medical illnesses. Quality of life in those with OCD is also largely affected by psychiatric and medical comorbidity. Comorbid symptoms, especially depressive symptoms, can be even more disruptive to QOL than OCD symptoms alone. However, QOL in OCD can improve with medications and with CBT and EX/RP psychotherapy. Efforts should be made to encourage treatment adherence and to address factors that may modify response to treatment and improvement in quality of life.

Clinicians and researchers should use QOL tools both at baseline and during follow-up in the treatment course in order to assess treatment efficacy and to monitor the progress toward recovery from OCD and recovery of quality of life. Research with larger samples should examine the effects of clinical factors such as symptom severity and subtypes. It may also be useful to develop a tool specifically designed to measure quality of life in OCD so that researchers and clinicians are more sensitive to this population [41]. Additionally, as has been studied in depression using the Individual Burden of Illness Index for Depression or IBI-D, tools that incorporate QOL, functioning, and OCD symptom severity could be designed [42] and randomized controlled large sample studies should be designed with three outcome measures: QOL, functioning, and OCD symptom severity. Interventional research to improve QOL in OCD is also needed. Exercise has also demonstrated effectiveness in reduction of OCD symptoms and in improving QOL [43]. and initial evidence shows the utility of the following interventions in other psychiatric disorders: exercise [44], nutrition [45], future-directed therapy,[46] meditation [47], massage [48], music [49], and humor [50]. Interventional research could further the notion that the ultimate success of treatment lays in more than remission of symptoms, but also the improvement of quality of life [51].

Conclusion

QOL assessment in both clinical and research settings should be assessed at baseline and during follow up so that we can examine the disease burden, monitor treatment efficacy, and strive for true relief. Thus, treatment providers should endeavor to reach symptom remission while also striving for improvements in functioning and quality of life.

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Chapter X

Investigation on Dimensions of Obsessive-Compulsive Symptoms in Patients with Tourette Syndrome

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Abstract

Tourette syndrome (TS) is a chronic neuropsychiatric disorder defined by multiple motor and vocal tics, and characterized by frequent comorobidity of obsessive-compulsive disorder (OCD) and obsessive-compulsive (OC) symptoms. OCD comorbid with chronic tic disorders including TS is specified as tic-related OCD. Dimensional approach of OC symptoms is recognized to be important not only in patients with typical OCD but also in patients with tic-related OCD. It is suggested that OC symptoms of aggression and symmetry dimensions are associated with tic-related OCD.

In this study, we described OC symptom dimensions in TS patients and investigated relationship between OC symptom dimensions, other clinical features including tics, anxiety and depression, and neuropsychological performance with focusing on aggression and symmetry dimensions.

OC symptoms of miscellaneous dimension were found most frequently, followed by those of symmetry and aggression dimensions. When change percent of Dimensional Yale-Brown Obsessive Compulsive Scale (DY-BOCS) severity score in each dimension from the worst ever period to the current survey was calculated, aggression dimension showed almost no change, and contamination dimension showed largest decrease,

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followed by symmetry and miscellaneous dimensions. When global functioning was compared between patients with OC symptoms of aggression dimension as prominent OC symptoms, patients with OC symptoms other than aggression dimension, and patients without OC symptoms, it was lowest in the patients with aggression dimension. DY-BOCS severity scores for aggression dimension correlated significantly with State Trait Anxiety Inventory (STAI) trait scores in adolescents and adults with TS. When findings of neuropsychological tests were compared between patients with OC symptoms of aggression dimension, patients without OC symptoms of aggression dimension, and controls, the patients with aggression dimension performed most poorly on perseverative errors of the Wisconsin Card Sorting Test (WCST). In contrast, OC symptoms of symmetry dimension had almost no significant impact on clinical features including tics, anxiety and depression, and neuropsychological performance.

In TS patients, OC symptoms of aggression dimension were associated with low global functioning, trait of anxiety especially in adolescents and adults, and poor performance on the WCST. It is suggested that OC symptoms of aggression dimension have an important role for TS and/or tic-related OCD.

1. Introduction

Tourette syndrome (TS) is a chronic neuropsychiatric disorder defined by multiple motor and one or more vocal tics, which start before 18 years of age, and persist for at least 1 year (APA, 2000). TS is characterized by comorbidity of various neuropsychiatric disorders and symptoms, especially obsessive-compulsive disorder (OCD) and obsessive-compulsive (OC) symptoms. Over a half of patients with TS have some OC symptoms (Hounie AG et al., 2006).

OCD comorbid with chronic tic disorders including TS is specified as tic-related OCD. In patients with typical OCD, compulsion occurs in order to neutralize anxiety caused by obsession. On the other hand, compulsion in patients with tic-related OCD tends to occur in pursuit of "just right" perception (Leckman JF et al., 1994) and to be related to difficulty of impulse control.

Dimensional approach of OC symptoms is recognized to be important not only in patients with typical OCD but also in patients with tic-related OCD. A meta-analysis of factor analytic studies of OCD has shown a four-factor structure of OC symptom dimensions consisting cleaning, forbidden thoughts, symmetry, and hoarding (Bloch MH et al., 2008). Recent studies of OC symptom dimensions in OCD patients have found that the symmetry/ordering (Labad J et al., 2008) and the aggressive/checking (Nikolajsen KH et al., 2011) dimensions were related to comorbid tic disorders.

Patients with both TS and OCD were suggested to have higher rates of anxiety and mood disorders than those with only TS (Wanderer S et al., 2012). A year-wise increase of mood disorders was found also in patients with TS regardless of comorbid OCD. OC symptoms and depression appeared to have a greater negative impact on QOL in patients with TS than tics (Eddy CM et al., 2011).

It is suggested that TS and OCD share a common dysfunction in cortico-striatal-thalamo-cortical (CSTC) circuits. Comorbid OC symptoms might affect neuropsychological aspects of TS (Eddy CM et al., 2009). As different OC symptom dimensions are suggested to be associated with a distinct pattern of neural activity (Mataix-Col D et al, 2004), different

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neuropsychological deficits might be found according to dimensions of comorbid OC symptoms.

In this study, we described OC symptom dimensions in TS patients and investigated relationship between OC symptom dimensions, other clinical features including tics, anxiety and depression, and neuropsychological performance with focusing on aggression and symmetry dimensions. For this purpose, we analyzed and examined datasets of our previous works again in study 1 and in study 3, and analyzed a new dataset in study 2. Then, we discussed results of the three studies with recent research of dimensional approach of OC symptoms.

2. Description of OC Symptom Dimensions and Relationship between OC Symptom Dimensions and Other Clinical Features (Study 1)

2.1. Description of OC Symptom Dimensions

Subjects were 40 patients with TS (29 males, 11 females; mean age: 18.8 years old) recruited from a specialty clinic at the Department of Child Psychiatry of the University of Tokyo Hospital (Kano et al., 2010). Assessment instruments included the Dimensional Yale-Brown Obsessive Compulsive Scale (DY-BOCS) (Rosario-Campos MC et al., 2006), the Yale Global Tic Severity Scale (YGTSS), and the Global Assessment of Functioning (GAF). DY-BOCS was developed for assessing presence and severity of 6 OC symptom dimensions: aggression, sexual/religious, symmetry, contamination, hoarding, and miscellaneous. Severity for each dimension is measured on three ordinal scales: frequency, distress, and interference, and its score ranges from 0 to 15. Total tic score of YGTSS consists of motor tic score and vocal tic score, and ranges from 0 to 50.

Out of 40 TS patients, 32 patients had some OC symptoms. OC symptoms of miscellaneous dimension were found most frequently (22: 55%), followed by those of symmetry and aggression dimensions (20: 50%, 18: 45%, respectively). Figure 1 showed mean severity scores of 6 dimensions at the worst ever period and at current survey. When change percent of score in each dimension from the worst ever period to the current survey was calculated only for those with scores >0 at the worst ever period, aggression dimension showed almost no change (mean: 1%), and contamination dimension showed largest decrease (mean: -40%), followed by symmetry and miscellaneous dimensions (mean: -35%, -33%, respectively).

2.2. Relationship between OC Symptoms, Tics and Global Functioning

2.2.1. Aggression Dimension, Tics and Global Functioning

We compared tics between 7 patients with OC symptoms of aggression dimension as prominent OC symptoms (aggression+), 25 patients with OC symptoms other than aggression dimension (aggression-), and 8 patients without OC symptoms (OC symptoms-). Mean total

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tic score of YGTSS was highest in the aggression+ group (29.3, SD: 9.4), and lowest in the OC symptoms- group (21.6, SD: 6.3). We found significant difference in YGTSS vocal tic score between the three groups ($p = 0.04$) by analysis of variance (ANOVA) and the aggression+ group had the highest one. As coprolalia and self-injurious behaviors (SIB) are likely to represent combination of obsessiveness and difficulty of impulse control, which are characteristic of TS, we compared their prevalence between the three groups. Coprolalia was found most frequently in the aggression+ group at current survey, and both coprolalia and SIB were found most frequently in the aggression+ group during lifetime (Figure 2).

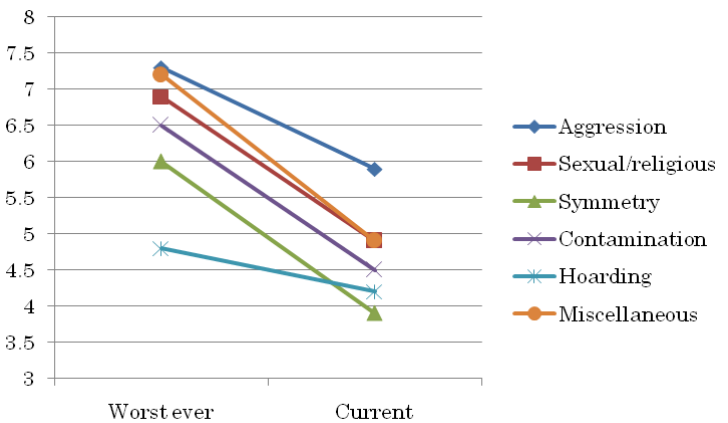


Figure 1. Mean Severity Scores of OC Symptom Dimensions.

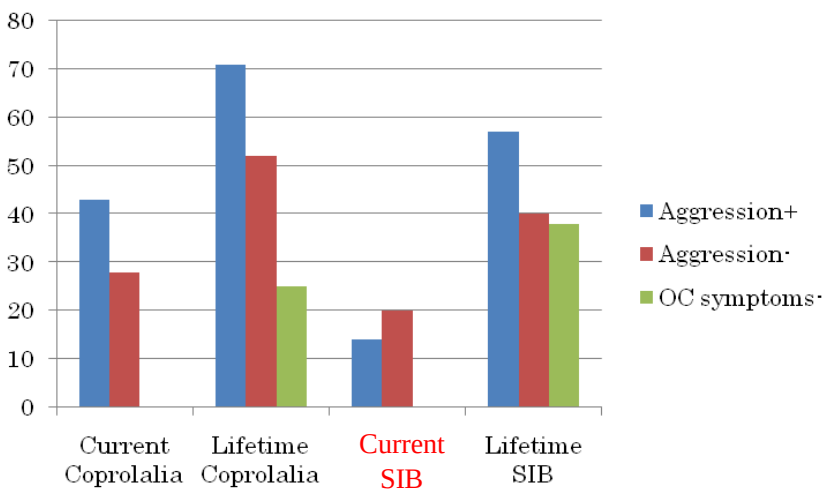


Figure 2. Prevalence of Coprolalia and Self-Injurious Behaviors between the Three Groups in Terms of Aggression.

When global functioning was compared between the aggression+ group, the aggression- group, and the OC symptoms- group, mean GAF score was lowest in the aggression+ group (53.7, SD: 6.3), and highest in the OC symptoms- group (60.6, SD: 14.2).

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2.2.2. Symmetry Dimension, Tics and Global Functioning

We compared tics between 8 patients with OC symptoms of symmetry dimension as prominent OC symptoms (symmetry+), 24 patients with OC symptoms other than symmetry dimension (symmetry-), and 8 patients without OC symptoms (OC symptoms-). Mean total tic score of YGTSS was highest in the symmetry- group (28.2, SD: 7.5), and lowest in the OC symptoms- group (21.6, SD: 6.3). We found significant difference in YGTSS vocal tic score between the three groups ($p = 0.02$) and the symmetry- group had the highest one. Coprolalia was found most frequently in the symmetry- group at current survey, and SIB was found most frequently in the symmetry- group at current survey and during lifetime (Figure 3).

When global functioning was compared between the symmetry+ group, the symmetry- group, and the OC symptoms- group, mean GAF score was lowest in the symmetry+ group (56.8, SD: 11.9), and highest in the OC symptoms- group (60.6, SD: 14.2).

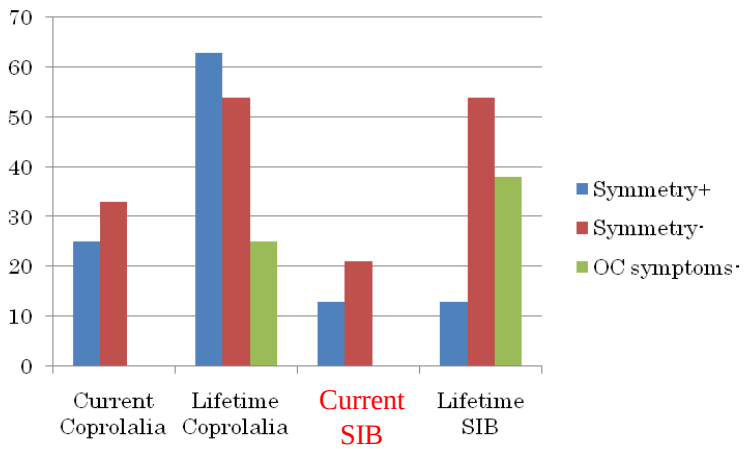


Figure 3. Prevalence of Coprolalia and Self-Injurious Behaviors between the Three Groups in Terms of Symmetry.

3. Relationship between OC Symptoms, Anxiety, and Depression (Study 2)

3.1. Anxiety, Depression, and Other Clinical Features

Subjects were 49 patients with TS (35 males, 14 females; mean age: 18.3 years old) recruited from the specialty clinic of the University of Tokyo Hospital and 43 healthy controls. The subjects were divided into four groups: junior high school students or younger individuals (the younger TS group and the younger control group), and individuals older than the younger group (the older TS group and the older control group). Assessment instruments included State Trait Anxiety Inventory for children (STAI-C) and Depression Self-Rating Scale for Children (DSRSC) for the younger groups, and STAI and Beck Depression Inventory (BDI) for the older groups, in addition to DY-BOCS, YGTSS, and GAF. Mean STAI-C-state score was 31.2 (SD: 8.8), mean STAI-C-trait score was 36.5 (SD: 6.4), and mean DSRSC score was 10.5 (SD: 5.5) in the younger TS group. Mean STAI-state score was

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41.9 (SD: 7.3), mean STAI-trait score was 52.9 (SD: 9.9), and mean BDI score was 8.2 (SD: 6.7) in the older TS group. When we compared anxiety and depression between TS patients and controls, mean BDI score in the older TS group was significantly higher than that in the older control group by t-test ($p = 0.035$). There was no significant difference of anxiety between the older groups. There was no significant difference of both anxiety and depression between the younger groups. When correlation between anxiety, depression, tics, and global functioning was examined in TS patients, DSRSC scores correlated significantly with both STAI-C-state and STAI-C-trait scores in the younger TS group by Pearson's correlation analysis ($r = 0.62$, $p < 0.01$; $r = 0.63$, $p < 0.01$, respectively). STAI-C-trait scores correlated significantly with YGTSS subscale scores for number of vocal tics also ($r = 0.45$, $p = 0.03$). BDI scores correlated significantly with STAI-trait scores, YGTSS subscale scores for complexity and interference of motor tics, and YGTSS motor tics scores in the older TS group ($r = 0.76$, $p < 0.01$; $r = 0.45$, $p = 0.02$; $r = 0.44$, $p = 0.02$, $r = 0.42$, $p = 0.03$, respectively). STAI-trait scores correlated negatively with GAF scores with marginal significance ($r = -0.37$, $p= 0.06$).

3.2. Aggression Dimension, Anxiety, and Depression

Correlation between anxiety, depression, and OC symptom dimensions was examined in TS patients with focusing on aggression dimension. Although no significant correlation was found in the younger group, DY-BOCS severity scores for aggression dimension correlated significantly with STAI-trait scores in the older TS group ($r = 0.47$, $p = 0.01$).

3.3. Symmetry Dimension, Anxiety, and Depression

Correlation between anxiety, depression, and OC symptom dimensions was examined in TS patients with focusing on symmetry dimension. No significant correlation between anxiety, depression, and OC symptom of symmetry dimension was found both in the younger TS group and in the older TS group.

4. Relationship between OC Symptom Dimensions and Neuropsychological Performance (Study 3)

4.1. Aggression Dimension and Neuropsychological Performance

Subjects were 33 patients with TS (26 males, 7 females; mean age: 18.0 years old) recruited from the specialty clinic of the University of Tokyo Hospital and 18 healthy controls (Matsuda et al., 2012). Neuropsychological tasks of this study consisted of the Wisconsin Card Sorting Test (WCST), the Stroop Test, and the Continuous Performance Test (CPT).

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Table 1. Performance on the WCST

	Agg + (N=11)	Agg - (N=23)	Controls (N=18)	F	*p*	*Comparison*	Effect Size
	Mean±SD	Mean±SD	Mean±SD				agg+ vs agg-
TE	43.2 ± 12.0	49.9 ± 8.7	56.6 ± 5.5	7.8	.00	NC > Agg-, Agg+	0.9
PE	40.4 ± 9.9	50.1 ± 7.8	54.7 ± 6.2	9.9	.00	NC, Agg- > Agg+	1.3
CLR	42.6 ± 11.5	49.8 ± 8.9	56.5 ± 5.3	8.4	.00	NC > Agg- , Agg+	1.0

	Sym + (N=15)	Sym - (N=19)	Controls (N=18)	F	*p*	*Comparison*	Effect Size
	Mean±SD	Mean±SD	Mean±SD				sym+ vs sym-
TE	45.0 ± 12.8	49.6 ± 8.0	56.9 ± 5.5	5.9	.01	NC > Sym-, Sym+	0.4
PE	43.7 ± 11.5	49.3 ± 7.7	54.9 ± 6.2	5.3	.01	NC > Sym-, Sym+	0.5
CLR	44.9 ± 12.7	49.3 ± 8.0	56.7 ± 5.3	6.1	.00	NC > Sym-, Sym+	0.3

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Table 2. Pearson's Correlation Coefficients between Clinical Variables and Subsets on the WCST and the CPT within TS Patients

	Age	Duration of Illness	YGTSS	DY-BOCS	Y-BOCS	"Aggression" OCS	"Symmetry" OCS
TE	-.42 *	-.45 *	-.22	-.04	.06	-.21	-.15
PE	-.36 *	-.34	-.19	-.19	-.14	-.36 *	-.30
CLR	-.42 *	-.45 *	-.15	-.05	.09	-.22	-.16
Commission (X)	-.32	-.23	.03	.04	-.13	.23	.04
Omission (X)	-.35 *	-.32	.15	.05	-.06	-.26	-.30

*: $p<0.05$.

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The WCST measures abstraction and cognitive flexibility. We calculated percentage of total errors made (TE), percentage of perseverative errors (PE), and percentage of cognitive level response (CLR) on the WCST. The Stroop Test measures ability to inhibit the tendency to follow a dominant response. Two error types were calculated on the CPT: omission errors linked to poor attention, and commission errors linked to impulsivity.

The TS patients were divided into two groups with or without OC symptoms of aggression dimension: the aggression+ TS group (n=11) and the aggression- TS group (n=22). Although there were no significant differences in IQ, tic severity, or gender distribution between the two TS groups, the aggression+ TS group was significantly older than the aggression- TS group, and had significantly more severe global OC symptom scores on the DY-BOCS than the aggression- TS group.

We compared the performance of the three groups: the aggression+ TS group, the aggression- TS group, and controls. Multivariate analysis of covariance (MANCOVA) was conducted using groups as the independent variable, age as covariate, and performance on neuropsychological subsets as dependent variables. Table 1 summarizes the group mean performance and statistical comparisons on the WCST. Results of MANCOVA controlling for age and comparing the three groups showed the effect of group was significant (Wilks-λ = .49, $F(18,78) = 1.9$, $p = 0.03$). Comparison between groups showed that main effects of group on WCST were significant. The aggression+ TS group performed poorer than the aggression- TS group on PE t-score of the WCST ($p = 0.003$), and the effect size of difference between the TS groups on PE of the WCST was 1.25. No significant main effects of group were detected on CPT and Stroop Test. Table 2 summarizes correlation analysis within TS patients. DY-BOCS severity scores for aggression dimension negatively correlated significantly with PE t-scores on the WCST ($r = -0.36$, $p < 0.05$). After controlling their ages, correlation between DY-BOCS severity scores for aggression dimension and PE t-scores on WCST was not significant.

4.2. Symmetry Dimension and Neuropsychological Performance

We also conducted a similar MANCOVA controlling for age between three groups: the symmetry+ TS group (n = 14), the symmetry- TS group (n = 19), and controls. Although there were no significant differences in IQ, tic severity, or gender distribution between the two TS groups, the symmetry+ TS group significantly older than the symmetry- TS group, and had significantly higher global OCS scores on the DY-BOCS than the symmetry- TS group. The second MANCOVA with the three groups showed no significant main effects of groups (Wilks-λ = .63, $F(18,78) = 1.1$, $p = 0.34$) (Table 1). DY-BOCS severity scores for symmetry dimension did not correlated significantly with any scores of each neuropsychological subsets in TS patients (Table 2).

5. Discussion

In TS patients, OC symptoms of aggression dimension didn't diminish the severity between the worst ever period and the current survey. This tendency seems inconsistent with

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recent study suggesting that forbidden thoughts dimension was likely to be associated with a waxing-and-waning course in patients with OCD (Kichuk SA et al., 2013). The subjects of that study included both tic-related OCD and non tic-related OCD, and were different from those of current study. Aggression dimension was relatively stable in TS and/or tic-related OCD, but might fluctuate in non tic-related OCD. Comparison between those with OC symptoms of aggression dimensions as prominent OC symptoms, those with other OC symptoms, and those without OC symptoms suggested that aggression dimension was associated with low global functioning. Severity of aggression dimension was associated with trait of anxiety in adolescents and adults with TS also. Recent study suggested that aggression dimension was associated with comorbid PTSD in patients with OCD (Torresan RC, 2013). OC symptoms of aggression dimension combined with trait of anxiety might give impact on global functioning in TS patients. These clinical features might be related with poorer performance on the WCST in TS patients with OC symptoms of aggression dimension.

Compared to aggression dimension, symmetry dimension had a smaller impact on global functioning regardless of high prevalence and possible association with "just right" perception (Ferrão YA et al., 2012) in TS patients. Symmetry dimension showed lower correlation with anxiety and depression, which might be related to less influence on both global functioning and neuropsychological performance.

Although this study has several limitations including small sample size and overlap between aggression dimension and symmetry dimension, we believe that OC symptoms of aggression dimension have an important role for TS and/or tic-related OCD. Further longitudinal investigation of OC symptoms with dimensional approach is necessary for TS and/or tic related OCD.

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